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child & adolescent health

term d

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PAEDIATRIC LIFE SUPPORT

Lecture: RESUS4KIDS Online Module

Danger

- Consider environmental dangers (electrical, sharps, bystanders) and infectious dangers (blood or airborne hazards)
- Eliminate the hazard or move the patient to safety
- Wear appropriate PPE

Response

- Ask "Are you alright?"
- Place a hand on their forehead and shake their arm

Send for help

- Shout for help to attract the attention of nearby people
- Press emergency response button at the patient's bedside
- Call for the arrest team (e.g. 22 22) or triple zero

Open airway

- Head tilt – chin lift manoeuvre can be used but may lead to further damage in situations of suspected spinal injury. In these situations, use jaw thrust without head tilt
- In infants, the neutral position is best
- Nasopharyngeal airway length is estimated from the tip of the nose to the tragus of the ear

1	Select an appropriate size.
2	Lubricate airway with water soluble lubricant and insert a large safety pin through the flange.
3	Insert the tip into the nostril and direct it posteriorly along the floor of the nose.
4	Gently pass the airway past the turbinates with a slight rotating motion.
5	Continue until the flange and the safety pin rest on the nostril.
6	Any difficulty inserting the airway, consider using the other nostril or a smaller tube.
7	Recheck airway patency.
8	Provide oxygen.

- Oropharyngeal airway length is estimated from the centre of the mouth to the angle of the mandible; only use this in unconscious or obtunded patient

1	Open the patient's airway using the chin lift
2	Select an appropriate sized oropharyngeal airway
3	Use a tongue depressor or laryngoscope blade to aid insertion of the airway
4	Recheck airway patency and look for improvement
5	If necessary consider a different sized airway
6	Provide oxygen and consider ventilation by pocket mask or bag and mask

Two initial breaths

- Use the look, listen and feel technique; ear is over the nose, cheek is over the mouth and eyes are looking at

the chest and abdomen. This should be for at least 10 seconds

- Give two effective rescue breaths
- With a self-inflating bag and mask (500 mLs for infant, 1000 mLs for a larger child)
- Ensure inflation only occurs until the patient's chest has risen – over-inflation can lead to air being forced into the stomach

Chest compressions

- Start chest compressions if, after 2 initial rescue breaths, there is no pulse felt over a ten second period OR there is a slow pulse OR the child is unresponsive and not breathing normally
- Finding a pulse: carotid, brachial and femoral
- Compressions should be given at a rate of 100-120 compressions per minute with a ratio of 15 compressions to 2 breaths
- Compress the chest by approximately 1/3 of the depth of the chest, at the lower half of the sternum
- Techniques: two fingers, two hands encircling, one hand

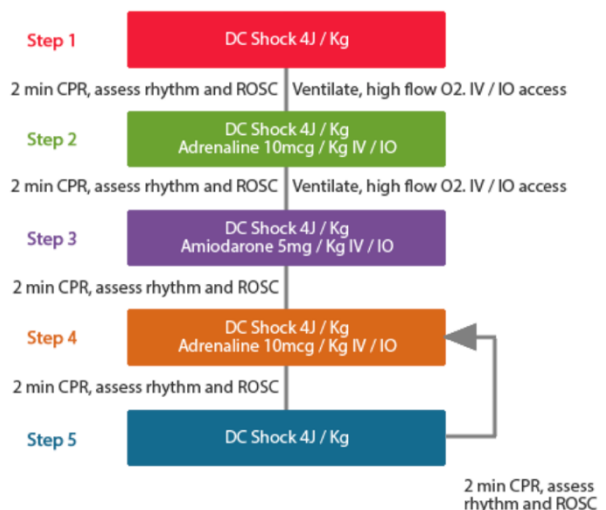
CPR for Infants vs CPR for Children

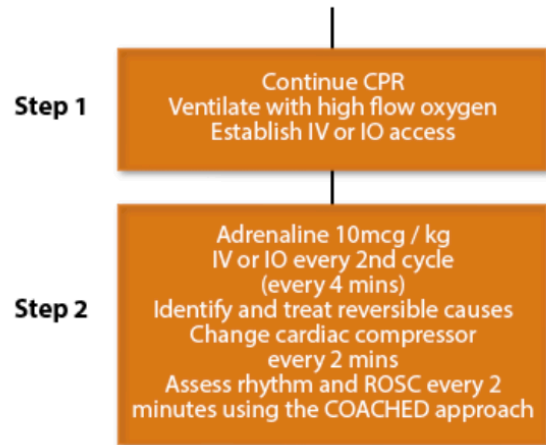
	<1 year old	1 year+ to older child
Head Position	Neutral	Sniffing
Effective Rescue Breaths	Two	Two
Pulse Point	Brachial or Femoral	Carotid, Brachial or Femoral
Compression Technique	Two fingers or two thumbs	One hand or two hands
Compressions per Minute	100-120	100-120
Compressions to Breaths - Two Rescuers with Paediatric Life Support Training	15:2	15:2

Defibrillator or attach monitor

- COACHED
- Cardiac arrest rhythms: shockable (VT, VF) and non-shockable (asystole – most common, PEA)
- Adrenaline at 0.1 mL/ kg of 1:10 000 solution (or 10 mcg per kg), given IV or IO
- Amiodarone at 5 mg / kg, given IV or IO

Shock Protocols and Drug Administration





Choking child

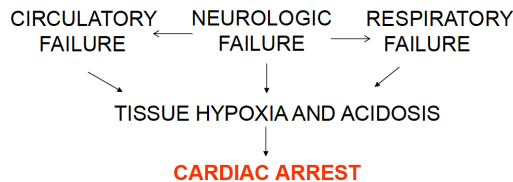
- If effective cough → encourage coughing, call for help
- If ineffective cough and conscious → give up to 5 back blows, then 5 chest thrust, and call for help
- If ineffective cough and unconscious → DRSABCD

EMERGENCY MEDICINE

Lecture: Recognition of the sick child: common ED presentations

Introduction

- In children, death usually occurs through two pathways: circulatory failure (30%) or respiratory failure (60%). This leads to cardiac arrest.



- We want to use the paediatric assessment triangle: Appearance, Work of breathing and Circulation to Skin
- There are some classic red flags are not to be missed
 - Purpuric rash – *Meningococcus*
 - Bulging fontanelle – Raised ICP, e.g. meningitis, SAH. Check at 35 degrees
 - Biphasic stridor – Glottic or subglottic lesion
 - High pitched scream – Meningitis or other head-based illnesses
 - Bile stained vomiting – Surgical emergency
 - Grunting respiration – End-stage of respiratory compromise
- There are serious illnesses with a quick remedy: anaphylaxis, poisoning, hypoglycaemia, adrenal insufficiency and meningococcaemia
- Vital signs are very important: temperature, heart rate (increases with screaming), respiratory rate, blood pressure, oxygen saturation – as well as growth in a child

Primary assessment

- Ask about activity (eating or sleeping) and alertness
- Airway: patency, look at the position child is in, drooling, secretions, noisy breathing, seesawing chest movement
- Breathing: divide into effort or respiratory distress (look at respiratory rate, recession, nasal flaring, peeping, accessory muscle use), efficacy (SpO₂) and effect (obtunded, drowsiness, tachycardia, pallor).
 - Inspiratory stridor = UA pathology (cross at)
 - Expiratory wheeze = LA pathology
 - Grunting = airspace pathology
- Circulation: HR, CRT, radial pulse, pallor, cyanosis
 - Hypotension is a pre-terminal sign
- Disability: AVPU (pain on nailbed, angle of mandible) + fluid intake/outtake + posture (decorticate/decerebrate)
 - Reduced fluid intake (= less than half of usual), reduced urine output (= less than 4 wet nappies in 24 hours)
- Exposure: Looking for rash, petechiae, bruising, # and check temperature
- Watch out for serious illness with normal primary assessment; this can occur through immunocompromised, neurodegenerative diseases and ?sepsis

Breathing difficulty

- Upper airway DDx: croup, epiglottitis, retropharyngeal abscess, oesophageal abscess, tracheal abscess
- Lower airway DDx: bronchiolitis, asthma, pneumonia, foreign body
- Croup
 - Caused by a virus, usually parainfluenza, affecting larynx, trachea causing subglottic oedema and air flow obstruction
 - Symptoms: low grade fever, night-time increase in oedema causing stridor, “seal bark” cough, respiratory distress, cyanosis. This recurs on several nights where they present over 3-5 days
 - Treatment
 - Dexamethasone 0.15 mg /kg
 - If more severe, oxygen, nebulised adrenaline
- Epiglottitis
 - Caused by *Haemophilus influenzae* type b, or other bacteria leading to rapid onset breathing difficulties within hours
 - Symptoms: high fever, intense sore throat, difficulty swallowing, drooling, stridor, sitting up / leaning forward / extending neck
 - Management
 - Intubation in theatres for definite treatment: nebulised noradrenaline
- Bronchiolitis
 - 80% of patients are younger than 1 year
 - Caused by virus, usually RSV or human rhinovirus affecting bronchioles causing a fixed obstruction
 - Symptoms: URTI symptoms, gradual onset of respiratory distress, expiratory wheeze, tachypnoea (RR 60 to 100), cyanosis
 - Signs: increased opacification on CXR with upper lobe collapse or consolidation
 - Treatment
 - Fluids
 - Respiratory support: high flow nasal prongs maintaining SpO₂ > 92
 - Nasopharyngeal CPAP
 - No role for bronchodilators (can cause hypoxia with VQ difficulties), antibiotics
- Asthma
 - Symptoms: signs of respiratory distress, coughing, expiratory wheezy, tachypnoea, cyanosis
 - Markers of acute severe asthma: pulsus paradoxus, SpO₂ < 90, accessory muscle use, decreased consciousness, hyperinflation of chest
 - Treatment
 - Oxygen via mask or NP
 - Nebulised salbutamol, then intravenous
 - Prednisone
 - Magnesium sulphate
 - Intubation and mechanical ventilation – last resort due to gas trapping

Vomiting

- Red flags

- Bilious vomiting
- Fever > 39 or persistent temperature < 36
- Colicky or crescendo-type abdominal pain
- Babies and all children < 2 years

Fever

- Assess for clinical signs of toxicity (and serious bacterial infection): pallor, drowsiness, irritability, decreased reactivity, poor feeding, poor tone
- Neonates are at risk of SBI: conduct full septic workup and empiric ABx with a temperature < 36 or > 38 , or if signs of toxicity
- In young infant (1-3 months), clinical signs of severe sepsis are subtle
- In the young child (< 2 years), a period of observation is required with the aim of identifying clinical features to indicate SBI
- In older children (> 2 years) treat in the same way to adults with management determined by history, exam and appropriate. A significant fever is > 39 degrees

Seizures

- Most common cause are febrile seizures, with family history and M>F
- Most seizures are short-lived and require no investigation. Management is thus to protect the airway in semi-prone positioning, ensure effective ventilation, identify immediate brain threats and paracetamol after the fit
- Need to rule out head injury, infection, hypoglycaemia or space-occupying lesion in situations of first afebrile seizure, prolonged seizure, recurrent or new neurological signs
- If highly febrile $\rightarrow$ meningitis or AMS encephalitis
- Management
 - Clear airway and lie on side
 - Give oxygen
 - Check for hypoglycaemia
 - Diazepam 0.2 mg/kg IV
 - Phenytoin 20 mg/kg or PB 20 mg/kg
 - Consider thiopentone, or muscle relaxation or intubation

Trauma

- Transient low BP suggest trauma and bleeding
- Always question non-accidental injury if the injury doesn't fit

ABDOMINAL PAIN

Lecture: Abdominal pain in children

Appendicitis

- History
 - Age – median 8-9 years, but should be considered in persistent abdominal pain in young children < 5 years
 - Pain history – SOCRATES features
- Physical examination
 - Inspection: position, vomit in bowl, vital signs, ask child to cough (check for localised pain)
 - Palpation: start with the LIF to RIF, use both hands to compare RIF/LIF guarding
 - Percussion: start with the LIF to RIF, dullness may indicate a mass
 - Auscultation: BS increased with gastroenteritis, silent with ileus + peritonitis, listen to the chest to exclude pneumonia
 - Rectal exam: not indicated
- Investigations
 - Bedside: UA
 - Bloods: FBC, CRP
 - Microbiology: MSU
 - Imaging: Abdo USS
 - NB: repeated careful physical examination is more reliable than USS in making a diagnosis of appendicitis, but can be useful in excluding ovarian pathology
- Differential diagnosis
 - Mesenteric adenitis
 - Constipation
 - UTI
 - Gastroenteritis
 - Ovarian pathology
 - Pneumonia
 - Omental torsion
- Management
 - Resuscitation + IV fluids
 - Antibiotics (aerobic + anaerobic, gent + ampi + flagyl)
 - Pain relief
 - Surgery (laparoscopic, open, or drainage & interval)

Malrotation

- Clinical features
 - Bilious vomit in first 4 weeks
 - Irritability
 - Abdominal pain
 - Peritonitis (due to bowel ischaemia)
- Diagnosis
 - Plain AXR showing lack of distal gas
 - Upper GI contrast study (to confirm) to show the duodeno-jejunal flexure should show an S shape
- Treatment
 - Surgery: Ladd's procedure, which untwists the bowel and divides peritoneal adhesions to broaden the base of the mesentery
 - Short bowel syndrome is a potential complication

Intussusception

- Intussusception is the invagination of one portion of the intestine into another. 95% are ileocolic (idiopathic, into caecum), while 10% have a pathological lead point (Meckel's, lymphoma, duplication cyst)
- Presentation
 - Child less than 1 year
 - Colicky abdominal pain +/- palpable abdominal mass
 - ?Bile stained vomit
 - Drawing up legs
 - "Red currant jelly" stools – late sign
- Examination
 - Only in late stage may you find abdominal mass (tender) or PR bleeding
- Investigation
 - AXR – mass, air- fluid levels
 - Abdo USS – more sensitive or specific, showing "target" or "doughnut" sign
- Treatment
 - Resuscitation and pain relief
 - Air enema reduction (85% success, esp. with long history and presence of lead point)
 - 5% recurrence within 24 hours, in 30%?
 - Surgery if treatment-resistant

ABORIGINAL AND TORRES STRAIT ISLANDERS

Lecture: Aboriginal & Torres Strait Islander health gap

Take home messages from case

- Cost of healthy food was up to 30% higher in rural area
- Welfare recipients need to spend 50% of disposable income to achieve a healthy diet (c.f. 15% nationally)

Health issues

- Iron deficiency
 - Aetiology multifactorial
 - 20-60% prevalence in remote settings
 - Leads to worse health and cognitive function
- Otitis media
 - Occurs earlier (1 week onwards c.f. 6 months or later)
 - Occurs more frequently (up to 94% in remote settings)
 - Lasts longer
 - More complications, e.g. perforations, chronic suppurative OM (perforation with 2 + weeks drainage), prevalence of 25-40% in remote settings
- Obesity
- MRSA, leading to empiric management of clindamycin (adults) and cotrimoxazole (children)
- Rheumatic heart disease
 - Acute rheumatic fever occurs at 3 per 1000 children, which is 100-fold increase
 - Prevention is primary = throat swab and empiric IMI penicillin, and tertiary = benzathine pen G IMI monthly
- Chronic kidney disease
 - Aetiology
 - In utero – foetal origins of disease hypothesis of decreased nephrons
 - In adults – see more proteinuria in young adulthood than adolescence or childhood, assumed to be acute post-strep GN associated
 - Prevalence
 - 45% all hospitalisations for Aboriginal people
 - CKD underlying 1/7 Aboriginal deaths
 - Prevention: treat scabies and skin infections, though routine screening is not indicated (except in diabetes)
- Foetal alcohol spectrum disorder
 - Indigenous women more likely to not consume alcohol, but of those who do, they consume more
 - Prevalence is unknown, but common in some remote settings (Fitzroy crossing)
 - Prevention: restrict supply, restrict demand, harm minimisation

The health-related gap

- Life expectancy for females = 76 (83)
- Life expectancy for males = 72 (80)
- Infant mortality 6 per 1000 (compared to 3 per 1000)

- Home ownership, Year 12 retention rate and current daily smokers improving
- Child mortality, maternal smoking decreasing
- Immunisation increasing

Lecture: Aboriginal & Torres Strait Islander other gaps

Demographics

- 700 000 Aboriginal and Torres Strait Islander people in Australia
- 25% live in remote settings; most live in NSW (31%) and Qld (28.5%)
- Younger age distribution

Identification

- 3 components: Aboriginal, identifies as Aboriginal and accepted by the community
- For patients, just ask!

Other gaps

- Funding gap: Despite the at least 2.5-fold burden of disease and disability that Aboriginal and Torres Strait Islanders face, the relative health spend is only a 47% increase
- Research gap: 60% of research is coming from remote areas, despite that not reflecting where Indigenous Australians live

Additional immunisations

- BCG at birth (NT, Qld, northern SA)
- Influenza 6 months +
- Hep A at 1 year, 2 doses (NT, Qld, SA, WA)
- 13v PCV (Pneumococcus) booster at 2 years
- Hep B as adults
- 23v PCV (Pneumococcus) at < 50 with risk factors, or 50 years +

Specific Aboriginal health initiatives

- Aboriginal community-controlled health services
- Aboriginal child health checks (growth, nutrition, ear health, immunisation)
- Closing the gap initiatives
- Specific guidelines, e.g. otitis media, alcohol use
- Immunisation differences

ADOLESCENT MEDICINE

Lecture: Adolescent health: the basic facts

HEADSS

- Home
- Education, eating, exercise
- Activities and peers
- Drugs including alcohol and tobacco
- Sexuality
- Suicidality and depression
- Sleep, safety, spirituality and social media

Developmental tasks of adolescence

- Establish realistic body image
- Establish a self-identity that is realistic, positive and stable
- Achieve appropriate independence from parents and other adults
- Integrate into the peer group
- Become comfortable with one's own sexuality and intimate relationship
- Acquire skills for a future vocation
- Develop a morality and value system
- NB: All of these tasks may be affected by physical or mental illness

Definitions

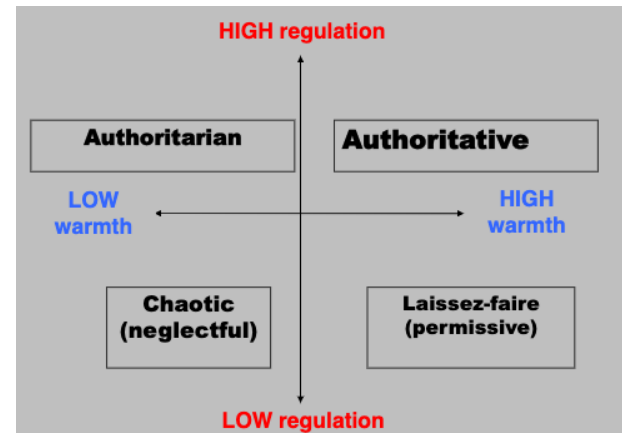
- Adolescence 10 – 19 years
- Puberty 10 – 16 years
- Youth 10 – 24 years
- Young adult/emerging adulthood 18 – 25, up to 30
- Australia has 4 million young adults (20% of population)

Adolescent health

- No increase in gains in health and wellbeing for adolescence as seen in childhood
- Drivers of morbidity and mortality include MH, injury, risk taking, reproductive health, some NCDs
- However, much of this morbidity is overlooked
- Some young people are more at risk from poor health outcomes, this includes
 - Lack of resilience (social network, connections, academic/social competence, outside activities, internal locus of control)
 - Isolation
 - Marginalisation
 - Rural and remote
 - Related to higher death rates
 - Less access to GPs and specialists
 - Less educated
 - More likely to be unemployed
 - Immigrants and refugees
 - Intellectual or physical disability
 - Living in chaotic families
 - Homeless youth
 - Gender or sexuality-diverse
 - Inadequate social skills, incl. ASD
 - Indigenous Australians
 - Mortality 2x (6x from assault, 4x from suicide)
 - 10x STI
 - 6x hepatitis
 - 6x teenage mother

- 6x child protection system
- 15x in juvenile justice, prison
- 2x unemployed
- 3x overcrowded housing
- 2-3x daily smokers

Authoritative parenting



- Authoritative parents are warm, involved and responsive to wellbeing needs
- They are consistent but responsive to development change
- They are firm and strict, and demands are around maturity and achieving best potential
- They foster and encourage psychological autonomy

Lecture: Adolescents and sleep

Changes to sleep during adolescence

- The required nightly sleep duration declines during adolescence
- The average 16-year-old requires 8.5 hours of sleep per night
- There is a sleep onset delay, which means they tend to fall asleep later and wake later in the morning
- There is a marked reduction in SWS during adolescence, with a 40% reduction from age 10 to 20 years → sleep is less refreshing
- REM sleep decreases in absolute terms, but remains constant as a proportion of total sleep time

Practical implications of adolescent sleep patterns

- Permanent jet lag
- Increased daytime sleepiness, due to delayed sleep onset and less restorative sleep
- Yo-yo sleeping, with large discrepancies between weekdays and weekends
- 17-20% of Australian adolescents do not get enough sleep
- Consequences of poor sleep quality and quantity
 - Risk of unintentional injury
 - Poor school performance
 - Low mood
 - Difficulty staying focused and completing tasks
 - Increased impulsivity, aggressive behaviour
 - Increased likelihood of stimulant use
 - Increased weight and obesity

How do we diagnose sleep disorders?

- *Sleep diary*: This records 24-hr sleep and wakefulness for 1-3 weeks, kept by young person + parents, and

helps identify trends and problems in the sleep-wake cycle

- *Actigraphy* (including personal health monitors): This is a device on wrist/ankle that records motion with a light sensor, where you can download recordings to determine periods of wakefulness and sleep and correlation with day/night-time. Use this with a sleep diary to increase ability to accurately define sleep-wake cycles
- *Sleep study or PSG*: Useful for OSA suspected or where symptoms cannot be explained by the above investigations. Sleep is monitored for 6 hours measuring EEG, EOG, EMG, ECG, nasal pressure + oro-nasal airflow, oxyhaemoglobin and end-tidal CO₂, movements of chest and abdomen, video recording

What are the common sleep disorders?

- Poor sleep hygiene
- Delayed sleep phase disorder
- Insomnia
- Sleep disordered breathing to obstructive sleep apnoea

Poor sleep hygiene

- Most common sleep problem in youth
- Environmental factors that contribute
 - Bedroom light, noise
 - Electronic devices
 - Homework late at night
- Behavioural factors that contribute
 - Irregular or yo-yo bedtimes
 - Eating dinner before bedtime
 - Strenuous exercise before bedtime
 - Stimulant consumption
 - Other anxiety and mood disorders

Delayed sleep phase disorder

- Diagnostic features (per ICSD)
 - Sleep onset and wake times intractably later than desired, which is incompatible with school and work attendance
 - Actual sleep onset time consistent each day
 - Little or no difficulty maintaining sleep once it has begun
 - Extreme difficulty waking at desired time in the morning
- This is a circadian rhythm disturbance – that it is unusually prolonged.
- Presents in adolescents and young adults with a reported prevalence of 7-16%, often with positive family history (40%)
- In severe cases, there can be a complete reversal of the sleep wake cycle (often associated with unstructured life, homelessness or psychiatric conditions)
- Management
 - Reschedule sleep time ½ hour each week
 - Improve sleep hygiene: bed = sleep, avoid caffeine, large meals and exercise, establish evening routine
 - Establish morning routine: waking at predetermined time, go outside, no daytime naps, incorporate exercise, use relaxation techniques
 - Consider melatonin in refractory cases

Insomnia

- At least 25% of adolescents describe at least one symptom of insomnia

- Diagnosable insomnia occurs in 4-5% of adolescents
- Diagnostic criteria
 - Difficulty initiating sleep and maintaining sleep
 - Early morning waking and non-restorative sleep
 - For at least 4 weeks causing significant impairment and/or distress
- Management
 - Treat comorbid mood disorders
 - Address situational issues (e.g. bereavement, bullying)
 - Reassurance
 - Avoid use of sedatives

Sleep disordered breathing

- This is a spectrum, ranging from snoring to OSA with severe desaturation
- Characterised by abnormally high upper airway resistance due to a combination of factors
 - Abnormal neuromuscular control e.g. cerebral palsy, Arnold-Chiari malformation
 - Congenital abnormalities of the airway e.g. laryngomalacia, subglottic stenosis, micrognathia
 - Airway shape e.g. obesity, tonsil enlargement, allergic rhinitis, infiltrative disorders
- The two most common causes in adolescence is obesity and adenoid and tonsillar enlargement
- Diagnosis requires a sleep study
- Management
 - Medical: weight management, treat allergic rhinitis, non-invasive PPV (nocturnal CPAP/BiPAP)
 - Surgical: adeno-tonsillectomy which is curative in a large proportion with mild-moderate OSA without other complicating factor

Lecture: Chronic illness & disability

What are the major morbidities in adolescent?

- Mental health
- Chronic illness and disability
- Health risky behaviours, which can impact health now and health in the future, e.g. obesity, sedentary behaviour, smoking, substance use

Epidemiology in young Australians

- In 10-14 year olds: asthma (20%), ADHD (15%), SD (9%)
- In 15-19 year olds
 - Females: anxiety and depression (25%), asthma (14%), eating disorders (5%)
 - Males: anxiety and depression (16%), asthma (14%), ASD (9%)
- 35% of 10-19 year olds have physical disorders

What is a chronic illness?

- A chronic illness is a permanent condition with residual disability, involving a long period of supervision, observation or rehabilitation
- 10-15% of young people have a chronic or disabling conditions; approximately 85% of children with these survive to adulthood
- Prevalence data

- Obesity 20-25%
- Asthma 16%
- Anorexia 2%
- T1DM 0.2%
- CF 0.04%
- Childhood cancer survivors 0.12% (generally, 70% survival rates)
- Epidemiology
 - Increasing survival rates (e.g. CF, CHD, ESKD)
 - Increasing prevalence (e.g. DM, IBD, ALL, asthma)
 - Conditions that appear earlier (e.g. obesity comorbidities, HIV/AIDS)

Risky behaviours with chronic illness

- Adolescents with chronic illness may have less robust peer networks, delayed independence
- They may have lower levels of knowledge about sexuality and contraception
- It is important to note that young people with chronic illness to not avoid risk behaviours, in fact they are more likely to
 - Smoke daily
 - Be cannabis users
 - Have performed violent or antisocial acts
 - Report 3 or 4+ risk behaviours

Transition care

- Transition is the purposeful, planned movement of healthcare from paediatric to adult services. This is a point where there can be a loss to healthcare
- The largest numerically transitioning group is diabetes and other endocrine disorders
- Adult medicine has less multidisciplinary care and expect patients to function more autonomously; they may have limited experience in adolescent and young adult diseases
- What can we do?
 - Start talking about transition early
 - Ensure increasing confidence in self-management
 - Re-education of the chronic condition
 - Shared paediatric and adult clinics
 - Specific transition support services
 - Adult clinics to adopt a proactive approach towards 'no show' patients
- What are the consequences to ineffective transition?
 - Loss of illness control
 - Presentation to ED with life threatening conditions e.g. DKA
 - Physical or mental health deterioration
 - Early appearance of secondary complications
 - No primary care access point if paediatric services have acted as a de facto GP service

Lecture: The brain and risk taking

Brain imaging in adolescents

- Modalities
 - Structural MRI: provides information on size, amounts of grey and white matter
 - Functional MRI: measures haemodynamic response in a specific brain area in response to stimuli, where the inference that increased blood flow is for providing increased oxygen for neuronal activity

- Changes seen in adolescence
 - Increase in brain size and white matter
 - Neuronal overgrowth and then subsequent pruning
 - Grey matter peaks for M/F at cortical (12/11), temporal (16.2/16.7) and parietal (11.8/10.2); prefrontal cortex is the last to mature
 - However, brain plasticity is a lifelong phenomenon

Brain development corresponds to stages of adolescence

- Early (10-14)
 - Puberty heightens emotional arousability, sensation seeking, reward orientation
 - Cognitive processes are maturing, improving reasoning, information processing and expertise
- Middle (14-18)
 - Cognitive processes become capable of planned, abstract, complex and multidimensional thought
 - Period of heightened vulnerability due to regulation of affect and behaviour → risk taking
- Late (18-21)
 - Maturation of frontal lobes leads to competency in regulation, however this takes time
 - This occurs late in development, with the consequent risk that trajectories may be sub-optimal

What influences neuronal development?

- Alcohol and other drugs, due to the activation of reward pathways and development of alternative neuronal connections
- Chronic stress may have effects on HPA axis, causing effects on learning and memory and PFC development
- Intellectual disability may increase the time to maximum cognitive maturity

Lecture: Adolescent substance use

Epidemiology

- Analgesics are the most commonly used substance, then alcohol, tobacco and cannabis
- Risk factors
 - Individual: impulsivity, early initiation, mental illness
 - Family: poor supervision, discipline, parental attitudes, low income, poor housing
 - School: early low achievement, aggression or bullying, lack of commitment to school, school disorganisation (fights, skipping)
 - Peer & community: alienation, friend use or condone, community disorganisation, availability of drugs, high turnover and lack of neighbourhood attachment

Smoking in adolescence

- One of 3 top risk factors for adult disease
- Epidemiology
 - 90% of adult smokers started as teenagers
 - Start between 12 and 16 years
 - 8% of 14-15 y.o. and 17% of 16-17 y.o. smoked in the last month

- Teenage smoking is influenced by peers, social anxiety, desire to control weight, cost, access and advertising
- If parents are smoking, the risk increases 3-fold (conversely, parental disapproval or quitting reduces this risk)

Alcohol in adolescence

- 11% of 12-13 y.o., 28% of 14-15 y.o. and 53% of 16-17 y.o. consumed alcohol in past month
- Teenage drinking leads to increased unintentional injury, risky sexual behaviours, risk of adult dependency (4x), mental health issues, other substance use and abuse. It also affects memory and concentration, even up to 1 week later
- Teenage drinking is influenced by access, cost and advertising
- Parents are the modellers of alcohol intake and the providers of alcohol for underage drinkers

Cannabis in adolescence

- Males use more than females, up to 10% males and 5% of females in 17-year olds
- Effects of cannabis
 - Altered perceptions
 - Impaired concentration, judgement, memory and cognition
 - Increased heart rate, lower blood pressure
 - Increased appetite
 - Conjunctival injection
 - Causes symptoms of anxiety, panic attacks, restlessness
 - Causes derealisation, depersonalisation, paranoia, hallucinations, psychosis
- May be used to self-medicate anxiety and depression
- Consequences
 - School and work drop out
 - Poor cognitive development
 - Gateway to other drugs

Energy drinks in adolescence

- Contain high levels of caffeine
- Can be used as mixers, used with other drugs, and can lead to cardiac/renal effects when combined with dehydration

Lecture: Adolescent Sexual and Reproductive Health

What is risky sexual behaviour?

- Sexual behaviour that leads to unplanned pregnancy or STI spread
- Early age of sexual intercourse debut
- Number of partners over a short time period
- Poor use of condom or contraceptives

What are the predictors of risky sexual behaviour?

- Problem behaviour in childhood and adolescence
- Family factors: father absence, poor parental monitoring, time home alone, poor quality of family relationships
- Individual: poor academic ability, doesn't attend school, low self-efficacy and self-esteem, poor mental health, other risky behaviour
- Early dating, non-intercourse sexual behaviours
- Intentions, perceptions, perceived norms

Epidemiology of sexually active teens

- Behaviours
 - About 25% of Year 10s and just under 50% of Year 12s have had sexual intercourse
 - Over 20% of Year 10s have reported 3+ partners in the last year, with more reporting oral sex. This tends to be more common than in Year 12s.
 - Majority have experienced some form of sexual activity by Year 10; 40% have experienced oral sex
 - Nearly 40% have reported not using a condom in their last sexual experience
 - Nearly 40% reported OCP
 - 20% reported being drunk or high in their last sexual experience
 - 6% same-sex attracted, 5% (M) and 15% (F) both-sex attracted
- Beliefs and perceptions
 - HIV knowledge is high, but STI knowledge is poor
 - Most report positive feelings after sex
 - Unwanted sex among females has increased
 - Half reported sending an explicit photo, 70% reported receiving one
- STIs
 - Chlamydia increasing over the last few decades, higher in females than males
 - Highest in the 20-24 group, then the 15-19 group

Teenage pregnancy

- Epidemiology
 - Most are unplanned and over 50% end in abortion
 - Where they continue, teenagers have higher rates of pre-term births, LBW or neonatal death
 - Mothers are more likely to experience PND, DV, low educational attainment, welfare dependence
 - Children are more likely to be abused, have learning difficulties, have emotional and behavioural problems and have early pregnancy themselves
- Client profile more likely to be
 - Indigenous
 - Single
 - Living in low SES
 - Smoker
- Consequences
 - Increased perinatal mortality
 - Increased risk of STI
 - Less likely to have antenatal care

ASTHMA

Lecture: Asthma diagnosis and assessment

Differences between paediatric and adult asthma

- Preschool wheeze may not become asthma
- Spirometry is not used as often in paediatric cases and can be less reliable, as patients may have difficulty producing reproducible results
- There are different signs to decide assessment of acute exacerbations, e.g. respiratory distress manifests differently, chest tightness may not be reported
- Paediatric asthma has the capacity to deteriorate quickly
- There exist issues of overuse of steroids

What is asthma?

- Reversible airflow limitation
- Resulting in recurrent episodes of wheeze, increased WOB, cough and chest tightness
- Due to chronic airway inflammation and airway hyperresponsiveness with intermittent airway narrowing due to bronchoconstriction, bronchial mucosal oedema and bronchial secretions

How do we diagnose asthma in children?

- Consistent clinical features
 - *Episodic* respiratory symptoms: wheeze, breathing difficulty, chest discomfort, cough
 - Examine triggers – URTIs (less likely) or multi-trigger wheeze
 - Atopic history
- Exclusion of other causes
 - Recurrent wheeze (50% of children will wheeze, but most will not be persistent by age 6 years)
 - Wheeze: infections (e.g. mycoplasma), inhaled foreign body, anaphylaxis, tracheomalacia and other structural airway issues
 - Cough: pertussis, CF, suppurative lung disease, protracted bronchitis
 - Dyspnoea: hyperventilation or anxiety, poor cardiopulmonary fitness
- Demonstration of reversibility
 - As spirometry is not always possible, can try clinical Ventolin or preventer response
 - FEV₁ > 12% or 200 mL
 - Recurrent viral-induced wheezes will respond to Ventolin

General asthma assessment

- HPC
 - Asthma diagnosis
 - Pattern – infrequent intermittent, frequent intermittent, mild/ moderate or severe persistent
 - Exacerbations – severity and frequency
 - Interval symptoms – daytime, exercising, night-time
 - Triggers
- Family and personal history of asthma, atopy, eczema
- Previous wheezing episodes
 - Flares frequency
 - Severity
 - Triggers

- Medications used
- Interval symptoms – nighttime, play or exercised induced, effect on sleep, effect on school or work

Lecture: Asthma management

Asthma medications

- Reliever:
 - Salbutamol (Ventolin and Asmol) are SABAs, used 2-6 puffs (< 5 years) or 4-12 puffs (6 years +)
 - Ipratropium (Atrovent) is an anticholinergic medication, usually only used in emergency situations, with 4 puffs (< 5 years) or 8 puffs (6 years +)
- Preventers
 - Fluticasone (Flixotide junior 50 mcg, 125 mcg, 250 mcg) is an inhaled steroid. The low dose (100-200 mcg/day) are used up to moderate persistent asthma, while the high dose (250-500 mcg/day) are used for severe persistent asthma
 - Montelukast (Singulair) is an oral leukotriene receptor antagonist, for frequent intermittent or for mild persistent asthma, OR exercise-induced asthma. This comes in 5 mg or 4 mg tablets. This can have psychological behavioural side effects
 - Sodium cromoglycate (Intal Forte 5 mg) inhibits mast cell degranulation, for frequent intermittent asthma or exercised-induced asthma. This has to be used 3x a day, and so adherence is not always adequate
- Emergency treatment
 - Oral prednisone (Redipred/Predmix) is a corticosteroid in 5 mg/mL solution, used 1-2 mg/kg/day depending on severity, for a max of 60 mg

What do we choose?

- This is determined by the pattern of asthma, age (dose of Ventolin and ICS) and weight (dose of prednisolone)
- 0-12 months – nothing
- 1-2 years – if persistent asthma, consider montelukast, or ICS if severe persistent
- 2-5 years – montelukast from frequent intermittent to mild persistent, then ICS if more moderate
- 6+ years – same as above

Lecture: Asthma action plans

Asthma action plan

- These are personalised asthma management guides for families and carers. They are given to family members, childcare and school services and sporting organisations
- There are four sections to direct treatment
 - When well
 - When not well
 - If symptoms get worse (acute flare)
 - In an emergency
- When well: Asthma is stable, no night-time symptoms, only occasional daytime symptoms, relievers used only occasionally or before exercise
- When not well: Night-time wheezing, morning symptoms, relievers are used more than usual, and symptoms are disrupting school/life

- If symptoms get worse: Increased wheezing, chest tightness and cough, waking up multiple times at night, and using relievers within 3 hours
- Danger signs: Severe shortness of breath, no relief from medications
- Always use brand names of medications, and tick 'use a spacer'

Medications used in AAPs

- Reliever
 - How many puffs
 - How often
 - Deliver device

	0-5 years	6+ years
When Well	2 puffs as needed	4 puffs as needed
When Not Well	4 puffs every 6-8 hours	6 puffs every 6-8 hours
Acute Flare-up	6 puffs every 3-4 hours	12 puffs every 3-4 hours
Exercise symptoms	2 puffs 10-15 min before exercise	2-4 puffs 10-15 min before exercise

- Preventer
 - How many puffs
 - How often
 - Dose – usually stays the same
- For a flare
 - If any child needs Ventolin more than every 3 hours, go to a hospital
 - Prednisone will be used as a rescue medication (syrup is 5 mg/mL of dose 1 mg/kg/day)
 - Usually 1 dose before medical review

Lecture: Asthma devices

MDI + Spacer

- Under school age children often use a small volume spacer with a face mask to ensure there is a good seal
- School age children usually use a small or large volume spacer
- Consider removing the spacer from age 8
- Advantages
 - Small size, portable
 - Available reliever + preventer
 - Placebo trainer
 - Wash in soapy water and air dry as needed
- Disadvantages
 - Coordination
 - Post-inhalation cough
 - No dose counters
- Space advantages
 - Placebo trainer
 - As effective as nebuliser
 - Inexpensive (compared to nebuliser)
 - Portable
- Disadvantages of spacer
 - Size of the large volume spacer
 - Static charge / issues around not cleaning correctly

Autohaler

- Advantages
 - Breath activated so requires less coordination
- Disadvantages
 - Lever can be difficult to pull into position for younger children
 - Children may stop inhaling after the "puff" or may not be able to inhale enough to

trigger the medication release especially during an asthma flare

- No dose counter
- More expensive than standard MDI

Turbuhaler

- Advantages
 - Small size and portable
 - Minimal taste
 - Cleaning: wipe mouthpiece as needed
- Disadvantages
 - Device must be upright for loading
 - May be difficult to use in an asthma flare-up
 - Post-inhalation cough

Accuhaler

- Advantages
 - Placebo trainer
 - Size
 - Dose counter
 - Cleaning: wipe mouthpiece as needed
- Disadvantages
 - Post-inhalation cough
 - Medication may lodge in the back of mouth and throat

Lecture: RCH videos

What is asthma?

- 1 in 4 children diagnosed at some point
- Children under 5 can have up to 6 puffs, 6+ can have up to 12 puffs
- In severe asthma episodes, use 4 puffs with 4 breaths every 4 minutes

Lecture: Assessment and management of acute asthma

Factors that make asthma more likely

- More symptoms, e.g. wheeze, difficulty breathing, feeling of tightness in the chest, cough
- Recurrent symptoms
- Symptoms worse at night or in early morning
- Symptoms triggered by exercise, exposure to pets, cold air, damp air, emotions
- History of atopy, e.g. allergies, atopic dermatitis, allergic rhinitis
- History in family, e.g. siblings, parents
- Widespread wheeze on auscultation
- Response to treatment trial of reliever or preventer

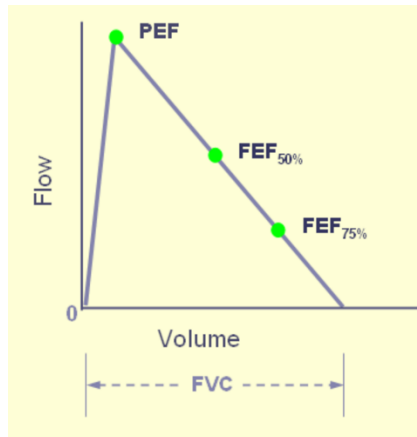
Factors that make asthma less likely

- Symptoms that occur only during viral illnesses
- Isolated cough
- History of moist cough

Lecture: Spirometry

What is spirometry?

- Spirometry is a test that is used to measure the ventilation of the lungs
- It measures dynamic lung volumes from a maximal expiration, by measuring the amount and the rate of air that comes out
- It includes: FVC, FEV1, FEF 25-75, PEF



- Our cut offs are around the lower limit of normal (mean – 1.64*SE)
- FEV1/FVC is > LLN but VC is low = restrictive
- FEV1/FVC is < LLN and VC is low = obstructive
- Classifying severity is using FEV1 % predicted

Spirometry graphs

- Volume-time is a spirogram
- Flow-volume (as above) can help us describe how well patients have done the test, and whether a patient has any abnormalities
- All volumes and flows are reported at body temperature and pressure saturated

How do we know spirometry is acceptable?

- The start-of-test criteria is determined by back extrapolation; back extrapolated volume should be <5% of the FVC or < 0.150 L, whichever is greater. This ensures there is a full inhalation before commencing expiration
- The end-of-test criteria is based about no change in volume (< 0.025L) for 1 second, and the patient has tried to exhale for >3 s in children aged <10 years, and for >6s in patients aged > 10 years
- The repeatability criteria are that there should be at least 3 technically acceptable blows, with FEV1 and FVC agreeing within 150 mL of each other

What are the common problems with spirometry?

- Submaximal effort
- Incomplete expiration or inspiration
- Not closing lips on mouthpiece leaks
- Posture – seated, upright
- Cough
- Glottic closure

How do we assess reversibility?

- Ideally, no SABA in last 4 hrs, no LABA in last 12 hrs
- Perform pre-BD spirometry
- Administer BD (e.g. 4 puffs salbutamol) with MDI + spacer
- Wait 10 minutes
- Perform post-BD spirometry
- Positive BD response is an increase in FEV1 of >12% or >200 mL

Types of abnormalities

- Obstruction: flow limited, e.g. asthma
- Restriction: e.g. volume limited, e.g. scoliosis
- Mixed

Interpretation of spirometry

- Patients results are compared to reference values: age, height, gender, and ethnic origin. HEIGHT has the greatest impact

BURNS

Lecture: Introduction and physiology

Burns holistic care

- A major burn injury causes major physical *and* psychological injury, so requires a multidisciplinary team approach
- This team involves: ICU, specialised nursing, surgery, anaesthetic and pain relief, physiotherapy, dietetics, SW, psychology, speech therapy, rehabilitation
- Scars of ANY size have long term physical and emotional repercussions and require ongoing support
- In NSW, we have 3 burns units in our state-wide burn injury service (SBIS)
 - All major burns referred to the three burns units (2 adult, 1 paediatric)
 - All minor burns referred to LHD, where they confer with burns units through digital photo consultation and refer according to NSW SBIS guidelines

Incidence of burns

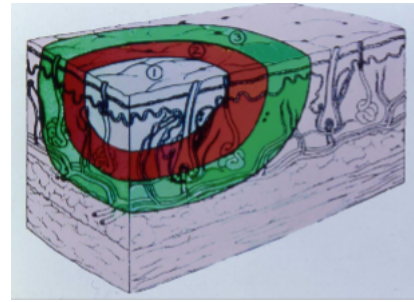
- 1% of Australia/New Zealand each year – of this, 10% require hospital and 10% are severe burns
- In the paediatric age groups, there are 2 peaks
 - <4 years: scalds or contact with hot surfaces, over 80% cover less than 5% of BSA so are treated as outpatients and day stay surgery
 - Adolescence: flame or contact predominate due to risk-seeking behaviour
- Severe burn care is expensive - \$700 000 for a 70% BSA burn, but also requires rehabilitation, time off work, loss of earning power, loss of lifestyle
- Reporting trend in burn patients have not changed over the last few years

Causes of children's burns

- Method of injury
 - Scalds 60%
 - Contact 30%
 - Flame 15%
 - Electrical and chemical rarer, as well as sunburn
- 75% of burns occur in the home
- Water temperature
 - Average temperature of domestic hot water is 70 degrees
 - At 60 it takes 1 second to cause a full thickness burn
 - At 55 it takes 10 seconds
 - At 50 it takes 5 minutes

Burn pathophysiology

- Jackson burn wound model: burn area of necrosis, area of stasis (where blood flow ceases and oedema occurs), area of hyperaemia
- Proper rehabilitation involves revascularizing the zone of stasis, and protection of the microcirculation at the burn site



- Hormonal response in major burns
 - Catabolic hormones increase, like glucagon, cortisol, catecholamines
 - Anabolic hormones decrease, like T4 and GH
- Immune response is depressed in major burns
- Inflammatory mediators are produced

What occurs in a systemic inflammatory response?

- Inflammatory mediators are produced locally in minor burns and systemically in major burns (> 10% BSA)
- These cause fluid and electrolyte imbalance due to increased microvascular permeability leading to production of oedema, loss of IV water and electrolytes leading to SHOCK
- Left untreated, the systemic inflammatory response will lead to other organ involvement, including:
 - oliguria and renal failure
 - paralytic ileus, bacterial translocation, gastroduodenal erosion
 - pulmonary oedema and ARDS
 - cerebral oedema
- The burn surface also provides an area for bacterial invasion → invasive sepsis. There is also immune deficiency as a result of nutritional depletion and exacerbates the risk of sepsis

How do burns affect the airways?

- Upper airway obstruction due to swelling of mucosa from direct heat, or soft tissue swelling from burns in the neck
- Lower airway affected by inhalational burn or chemical burn
- Systemic intoxication of CO or cyanide

How do burns affect the circulation?

- Increase in capillary permeability causing local oedema and general loss of intravascular fluid causing hypovolaemic shock
- In limb or around chest, circumferential full thickness burns can cause extravasation of fluid to the subcutaneous compartment leading to rising pressure in the chest or limb → ischaemia or chest expansion
 - We can prevent this from elevating the limb, and considering doing an escharotomy
- Circumferential burns of the chest can also cause respiratory difficulty from a Cuirasse effect → ventilation and escharotomy
- Major burns can also cause loss of fluid into the peritoneum, or burns of the abdominal wall can produce an abdominal compartment syndrome → escharotomy and laparotomy

Lecture: Evaluation and treatment

Evaluation of a burn

- Primary survey
- First aid – if not given prior, give after primary survey within 3 hours of the burn
 - Remove patient from source of heat
 - Remove clothes
 - Cool the burn with cold running water (15 degree) for 20 minutes
 - Avoid hypothermia
- Fluid resuscitation if burns > 10%
 - IV or IO Hartmann's or N/S
 - Use Parkland formula 3 mLs/ kg / % burn over first 24 hours (first ½ in 8 hrs and second ½ in 16 hours)
 - Aim for urine output 0.5-1 mL/kg/hr
 - Maintenance fluid N/S + 5% dextrose
 - 100 mL/kg/24hrs up to 10kg
 - 50 mL/kg/24 hrs up to 20kg
 - 20 ml/kg/24 hrs > 20 kg
 - Begin ASAP and calculate amount from time of burn
- Assess total BSA of burn
 - Head 18% (up to the age of 1, for each year after, reduce by 1% and add ½ to each leg)
 - Leg 14% each
 - Front/back 18% each
 - Arms 9% each
 - Palm 1% each
- Assess depth of burn, which is related to temperature, length of exposure, first aid and skin thickness
 - Burns are dynamic and heterogeneous, so clinical assessment is unreliable in first 5 days
 - Objective measurement using laser doppler imaging is used, for a validated assessment of depth within 48-72 hours
 - It measures the flow through the superficial capillary dermal plexus

Classification of depth of burns

- Epidermal
 - No blisters, regeneration from intact basal layer
 - Caused by flash flame, sunburn or scald
 - No intervention
- Superficial dermal
 - Caused by sunburn, scald and flash flame
 - Blister – fluid that accumulates after epidermis shears off, pink, blanches
 - Epidermis regenerates from dermal papillae, hair follicles, sweat glands
 - Will spontaneously heal <14 days
- Mid dermal
 - Red non blanching and blotchy (as dermal capillary plexus is damaged). Redness due to extravasation of Hb into dermis
 - Epidermis regenerates from hair follicles and sweat glands
 - Will variably heal in 14-21 days
- Deep dermal
 - Few scattered hair follicles
 - No spontaneous healing – heal by contraction and scarring
- Full thickness – white, charred and waxy

Lecture: Surgical treatment and follow up

Pain relief

- Acute presentation: IV opioids until well analgesed (titrate to effect)
- Outpatient: Pre-dressing preparation with distraction therapy with child-life therapist, conscious sedation and analgesic with oral narcotic + anxiolytic, anticipate need for GA

Initial management of the wound

- If contaminated, wash with soapy chlorhexidine
- Irrigate wound with water and normal saline
- Temporary dressings for transfer
 - <8h, glad wrap
 - >8h, consider antibacterial (Bactigras)
- No place for prophylactic antibiotics

Dressings

- Superficial – moisturise
- Dermal – silver containing antimicrobial dressing (e.g. Acticoat 3 day or 7 day)
- Blisters – should be de-roofed if there is good analgesia, aspirate if not
- No blisters – apply dressing directly to the wound
- Make a decision to graft by d10 in mid or deep dermal burn, because the longer it takes the greatest chance of hypertrophic scars

Skin grafting

- Excise eschar to viable tissue, using a sharp tangential excision (Jankovic). This used to be done with knives but now with Versajet for a Venturi effect
- Apply autograft split skin (epidermal only) using a dermatome
- Sometimes full thickness is appropriate (e.g. palms, ears), but only for small areas
- A major consequence is hypertrophic scars, which are influenced by time to operation and the patient's genetic makeup.

Management of burns scars

- Pressure garments
- Silicone gel applied to the burn
- Splinting to prevent contracture
- Steroid injection to prevent hypertrophic scars
- Laser therapy
- Microneedling
- Surgical excision regrant using dermal template – rare

Psychological support

- Early parental involvement
- Long term psychosocial support, with an emphasis on short term rehab and long-term review at adolescence
- Social worker
- Burns support foundation: family camps and burns camps for all survivors

Lecture: Wound management

Pharmacological management

- Analgesia: oral morphine, paracetamol
- Sedation: oral midazolam

Non-pharmacological method

- Presence of a parent or non-threatening helper
- Distraction with game or screen
- Explaining to the patient and parent what will happen – preparation helps ease trauma

Management

- Cleaning and debridement
- If face & head: eye review, early intubation, airway burn assessment, scalp assessment
- Wound assessment
 - Mode of injury
 - First aid
 - Colour, texture, capillary return
 - Signs of infection
- Antibiotics only for proven sepsis
- Dressing
 - Acticoat: nanocrystalline silver in layers of dressing – comes in Acticoat and Acticoat 7, painful to apply and remove
 - Mepilex Ag+: antimicrobial soft silicone foam dressing with silver, pain free to apply and remove, for mid-dermal burns
 - Flamazine: silver sulphadiazine cream for broad spectrum antimicrobial protection, must be changed every 24 hours
 - Mepilex and mepilex lite: hydrophilic polyurethane foam with soft silicone layer, remains for 7 days, for superficial to mid-dermal burns
 - Bactigras: tullegras with soft paraffin and chlorhexidine, gives antimicrobial cover for Staph, change every 24-48h
 - Paraffin: for superficial burns to the face
- Healed burn care: sorbolene cream with 10% glycerol, apply 4+ times a day

Itch

- Itch is hard to control
- Can use moisturisers, antihistamines

Lecture: Burns management (rehabilitation)

Aims of rehabilitation

- Maintain joint mobility
- Prevent contractures
- Initiate scar management

Techniques of rehabilitation

- Exercises
 - Maintain joint mobility
 - Prevent skin and joint contractures
- Splinting
 - Begins immediately and adjusted regularly
 - Continue until we are confident no joint will contract
 - The position of comfort is the position of contractures
- Pressure
 - Pressures control hypertrophic scars, retain silicon products, prevent contractures and improves cosmetic appearance
 - Can also help control itch and hypersensitivity
 - Worn for 23 hours of day for 12 months
- Contact media
 - Silicone products minimise scarring, soften contractures, improve cosmetic appearance
 - Can be used alone or with pressure

Hypertrophic vs keloid scars

- Hypertrophic scars are a physiological response to an injury, a rapid growth phase, regression increase in collagen formation → scar is elevated
 - Can restrict ROM, affects psychological health
- Keloid is an inherited metabolic alteration to collagen, and grows beyond peripheral border of wound

How do scars develop?

- Healed burn goes into a state of 'overdrive' where there is increased vascularity and new tissue develops in a 'distorted, knotty' fashion
- Scarring depends on TIME to healing (<14 days no scar, >21 or graft definitely scars)
- Scar risk factors
 - Younger age
 - Darker skin
 - Genetic predisposition
 - Length of time to heal
 - Infection
 - Concave areas
 - Areas of high mobility
- Problem areas: flexor surfaces, hands, feet, axilla, face, neck, elbows and knees

When do we stop treatment?

- When the scar is mature and there is optimal ROM
- When the scar is soft, flat and pale
- Surgical scars take around 6 months, while burns scars take 12-18 months to mature

Lecture: Psychological response to a burn injury

Burns treatments adds trauma to trauma

- The trauma of the burn incident experience
- Adjustment to physical injury
- Physiotherapy demands, e.g. exercises and splinting
- Impact on self-esteem and body image

Common emotional responses expressed by parents/carers

- Guilt or responsibility
- Anticipatory grief – about loss of physically perfect child or child might die
- Anger and blame towards primary carer

Mental health risk

- Burned children may have difficulties with emotional development due to living each day with a scarred and altered body
- They are vulnerable to depression, which can distort their body image and world view
- Those who suffer facial disfigurement may be at greatest risk, but the size of burn and its visibility does not always predict outcome

What do psychologists due with burn patients?

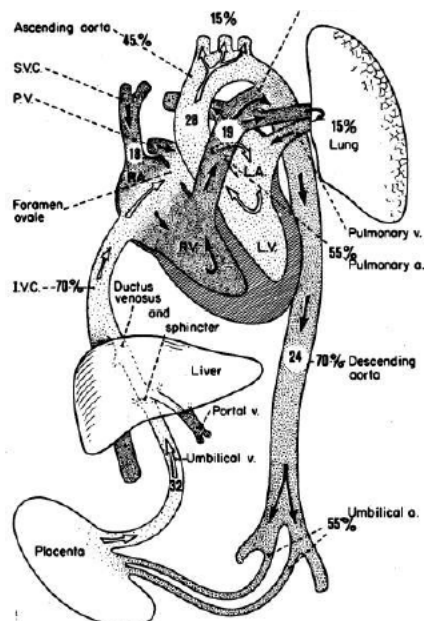
- Explore the concept of self, built on body image, self-awareness and imagined opinions of others
- Incorporating conflicting opinions of others into adolescent's self-perception can be difficult when they are pre-occupied with the opinions of others

CARDIOLOGY

Lecture: Paediatric cardiology

Age related change

- There are foetal communications to connect to the placenta
 - Ductus arteriosus – PA and aorta
 - Foramen ovale
 - Ductus venosus
- Pulmonary vascular resistance is very high as the lungs have high resistance and are filled with fluid
- 6-week baby check is important for detection of CHD, as by then the PVR should not be elevated
- Changing ventricular dominance (from R to L)



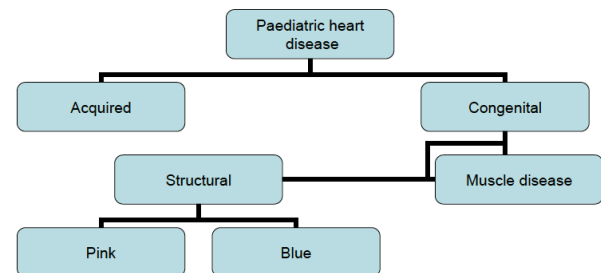
Foetal circulation

- Placenta provides oxygenated blood that is delivered to the heart via the ductus venosus
- More oxygenated blood is diverted to the developing brain and heart via the PFO
- 2/3 of ventricular output from RV
- 85% of RV output goes via ductus to descending aorta, 15% enters the lung via pulmonary arteries
- Blood flow to the lungs is for nutrient supply (15%)
- There is very low oxygen tension; foetal Hb has a higher O₂ affinity which means the partial pressure coming from the placenta is low
- Blood in descending aorta goes via internal iliac, then umbilical arteries to placenta
- Blood returning from head and neck enters RA from SVC, joined by coronary sinus before entering RV

Normal transition

- As breathing commences, there is a decrease in PVR and increase in SVR
- There is closure of the three foetal systemic-pulmonary shunts
 - PDA: closes functionally 10-15 hours after birth, related to increase in oxygen tension (PaO₂) and prostaglandins, and closes permanently within 2-3 weeks by thrombosis, intimal proliferation and fibrosis

- Foramen ovale: closes functionally at birth by combination of increased LA pressure, decrease in RA pressure. 25% of adults have a 'patent' FO
- Ductus venosus: commences shortly after birth due to a cessation of umbilical venous return, passively due to reduced flow and pressure, complete closure within 3-7 days after birth
- RV dominance SLOWLY diminishes throughout childhood years
- Heart rates decreases: maximal at 3 weeks to 3 months of age (of about 140)



Paediatric heart disease: overview

- Acquired
 - Rheumatic fever
 - Kawasaki disease (vasculitis)
 - Endocarditis
 - Some cardiomyopathies
- Congenital
 - Muscle disease
 - Pink structural
 - Blue structural

When is endocarditis prophylaxis needed?

- Prosthetic cardiac valves
- Post cardiac transplant with cardiac valvulopathy
- RHD with valvular pathology
- CHD: unrepaired cyanotic heart disease with palliative shunts/conduits
- Previous IE

Incidence of congenital heart disease

- CHD is the commonest of all congenital defects
 - VSD 35%
 - PDA 10%
 - ASD 8%
 - Pulmonary stenosis 6%
 - Coarctation 4%
 - Aortic stenosis
 - Tetralogy of Fallot 3%
 - TGA 3%
- 8 of 1000 live births (~ 1 in 100), higher incidence with stillbirths
- Associated with syndromes
 - Down syndrome (Trisomy 21) – AVSD, T13, T18, Turner
 - Velocardiofacial or DiGeorge (22q11)
 - Marfan (fibrillin, 15q21)
 - Williams (elastin, 7q11)
 - CHARGE syndrome
- Associated with toxins: FASD, maternal diabetes warfarin
- Associated with maternal SLE, foetal rubella
- Most cases without clear explanation
- 75% are curable or mild

Work up of congenital heart disease

- History
 - Cyanosis
 - Increased work of breathing
 - Sweating (activated SNS)
 - Feeding problems
 - Growth problems
- Examination
 - Dysmorphic features
 - Growth
 - HR, BP, RR
 - Clubbing
 - Pulses, esp. RF delay as coarctation is a clinical diagnosis
 - Heaves, thrills
 - Heart sounds (aortic before pulmonary), murmurs
 - Liver, spleen
- Investigations
 - CXR
 - ECG
 - Echo
 - CT/MRI

Grades of murmurs

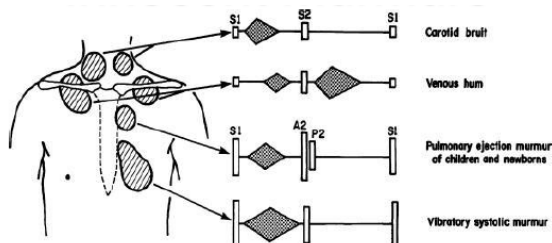
- Grade 1: scarcely audible
- Grade 2: soft
- Grade 3: loud
- Grade 4: loud and faint thrill
- Grade 5: very loud and easy thrill
- Grade 6: can hear without stethoscope

Features of an innocent murmur

- Normal peripheral exam
- Ejection
- Systolic
- Musical and vibratory
- Grade 1-3
- Changes with body position (4th left ICS?)
- Varies with review, augmented by illness

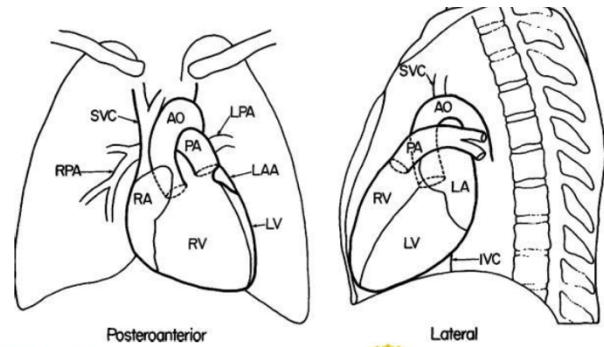
Exclusions of an innocent murmur

- Symptomatic
- Diastolic murmur or loud systolic murmur
- Cyanosis
- Abnormally strong or weak pulses
- Abnormal heart sounds, e.g. clicks
- Abnormal CXR or ECG



What are innocent murmurs?

- Still's
- Pulmonary flow murmur
- Carotid bruit
- Venous hum



What to look for on CXR?

- Position
- Size
- Contour
- Pulmonary vascularity

What to look for on ECG?

- Right ventricular dominance
- Right axis deviation

Acyanotic CHD

- Left to right shunt
 - ASD – RA, RV, PA expands
 - VSD – LA, LV, PA expands
 - AVSD
 - PDA
- Obstructive lesions
 - Aortic stenosis
 - Pulmonary stenosis
 - Arch obstruction or coarctation

Cyanotic CHD

- Tetralogy of Fallot: VSD, pulmonary stenosis, overriding aorta, RV hypertrophy
- Transposition of the great arteries
- Truncus arteriosus
- Tricuspid atresia
- Ebstein anomaly
- Single ventricle
- NB: Hypoplastic left heart syndrome is fatal

Consequences of R to L shunts

- Hypoxia and “tet spells”
- Myocardial or somatic tissue dysfunction
- Paradoxical embolus
- Polycythaemia
- Stroke
- Poor exercise intolerance
- Bleeding risk

Arrhythmias

- Ectopic beats – some focus in the ventricle, very common
- WPW syndrome – delta waves and short PR
- LQTS – QT interval is long

CHILD AT RISK

Lecture: Prevention and reporting

Who are mandatory reporters?

- All doctors are mandatory reporters – who are required by law to report to FaCS if they suspect that a child is at risk of significant harm
- Other mandatory reporters include
 - All health professionals
 - Teachers
 - Police
 - Childcare workers
 - Welfare workers
- Besides the legal obligations, there is an ethical and moral duty to report child abuse and neglect
- Chapter 16A contains the exchange of information legislation, that allows exchange of information with other prescribed bodies working with the children for whom there is concern about harm – this overrides any patient privacy legislation

When is a child at risk?

- Physical abuse
- Sexual abuse
- Neglect – general, medical, educational
- Psychological or emotional harm
- Exposure to domestic or family violence
- Prenatal harm e.g. maternal substance abuse
- Factitious disorder or Munchausen's by proxy

How do you report?

- Determine the risk of harm: type of harm, and level of risk
- Use the Mandatory Reporter's Guide: a tool to help with decision making about level of risk and what initial actions should be taken, and helps decide whether a child protection report is required
- Discuss with someone who can help, e.g. a senior colleague (ED consultant, paediatrician), Child Protection Unit
- Option 1: Immediate report to Child Protection Helpline (132 111)
 - Need to make a Risk of Significant Harm report (or Child Protection Report, or FaCS report)
 - Details needed: child's details, family background, incident details (date, type of risk, person causing harm)
 - E-report forms can be used for non-imminent suspected risk of significant harm reports (e.g. chronic neglect, not being brought to appointments)
- Option 2: Call the Child Wellbeing Unit (CWU) Monday to Friday within work hours
 - Health, Education and Police have their own CWU
 - Staffed only within office hours, and takes reports that do not meet *significant* risk, but is still concerning
 - Considers the idea of "accumulative" risk of harm, and when level of concern is sufficient, this can be escalated to the Helpline
- Option 3: Document your concerns in the patient's record and continue working with patient

- No reporting is necessary and keep working with family
- Consider community supports, e.g. NGO
- If situation changes, do MRG again
- In all cases you can consult with
 - CPU
 - Hospital social worker
 - General paediatrician

What is the CPU?

- CPU is staffed by child protection paediatricians and forensic paediatricians and senior social workers
- CPU can undertake forensic medical assessment of children for suspected cases of child maltreatment
- Not the same as FaCS or CWU!

What about suspected child sexual abuse?

- Every geographical area in metropolitan Sydney is covered by a sexual assault service
- Can call CPU or local hospital for advice

Lecture: Physical abuse

Differences in forensic history

- History
 - No history if not witnessed
 - Unreliable if self-inflicted, drug affected or head injury
 - Inaccessible, e.g. infant
- Patient
 - May be agitated, crying and confused
 - Follow up is needed re: surgery, bandaging and late appearance of bruises
- Scrutiny
 - Can be hostile, can include press, internet, justice system
 - Should be impartial, free from prejudice or bias
- Consent is vital even though findings may not benefit patient; if carer is the suspected offender, may need permission for investigations from FaCS

What is the history of the injury needed?

- Record all sources of information, but firsthand witness preferred
- When, where, what, what the child was doing, wearing
- Witnesses present
- Injury mechanisms, objects involved
- Condition immediately after injury and over time, symptoms and their time course
- Previous history of injury
- Current developmental level
- Delay in presentation

What do we need to determine in the history?

- Can the child precipitate injury? (Need to know development stages)
- Was the weight of child sufficient to generate injury?
- Was there additional movement needed to increase force?
- Were they protected by clothing?
- Did they land on objects?
- Draw the positions of relevant people and objects

What explanations are concerning?

- No explanation, vague explanation

- Firsthand witness present but did not see what occurred
- No firsthand witness but parent/carer claims they knew when it happened
- Important detail of the explanation changes dramatically
- Explanation inconsistent with pattern, age or severity of injuries
- Explanation inconsistent with the child's physical or developmental capabilities
- Different witnesses provide different explanation for the injury

Injury terminology

- Blunt force trauma
 - Abrasion
 - Bruise
 - Laceration
- Sharp force
 - Incision
- Types of injuries
 - Cutaneous: Wheals, Bruises, Burns, Bites, FTT
 - Fractures
 - Inflicted head injuries
 - Abdominal injuries
 - Abdominal injuries
 - Ingestion
 - Neglect of medical needs
 - Factitious illness

Lecture: Fractures

Transverse fracture

- This is a result of direct or indirect forces and is a suspicious fracture
- Indirect = bending load (e.g. falling while carrying child)
- Direct force to upper limb

Suspicious fractures

- Linear parietal skull # is the most common fracture from a fall as it is the thinnest; not always suspicious
- Can get a negative XR and then a periosteal reaction can occur on D10, a bone scan can be done as well to show this
- Multiple fractures
- Fractures of different ages
- Fracture not adequately explained
- Fracture in an infant <1-year-old
- Classic metaphyseal lesion: periosteal reaction and shearing forces that cause little bits of bone to fall off
- Posterior rib fractures: by compressing chest, where posterior rib presses against vertebra
- Acromion
- Spinous process
- Sternum
- Hands and feet
- Vertebral body fractures and subluxation: caused by forceful impact or shaking
- Complex skull fractures

Fractures: abusive or not?

- Abuse: young and multiple fractures
 - 55-70% of fractures occur before age 1
 - >50% have multiple fractures

- Non-abused children are older than 18 months, and multiple fractures are uncommon
- Normal variants can be mistaken for abusive fractures, and multiple fractures are uncommon
 - Periosteal new bone along shafts of long bones
 - Spurring and cupping of the metaphyses
- Medical causes of #s that could be mistaken
 - Birth trauma:
 - Prematurity: osteopenia, rickets, prostaglandin treatment
 - Neuromuscular defects: osteoporosis, contractures, decreased or absent pain perception
 - Neoplasm
 - Low phosphate
 - Scurvy
 - Rickets
 - Drug toxicity: methotrexate, prostaglandin E, hypervitaminosis A
 - Menkes kinky hair syndrome
 - Defect in Cu metabolism
 - Wormian bones in skull
 - Periosteal reaction, metaphyseal-epiphyseal #s
 - Sparse and kinky hair
 - FTT
 - Developmental delay
 - Osteogenesis imperfecta (OI)
 - Deficiency in Type I collagen, biochemical collagen test + in 80% of OI
 - 12 types: blue sclera, FHx, osteopenia, Wormian bones
 - 1 of 50 000 live births
 - Congenital syphilis and osteomyelitis
- Fractures of concern
 - Femur 28%
 - CML
 - Humerus 48% if <3 years
 - Skull 30%
 - Rib 71% highest specificity for abuse

Lecture: Inflicted head injury

Mechanisms in inflicted head injury

- Violent shaking, if with impact associated with skull fractures
- Acceleration or deceleration forces with rotation → shearing forces → SDH
- Retinal haemorrhages are common in shaking injuries
- Short falls do not explain the injuries seen in IHI

Discussion of non-accidental head injury

- Most common cause of child abuse deaths, with high morbidity with severe neurological outcomes
- 50% of cases have #s on skeletal survey, including rib and metaphyseal fractures
- Highest risk is infants less than 6 months,
- The first child is commonly abused
- Often the child has FTT, was premature or is perceived to have a difficult temperament
- Shaking baby peaks with the change in cry (after 6 weeks)
- Perpetrator often has a history of abuse; most often is the father or male figure in household
- Must assess for substance abuse and DV

- Can be young parents who have poor social supports, financial and housing problems
- Must screen siblings for risk of physical injury

What are the long-term sequelae?

- Death 10%
- Severe neurological impairment 30%
- Mild neurological impairment 10%

Lecture: Burns

Distinguishing accidental from inflicted burns

- Avoidable vs intentional 9:1
- 6-20% burns are inflicted
- Usually 0-4 years, mean age 24-36 months
- Cigarettes are common
- Sites: buttock, perineum, feet and hand
- If a child is <2 and suspected inflicted burn, consider a skeletal survey
- Burns can leave typical patterns, e.g. donut sparing

Differential diagnoses

- Impetigo
- Fixed drug reaction
- Phytophotodermatitis
- Allergy
- Mastocytosis

Lecture: Sexual assault

Presentations of child sexual abuse

- Disclosure – most common, to another child, parent or relative, at school
- Witnessed (by parent, relative, friend, siblings)
- Behavioural difficulties – sexual behaviour, changes in mood, relationships, sleep, appetite
 - 1/3 of children do not exhibit any behavioural changes
- Genital symptoms – soreness, redness, bleeding
- STIs – Chlamydia, genital herpes, gonorrhoea
- Chronic or recurrent UTIs
- Abdominal pain
- Unexpected pregnancy
- Pornographic material found

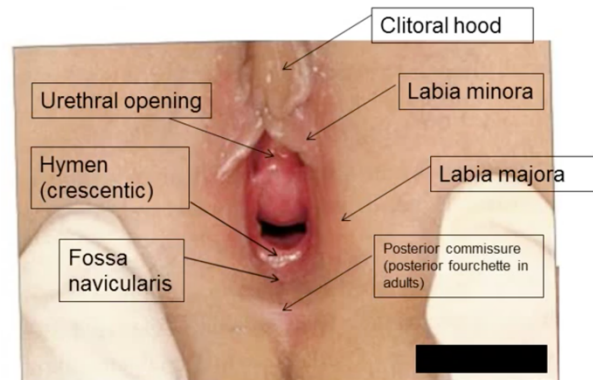
Types of child sexual assault

- Inappropriate genital touching
- Digital penetration of child's genitals (digital-vaginal, digital-anal)
- Penile-vaginal penetration
- Penile-anal penetration
- Penile-oral penetration
- Exposure to pornography
- Grooming of the child
- NB: Most perpetrators are people that the child is close to
- NB: Legal age of sexual consent is 16

Hymenal anatomy

- Pre-pubertal anatomy in the above image (see above)
- The hymen is a cuff of tissue at the entrance to the vagina; not a piece of skin that seals off the vagina
- Every female is born with a hole in the middle of the hymen

- Every hymen is different in shape and size, normal types of shapes are
 - Redundant/fimbriated
 - Crescentic
 - Annular
- In puberty, the hymen increases in size and stretches under the influence of oestrogen
- Activities such as horse riding and gymnastic do not cause changes to the hymen or any injuries; even falling on the bar of a bike is unlikely to cause injury to the hymen but may cause bruising to the area (e.g. saddle injury)



After sexual abuse

- Even with vaginal penetration, the hymen often appears normal. There may not be any signs of injury and even if there are tears or bruises, they will heal very quickly (hours to days)
- Some of these "injuries" are very subtle and not easily to see without a colposcope
- Some findings that can appear
 - Bruising, petechiae or abrasions at the posterior hymen at 6 o'clock
 - Specific injury to hymen between 3-9 o'clock, with hymenal laceration, bruising or petechiae
 - Bleeding, discharge
 - Signs of healed injury

Differentials to child sexual abuse

- Normal variant
- Skin conditions
 - Vulvovaginitis
 - Nappy rash
 - Strep infection
 - Lichen sclerosis: subepithelial haemorrhage, hypopigmentation and thinning of skin, bruising
 - Haemangioma/urethral prolapse
- Accidental trauma
 - Straddle injury
 - Vaginal foreign body

What to do if concerned about CSA?

- Report CS helpline which ensures appropriate interagency response
- Arrange for medical assessment
 - By local sexual assault service
 - Timeframe for examination varies
 - Do not send the child to ED without calling first

CHILDREN WHO DON'T FIT IN

Lecture: Language

Principles of communication and developmental problems

- A disorder (rather than a difference) has features of organisation that is immature, inconsistent and inappropriate – causing an imbalance of demands and resources
- Learning is a permanent change in behaviour or knowledge due to experience, and learning affects behaviour but not all behaviours are caused by learning. Behaviour, however, is the only evidence we have of learning
- Communication is central to social success, to diagnosis and management
- Nature and quality of communication is important, as it involves different parts, not just speech
 - Words
 - Speech
 - Non-verbal
 - Inner language – abstract language
 - Emotion and empathy

Overlap of communication problems (students who struggle)

- Learning/organising/output/behaviour
 - Language, dyslexia, motor skills, ADHD
 - Intellectual disability
 - Emotional or psychosocial disorder
- Gifted underachiever
 - Subtle qualities of language disorder
 - ADHD
 - Motor
 - Prone to anxiety
 - ASD
- Poor social communication or behaviour
 - Language, ADHD, ASD, anxiety
 - Intellectual disability
 - Emotional or psychosocial disorder

Prevalence of communication disorders

- Language impairment 10% (deficits, dyslexia)
 - Specific learning disabilities or specific language difficulties are most common (1 in 10)
- ADHD 8%
- Dyspraxia / DCD 6-10%
- Intellectual disability 3% mild, 1% moderate
- ASD 1%

Lecture: ADHD and ODD

Attention-deficit hyperactivity disorder

- The most common psychiatric diagnosis in childhood, affects 5-8%
- The features of ADHD occur in people of all levels of intellectual ability
- It is commonly associated with other diagnoses, e.g. ODD, CD, anxiety, learning difficulties
- Inherited condition (70% heritable), with males affected more than female
- Features
 - Inattention, hyperactivity, impulsivity to an extent that is developmentally inappropriate and leads to functional impairment

- Easy fatigued mentally: short attention span, avoiding tasks that are 'too hard' mentally
- Taking 'mental short cuts': saying "I don't know" or "I don't remember" because thinking of the answer requires too much effort, listening to only part of a sentence and guessing the rest, pretending to be listening
- Assessment
 - Look for functional impairment (5 modalities, as follows)
 - Academic underachievement
 - Unreasonable levels of stress/disruption in school
 - Unreasonable levels of stress/disruption at home
 - Behaviour significantly impacts on peer relationship
 - Child is aware of difficulties and has low self-esteem
 - Homework – how long does it take?
 - How much time do they stay on task on homework?
 - How much time in class do they listen?
 - Ask about mood/temper outbursts
- DSM-5 criteria (need 6/9 criteria of inattention, and 6/9 of hyperactivity/impulsivity)
- Management
 - Easily fatigued mentally: keeps tasks short, remind child, avoid environmental distractions
 - Taking 'mental short cuts': keep instructions short and simple
 - Ensure adequate diet, sleep, vision, hearing
 - Make sure the tasks are not beyond the child's ability
- Medications
 - Methylphenidate, dexamphetamine
 - Works on DA/NA balance
 - Onset 3-40 min, lasts 3-4 hours
 - Can cause issues with appetite, tics

Oppositional defiant disorder

- Associated with ADHD in at least 40% of cases
- Features
 - Often loses temper
 - Seems angry, resentful, touchy, easily annoyed
 - Deliberately annoys people
 - Trying to induce negative response in others
- Assessment
 - Ask what proportion of time the child is out of sorts or moody
- Management: Stimulant medication

Lecture: ASD

Autism spectrum disorder

- There is an increase in mental health disorders with Intellectual Disability or ASD, esp. ADHD, anxiety
- Features
 - Social impairment
 - Restricted intense interest
 - Over-literal language
 - Family history of traits to disability

COMMON PROBLEMS PRESENTING TO GP

Lecture: Constipation

What is constipation?

- Delay or difficulty passing stool present for >2 weeks
- 4x times per day in first week of life, which reduces to 1x per day by two years of age
- Normal for some breast-fed babies not to have stools for several days or even longer

Functional constipation

- This is the *most* common cause of constipation
- The child has painful bowel motions so withholds stool to avoid unpleasant defaecation
- Caused by toilet training, changes in routine or diet, stress, illness, unavailability of toilets or child is too busy
- Diagnosis (Rome III)
 - <2 bowel motions per week
 - At least 1 episode of incontinence per week after being toilet trained
 - History of excessive stool retention
 - History of painful or hard bowel movements
 - Large faecal mass in rectum
 - Large diameter stools that may obstruct toilet
 - For at least 1 month (<4 years), or at least once per week for 2 months (>4 years)
- May lead to soiling, abdominal distension, cramps and reduced oral intake
- Management
 - Education – parents should be positive and supportive
 - Disimpaction with oral & rectal medication
 - <1 year: lactulose + suppository/enema
 - >1 year: lactulose or mineral oil + suppository/enema
 - Maintenance therapy
 - Dietary interventions: fluid, fruit and veges
 - Behavioural modifications
 - Medication if necessary: lactulose first line (otherwise osmolax, Movicol, Parachoc). Should be given for at least 2 months then weaned if free of constipation for 1 month
 - In children with intractable constipation a 2 to 4-week trial of cow's milk protein avoidance may be indicated

Red flags

- Onset <1 month of age
- History of delayed passage of meconium >48h
- Fever
- Vomiting
- Bloody diarrhoea
- Failure to thrive
- Anal stenosis
- Tight empty rectum
- Impaction
- Abdominal distension
- Sacral dimple, tuft of hair on spine, gluteal cleft deviation

- Reduced tone, reflexes and strength in lower limbs

Lecture: Lactose intolerance

What is lactose intolerance?

- Lactose is a disaccharide and is found exclusively in mammalian milk, and absorption requires lactase activity in the small intestine brush border
- Features
 - Abdominal pain
 - Diarrhoea
 - Nausea
 - Flatulence or bloating after ingestion of lactose containing food
- Types of lactose intolerance
 - Primary lactase deficiency
 - Relative or absolute absence of lactase
 - Develops in childhood, uncommon <2 years old
 - Most common cause
 - Secondary lactase deficiency
 - Results from small bowel injury, e.g. gastroenteritis, bacterial overgrowth, chemotherapy
 - More common infancy
 - Can be transient
 - Congenital lactase deficiency
 - Extremely rare, with only a few case reports
 - Would not survive without lactose free human milk substitute
 - Developmental lactase deficiency
 - Relative lactase deficiency
 - Observed in infants < 34 weeks' gestation, but is overcome after some time
- Diagnosis
 - 2-week trial on strict lactose free diet leading to resolution of symptoms with recurrence after lactose is reintroduced is diagnostic
 - If subtle case, hydrogen breath test
- Investigations
 - Stool can be tested for low pH and faecal reducing substances; however, this is a difficult specimen to collect properly. Stool specimens can also look for Giardia or cryptosporidium
 - Intestinal biopsies can measure lactase activity in the gut
 - Antibodies for coeliac disease
- Management
 - Breastfed infants continue breastfeeding
 - Lactose free formula in infants
 - Avoidance in milk and other dairy products in children
 - May tolerate yoghurt, as bacteria partially digests the lactose, as well as aged cheeses

Frothy stools in the breastfed infant

- This is due to relative lactose malabsorption, and can be due to oversupply of carbohydrates; not allowing the baby to feed fully from one breast causes excess carbohydrate fore milk and not enough fat-rich hind milk to be consumed
 - Hind milk slows absorption and allows time for digestion

- Advise mother to feed the baby from one breast only for a few feeds
- Congenital lactose intolerance is *exceptionally rare* therefore change to lactose free formula is not generally indicated

Lecture: Nocturnal enuresis

What is nocturnal enuresis?

- Very common, affects 10% of 5 years
- More common in boys, more common with family history
- Types
 - Primary: never dry for more than a few months
 - Secondary: was dry for >6 months, then wets again
- Caused by impaired cortical arousal with reduced bladder capacity, and some have nocturnal polyuria
- The child has no conscious control of it

What to ask on history

- Daily fluid intake
- Frequency and volume of daytime voiding
- Daytime symptoms, such as incontinence or urgency
- Symptoms of constipation
- Identifying other contributing factors, e.g. toilet phobia, fear of dark

Assessment

- Time and volume chart – which provides information about frequency and volume of urine production, can help diagnose bladder spasm and reduced bladder capacity
- Assess for UTI, glucose in urine

Management

- Reassurance, as most grow out of it
- Treat underlying factors
- Avoid caffeinated drinks, ensure adequate fluid intake
- Encourage to pass urine regularly through day
- Restricting fluids and waking during night do not work
- Needs professional help if still wetting >6 years
- First line: bed-wetting alarm
- Second line: desmopressin (reduces number of wet nights, but effect not sustained), advise child not to drink more than 240 mL of fluid 1 hour before and up to 8 hours after treatment to prevent water intoxication

Lecture: Plagiocephaly

What is plagiocephaly?

- Positional plagiocephaly is posterior skull flattening, usually on one side with an ipsilateral frontal bulge, forward ipsilateral ear, and the skull looks like a parallelogram from above
- It is usually unilateral, if bilateral this develops postnatally
- Unilateral plagiocephaly is more common in males, on right side and present from birth
- Exacerbated by sleeping on back
- Prevalence estimated from 5% to 48% of newborns
- Differential diagnosis
 - Craniosynostosis is the premature fusion of one or more cranial sutures.

- Also feel palpable ridging over fused suture with compensatory overgrowth of other sutures
- Associated with raised intracranial HTN, developmental delay
- Ultrasound can be useful

- Congenital muscular torticollis
- Assessment
 - Head circumference
 - Palpation of fontanelles, sutures for ridging
 - Looking at head shape from above and in front
 - Looking for torticollis
 - Examine the spine for curvature and sign of muscular asymmetry (in some cerebral palsy)
- Management
 - Observe, as most improve with time (2+ years)
 - Physiotherapy for full neck ROM
 - Refer to paediatrician or paediatric neurosurgeon if severe, not improving with time or features of craniosynostosis

Lecture: Gastro-oesophageal reflux

What is regurgitation and reflux?

- Regurgitation is very common
 - 2/3 report regurgitation at 4 months, which is usually outgrown by 1 year
 - Possetting is common, particularly when the stomach is full after a feed
 - Rarely causes problems for a child
- When does it become pathological?
 - GOR begins in first 2 months, improves and usually resolves by 1 year and growth is normal
 - GORD may have features of haematemesis, melaena, dysphagia, ongoing regurgitation past 1-year, poor growth, pain, Sandifer's syndrome (posturing of neck) or back arching
- Who is at risk?
 - Premature infants
 - Parental history
 - Obesity
 - Hiatus hernia
 - Congenital diaphragmatic hernia
 - Repaired oesophageal atresia
 - Neurodisability
- Management
 - Conservative measures for GOR
 - Not positional changes
 - Consult breast fed infant
 - Reduce feed volume, smaller more frequent feed, thickened feed if formula fed
 - Pharmacological for GORD
 - 4-week trial of PPI or H2A
 - Do not give metoclopramide, domperidone, erythromycin
 - Tube feeding if severe FTT
 - Surgery: fundoplication

Red flags

- Forceful, projectile vomiting < 2 months → pyloric stenosis
- Bile stained vomiting and abdominal distension → bowel obstruction, e.g. malrotation

- Haematemesis → serious bleeding
- Onset after 6 months, or persisting after 1 year → consider other diagnosis, e.g. reflux, cerebral space-occupying lesion
- Blood in stool or chronic diarrhoea → cow's milk protein intolerance, bacterial gastroenteritis
- Rapidly increasing HC or headache with morning vomiting → raised ICP caused by hydrocephalus or brain tumour
- Acutely bulging fontanelle +/- altered LOC → raised ICP, e.g. meningitis

Investigations

- For GOR, not indicated
- GI contrast study: for unexplained bile stained vomiting, reflux with dysphagia
- Ultrasound: projectile vomiting if <2 months
- Oesophageal pH study +/- impedance monitoring to confirm diagnosis of reflux
- Urine culture
- Upper GI endoscopy

DENTISTRY

Lecture: Oral soft tissues: anatomy and general health

Different things to note on inspection

- Gum pads are full of teeth, frenulum in midline
- Eruption cysts, filled with old blood, and herald the arrival of primary teeth. These can be deflated if it is causing irritation to the child
- Neonatal tooth is loose and can be rough to the ventral surface of the tongue, forming ulcers. This tooth can be removed if irritating to the child
- Gingival-mucosal barrier should be very marked
- Anterior open bite can be caused by pacifier use, where upper and lower anterior teeth do not contact
- Dental caries of the lower incisors can appear when there is lack of saliva from the submandibular and sublingual glands
- Stained teeth associated with dietary habit and iron supplementation
- Tongue tie (ankyloglossia) can inhibit feeding or speech but it is not necessary to be cut

Development of teeth

- Primary teeth develop in utero, completed by 6 months
 - They erupt between 6 and 36 months of age
 - There are 20 primary teeth
 - Variations in size, shape, colour, position and eruption are diagnostic
- Permanent teeth commence 3-4 months in utero
 - They begin to calcify from birth
 - Upper and lower anterior teeth and first permanent molars are completed by 3 years
 - They erupt between 6 and 18 years (for wisdom teeth)
 - Again, variations in size, shape, colour, position and eruption are diagnostic

Dental caries

- High caries rates run in family, so children with mothers with high decay rates have a greater risk
- 75% of all caries is found in 20% of children
- Often associated with other chronic diseases
- Prevention
 - Clean for 2 minutes twice a day
 - Use fluoride toothpaste
 - Gums should not bleed
 - Give teeth 2 hours rest between meals
 - Limit sugary snacks and drinks
 - Lots of water!

Lecture: Oro-facial infections and non-accidental injuries

Oro-facial infections

- These infections can be odontogenic infections (damage to teeth), oral ulcerations (viral, bacterial, traumatic) and bleeding
- There are many causes of facial swellings, but odontogenic infections should be on the differential list
- Teeth should be considered like little bones (marrow covered by bone), so trauma and caries can lead to marrow infection. This infection occurs in a closed space so the inflammatory exudate leaves via apex of tooth into soft tissues or into the mouth

- Many children have repeat antibiotics for facial infection without a dental follow-up

Facial swelling

- These are usually a facial cellulitis rather than an abscess, and can be associated with fever and dehydration

Non-accidental injury

- Consider when bruising in the hard palate, gingiva
- Could be due to burns, oral sex, etc

Lecture: Medical-dental interface

What happens at the medical-dental interface?

- Maxillary frenum – normal
- Ectodermal dysplasia
- Single upper central incisor – a feature of holoprosencephaly, and the child may have other midline features
- Extra teeth can occur with cleidocraniodysplasia, where the child might have hypoplastic clavicles, supernumerary teeth and failure to arrive of permanent teeth
- Dentinogenesis imperfecta (pale teeth) is associated with OI and blue sclera
- Primary immune deficiencies such as Langerhans cell histiocytosis can cause bleeding gingiva, ulcers, mobile teeth and pain
- Neutrophil function disorder can cause inflamed gingiva and loose teeth presenting as precocious aggressive periodontitis and palmoplantar keratoma. This is Papillon-Lefèvre syndrome
- Erosion of teeth structure can be caused by GORD
- Bifid uvula can be associated with other midline features. Consequently, this can mean that the soft palate isn't working, causing issues with swallowing and breathing. This can be seen in velocardiofacial syndrome

Lecture: Oro-facial trauma

Oro-facial trauma

- Many children damage their primary teeth (e.g. falling over, running, bicycles)
- Dental injuries
 - Degloving – mucosa is removed off alveolus
 - Luxation – tooth moved in a direction
 - Fracture
 - Avulsion
 - Laceration
- Primary dentition by age 5: 30% of children have damaged their teeth
- Permanent dentition by age 14: 22% of children have damaged their teeth
- Male > female ratio 2:1
- Peaks at 2-4 years and 8-10 years
- Usually a single maxillary anterior tooth, and includes both hard and soft tissues
- Management
 - Do not replace avulsed primary teeth, due to damage to permanent, longevity
 - Replant avulsed permanent teeth immediately
 - If not sure what tooth is out or if not possible to replant immediately, place the tooth in milk or saline

DERMATOLOGY

Lecture: Infections of the skin

Impetigo

- Impetigo is commonly known as “school sores”, and the most common bacterial organisms implicated are Staph (aureus) and Strep (pyogenes)
- The bacteria form toxins, which cause damage to the stratum corneum. As a result, it causes (yellow) crusting and flaccid blisters, and spread is by direct contact or fomites
- This typically occurs in face, diaper area and exposed areas
- There are two forms: non-bullous (70%) and bullous (30%) impetigo. Bullous impetigo is invariably caused by Staph aureus
- If this remains untreated, can lead to sepsis, septic arthritis, lymphadenitis and pneumonia. There are also autoimmune implications, such as guttate psoriasis, scarlet fever and post-strep GN.
- Treatment
 - Clearing the crust using saline or warm water
 - Topical, oral or IV antibiotics
 - Treat the underlying carriage – intranasal mupirocin (Bactroban)

Staphylococcal scalded skin syndrome

- Haematogenous spread of Staph toxin and attacks the skin throughout the body → painful blisters on the whole body
- Management: pain, antibiotics, managing temperature, fluid and electrolyte imbalances

Molluscum contagiosum

- Caused by molluscum pox virus, which lives in the central pore of the papules surrounded by a protective membrane
- Spread by close contact, usually in context of water, e.g. baths and swimming pools
- Features: skin-coloured papules with umbilicated centres, eczema around lesions
- Can last for several years
- Treatment is with
 - Reassurance that it will resolve spontaneously
 - Cantharidin (from blister beetles)
 - Deroofing lesion
 - Cryotherapy – painful

Warts

- Warts are caused by HPV, which has over 70 genotypes
- Spread by skin transmission or fomites
- Incubation period 1-6 months
- Treatment
 - Spontaneous resolution 2/3 lesions in 2 years
 - Cryotherapy
 - Wart paste (of salicylic acid)
- Type of warts
 - Common warts – keratotic on the surface, not umbilicated
 - Plane warts – flat, on the face

- Filiform warts – finger like projections usually on the face
- Plantar warts – usually painful on the foot
- Periungual warts – on the finger
- Anogenital warts

Therapy

- Also known as “ringworm”, caused by dermatophytes
- Dermatophytes can be from animals, soil or humans and affects anything that has skin
 - *Microsporum canis* = exposure to dog
 - *Tricophyton tonsurans* = spread by close living quarter
- Typical tinea causes a ring like rash, scaly at the edge, clear in the centre with another ring
- Tinea incognito = tinea attempted to fix with steroid use that doesn't let it resolve properly
- Can lead to a kerion, an abscess caused by fungal infection. This most often occurs on the scalp (tinea capitis) but can arise on any site exposed to the fungus
- Treatment: antifungals

Scabies

- Scabies are caused by the mite, *S. scabiei*, which is an obligate human host and forms burrows under skin
- Causes intense pruritis, as well as papules, nodules, vesicles and burrows

Lecture: Dermatitis and psoriasis

Atopic dermatitis

- Atopic dermatitis = eczema
- Pathogenesis
 - There is an underlying epidermal barrier dysfunction with immune dysregulation of Th2 imbalance (with genetic factors overlying this)
 - There is often a filaggrin deficiency, which is a protein found in the epidermis that helps maintain the integrity of the skin
 - Environmental factors can contribute to flares of atopic dermatitis
 - Flares are treated with regular moisturisers, anti-inflammatory therapy and avoid environmental factors
 - There is no cure; it is a chronic inflammatory disease
- Prevalence in children is 10-20%, while in adults is 1-3%
- In children with atopic dermatitis, it presents < 6 mo in 45%, by 1 year 60%, by 5 years 85%
- There is an association with atopic asthma (50%) and allergic rhinitis (50-80%) and food allergy (30%)
- Clinical features
 - Eczema on cheeks in babies – erythema with poorly demarcated margins and scaly
 - Widespread in babies
 - Typically spares nappy area
 - In older child, often in the flexor folds
 - Lichenification can be a consequence of chronic itching
- Environmental irritants
 - Drying agents: soap, shampoos, detergents
 - Abrasive clothing: wool, nylon, acrylic
 - Abrasive surfaces: carpets, rugs, sandpits
 - Water: frequent bathing, pools
 - Heat
 - Dry ambient air

- Chemicals: fragrances, preservatives
- Eczema predisposes to infections
 - *S. aureus* or *S. pyogenes* – secondary infection
 - Colonisation with *S. aureus*, which can exacerbate the inflammatory process via superantigen effect. This can be controlled with bleach baths, anti-microbial washes or in some cases, low dose antibiotics (Bactrim)
- Allergens that cause eczema
 - Airborne: HDM, pollens, animal
 - Ingested: cow's milk, wheat, egg, peanut, seafood
 - Use skin prick test or RAST
- Management
 - Regular moisturisers
 - Anti-inflammatory therapies: topical steroids, topical calcineurin inhibitors (Elidel, tacrolimus), systemic phototherapy or systemic immunosuppressive drugs if nonresponsive
 - Need to choose appropriate strength, vehicle (ointment, cream, lotion) for the topical steroids
 - Sigmacort (mild) for face, Aristocort (moderate) for trunk, Diprosone (potent) for limbs
 - Avoid environmental triggers

Food intolerance

- Natural chemicals (e.g. salicylates in tomatoes, amines, glutamates)
- Food colours
- Food preservatives
- The only way to diagnose is with food challenge and an elimination diet

Childhood psoriasis

- Prevalence of 2-3% of population, and of adults with psoriasis, 30-45% of them had onset < 20 years of age
- Triggered by trauma, Strep infection and stress
- Clinical features
 - Erythematous, well-demarcated plaques
 - Silvery-white scale
 - Symmetrical
 - Classically located in the extensor knees, elbows, lumbosacral in adults
 - In children: guttate, facial, flexural, scalp
- Management
 - Address triggers
 - Topical therapy: corticosteroids (1st-line), tar, calcitriol (Vit D analogue), dithranol
 - Phototherapy of narrowband UVB, sunlight
 - Systemic therapy: MTX, cyclosporin, acitretin
 - Biologic agents for paediatric psoriasis: Etanercept

Lecture: Birthmarks

What are birthmarks?

- Birthmarks are important because they can cause distress to parents
- They are too much (or not enough) of a particular component of the skin but they may not be evident at birth
- Café au lait – overactive melanocytes that produce pigment

- Port wine stain – a capillary malformation or increased capillary dilatation
- Hypopigmented macule – underactive melanocytes producing pigment (can be associated with ash leaf spots in tuberous sclerosis)
- Epidermal naevus – linear ray of extra keratinocytes

Infantile haemangioma

- Often referred to as strawberry haemangiomas
- This is the most common benign soft tumour of childhood, that occurs in 1-2% of newborns
- At 1 year of age, affects 10-12% of infants
- They are not usually present at birth, though there may be a “precursor” lesion noted at birth in 30%. In the first 2-3 weeks of life, there is the proliferative phase where they grow in 3-6 months and continue for another 6 months. Then they reach a plateau phase where they stabilise. Finally, there is an involution phase (50% by 5 years, 90% by 9 years)
- Types
 - Superficial haemangioma
 - Deep haemangioma
 - Mixed haemangioma
- Complications
 - Vast majority don't cause any problems
 - Ulceration (most distressing, slow to heal)
 - Location involving other important organs (e.g. eyes)
 - Cosmetic deformities
- Sometimes they can be heralding an underlying or regional problem, e.g. sacral haemangioma may suggest spinal defects or urogenital defects, mandibular haemangiomas may be associated with laryngeal haemangioma (can cause stridor, worsened croup)
- Sometimes they may be related to PHACES syndrome
 - Posterior intracranial fossa anomalies, e.g. Dandy-Walker
 - Haemangioma
 - Aortic and arterial anomalies: coarctation of aorta, anomalous arteries, aneurysms
 - Cardiac abnormalities: valvular atresia, VSD, cor triatriatum
 - Eye anomalies: coloboma, glaucoma, cataracts
 - Sternal defects
- Treatment
 - Only if complications or at risk of complications
 - Oral propranolol (first line)
 - Oral corticosteroids

DYSMORPHIC SYNDROMES

Lecture: Congenital malformations and dysmorphic syndromes

Definitions of congenital malformations

- Major malformations: produce significant long-term disability or death
 - 15-20% of stillborn babies, 1/3 due to chromosomal abnormalities
 - 2-3% of live born babies
 - Extra 2-3% of children within first 5 years of life
 - Examples: craniosynostosis (fusion of the fingers), scaphocephaly (premature fusion of the sagittal suture), FASD (smooth philtrum, thin upper lip, short palpebral fissures), microcephaly, holoprosencephaly, cyclopia
- Minor malformations: unusual morphological features of no serious medical or cosmetic consequence
 - Single transverse palmar crease
 - Unilateral cleft lip
 - Bifid uvula
 - Haemangioma
 - Clinodactyly
- With more minor malformations, you will have increased risk of having a major malformation\

Types of congenital malformations

- Malformations are morphological defects of specific anatomical structures occurring during organogenesis, this includes:
 - Agenesis
 - Aplasia: absence of cellular proliferation
 - Hypoplasia: insufficient cellular proliferation
 - Hyperplasia
 - Dysplasia
- Deformation is an abnormal shape/position of body part due to anomalous mechanical forces, usually due to extrinsic causes (e.g. co-twin, uterine fibroids, oligohydramnios) or can be intrinsic (e.g. compressive forces from oedema)
 - This usually occurs after organogenesis
 - No underlying tissue defects
 - Features include loss of symmetry, altered alignment, abnormal positioning, distorted configuration
- Disruption are a result of a destructive process that alters structures after their normal foetal development
 - Features include altered shape, division or fusion or amputation of parts
 - Vascular occlusion is a common mechanism

Terminology of congenital malformations

- Syndrome: group of features that occur together, implies a common specific cause
- Association: non-random concurrence of features that occur together more frequently than expected by chance, with no known cause
- Complex: general term for a composite of features
- Spectrum: entities with multiple features, especially where the prominent features are variable in expression
- Sequence: pattern of anomalies resulting from a single primary anomaly or mechanical factor

- Developmental field: finite areas of embryonic tissue that develop into multiple, related morphologic structures; helps explain why certain malformations occur together
 - Monotopic: malformation of a single anatomic structure
 - Polytopic: multiple anomalies in different body areas from disturbance in a single developmental field

Clinical approach to the dysmorphic child

- Pregnancy
 - Parental age
 - Mode of conception
 - Drugs: alcohol, cigarettes, recreational
 - Illnesses, infections, infertility
 - Foetal movements, ultrasound data
 - Previous foetal loss, stillbirths
 - Complications: bleeding, growth retardation, premature rupture of membranes, LGA, poly/oligo-hydramnios
- Family history
 - Ethnic origin
 - Consanguinity
 - Relatives with similar malformations, congenital anomalies, developmental abnormalities, intellectual disability, miscarriages, stillbirths
 - Correct family pedigree
 - Family photographs
- Birth history
 - Gestation
 - Singleton, multiple pregnancy
 - Mode of delivery
 - Resuscitation
 - Immediate problems: feeding, tone
 - Other: jaundice, hypoglycaemia, seizures
- Medical history
 - Developmental history
 - Physical problems
 - Vision/hearing
 - Behavioural problems
 - Sleeping and eating patterns
 - Menarche, puberty

Aetiology of congenital malformations

- Genetic
 - Chromosomal, e.g. Turner, trisomy, microdeletions (VCF, Williams syndrome)
 - Single gene mutations or mosaicism
 - Mendelian, e.g. AD, AR, X-linked
 - Other: uniparental disomy, mitochondrial, triplet repeat disorders
- Teratogenic/environmental
 - Drugs, e.g. alcohol, anticonvulsants, antibiotics, isotretinoin, antimalarials, anticancer drugs, androgens, cigarettes, cocaine
 - Radiation
 - Maternal factors, e.g. epilepsy, PKU, diabetes, iodine deficiency, folate deficiency, hyperpyrexia
 - Organic solvents, mercurial
 - Intrauterine infection, e.g. TORCH, VZV
- Multifactorial

Molecular basis of birth defects

- Early positional information: embryo changes into layers, then segmented. This progress involves expressing genes that direct orientation and pattern formation
- Differentiation: specialisation and migration of cells, which depends on complex signalling
- Cell death: Apoptosis is important for 'modelling' of body parts. Teratogens can disrupt this process
- Developmental timing: Altered timing may occur, due to intrinsic or extrinsic factors

Lecture: Approach to genetic testing

Why do we do genetic test?

- Confirm a diagnosis or clinical suspicion, either in the pre-natal setting or a clinically diagnosed condition
- Find an answer
 - Prenatal setting – why there are multiple congenital abnormalities
 - Postnatal setting – why there is dysmorphism, DD, MCA
 - Familial conditions

How does an answer help?

- Prenatal: family planning options
- Postnatal: changes delivery or management of child, stops the 'diagnostic treadmill'

Standard G-banded karyotype

- Looking for translocations, aneuploidy, large deletions and duplications
- Rapid turnaround: 36 hours to 1 week
- Pickup: 3%

FISH & MLPA

- FISH for specific conditions, e.g. 22q11, Prader-Willi etc
- Sub-telomere screen
- Must specify the region you are *fishing* for, as you are looking for specific conditions (e.g. microdeletion screen, craniosynostosis)

Other genetic tests

- Methylation studies – for Prader Willi, Beckwith Wiedemann, Angelman, Russel-Silver
- Sanger sequencing (** most common) – where you spell out individual genes and look for changes in 'spelling', \$\$\$ (\$1000-2000) and time consuming. Examples: SMN gene (spinal muscular atrophy), MECP2 gene (for RET syndrome), Fgfr3 gene (for achondroplasia and others)
- Repeat studies – for triplet repeats in genes, e.g. Fragile X, Myotonic dystrophy
- Chromosomal fragility studies – rarely used, for something called Fanconi anaemia

Chromosome microarray

- Benefits
 - Higher yield – 15%
 - 10-100x resolution
 - Genome wide
 - Delineates deletions/duplications more clearly
- Disadvantages

- Cannot detect balanced translocation or map imbalances
- Misses point mutations and small mutations
- Can 'unmask' carriers of recessive conditions and consanguinity
- Can be hard to interpret
- Prenatal microarray can diagnose aneuploidies, small deletions/duplications, 'common' microduplication/deletion syndromes: VCF, Williams

Next generation sequencing

- Miniaturised and massively parallel platforms
- Sequencing millions of reads at once
- Much cheaper and faster than Sanger sequencing
- But this can potentially introduce more artefacts, so needs more bioinformatic analysis
- What can we do with this?
 - Help complete diagnosis of Mendelian disorders
 - Start to utilise this in non-Mendelian, complex, common disorders (e.g. asthma, heart disease)
 - It is more cost effective and quick
 - Help screening for carriers, newborns, prenatal

Non-invasive prenatal testing

- NIPT can screen pregnancies for aneuploidy, performed from 9-10 weeks
- Uses NGS to count amount of free-floating foetal DNA (from placenta) in maternal blood
- May help avoid unnecessary procedures like CVS, amnio which have a risk of miscarriage
- Can determine sex, trisomies for now
- Only privately available, costs around \$450

Lecture: Genetics of intellectual disability

Terminology of intellectual disability

- Developmental delay < 5 years
- Intellectual disability > 5 years

What is intellectual disability?

- It is a state of functioning characterised by limitations in both intelligence and adaptive skills, presenting before 18 years of age
- Global developmental delay is the preferred term until > 5 years, when formal IQ testing is more reliable
- Affects 1-3% of the population
- Most ID is in the mild range (85%)
 - 30% more common in males
 - Mild is more often familial
 - Mild is more prevalent in developing countries and low SES
 - Mild ID is likely multifactorial
- Associations
 - Can be isolated
 - ID with multiple congenital malformations
 - ID with dysmorphic features
 - ID with other neurological features
 - ID with autism spectrum disorder
- Risk of recurrence is higher in males, than females at the moderate-severe level, and for mild depends on family history

Most common genetic causes of ID

- Trisomy 21

- Fragile X syndrome (FMR1 gene)
 - Affects 1 in 5000 male
 - Most common single mutated gene (1-2% of ID diagnoses)
- Microdeletion/microduplication copy number variations
 - Found in 10-15% of ID, higher pickup in ID + dysmorphism
- The cause of ID in most remains unknown

Approach to diagnosis of ID

- Clinical assessment
 - 3 generation family history
 - Consanguinity, miscarriages
 - Perinatal history
 - Development, speech, hearing and vision
- Examination
 - Growth parameters
 - Dysmorphic features (“atypical” features)
- Baseline tests
 - FBC, EUC, LFT, TFT, CK, UMS... other metabolic?
 - Cerebral imaging
- Genetic tests
 - Chromosomal microarray
 - Fragile X DNA
 - Other targeted single gene test, small or large gene panel
 - Rare – non-targeted “whole exome”

Fragile X syndrome

- Fragile X syndrome is the most common inherited form of intellectual disability and autism
- It is caused by a triple repeat mutation in the FMR1 gene at Xq28, with full mutation of >200 CGG repeats
- Should be performed in all DD/ID of unknown cause, unless major congenital abnormalities
- ~1-2% of boys with ID have full mutation, while ~0.5-1% of girls with ID will have full mutation
- This WON’T be picked up by chromosomal array, or NGS sequencing: important
- Important that FMR1 mutations are always inherited, so need for family cascade screening

Chromosomal microarray

- Chromosomal microarray is a first-tier clinical diagnostic test for individuals with developmental disabilities or congenital anomalies
- Doesn’t detect balanced rearrangement, low level mosaicism
- Does not detect mutations in genes (including triplet repeat mutations), as it is a segmental copy number variation test, not a sequencing test
- What does it detect?
 - Major chromosomal imbalance (>5-10 Mb)
 - Microdel/microdup syndromes, such as VCFS 22q11, Williams 7q23, Prader-Willi 15q12
 - Neurodevelopmental susceptibility CNV
 - Variants of unknown significance
 - Normal, which does not rule out other genetic abnormalities like gene mutations and triplet repeat

ID gene panel approach

- NGS method (WES or WGS) restricting analysis to a selected group of genes implicated in ID is 2nd line testing
- These test panels differ by the genes included, test methods, reporting stringency, interpretation and cost
- They are not covered under Medicare
- Ethical issues: They will identify mutations that are unrelated to the question being addressed, there may be adult-onset disorders for which no treatment exists or variants of unknown significance/expression

ID due to single gene disorders

- X chromosomal linked
 - Males > females
 - FMR1 gene (Fragile X gene)
 - MECP2 gene (Rett syndrome)
 - DDX3X gene, female limited phenotype
 - No mutations individually explain more than 0.1% ID (except Fragile X), but collectively explain up to 10% of ID in males
- Autosomal dominant
 - M=F
 - Familial transmission possible, but often de novo
 - But variable penetrance, expressivity
- Autosomal recessive
 - M=F
 - Sibling recurrences are common, and consanguinity increases the likelihood

DNA facts

- Only 1.5% of DNA is coding
- We have 20 000 genes, but only 5000 have a known function
- 800 genes have been implicated in intellectual disability

Diagnosis rates (not very important)

- ID panel of 560 genes, perhaps 11%
- Trio of whole exome sequencing 15-30%
- Whole genome sequencing 40%

ENT

Lecture: The ear

Foreign bodies in ear canal

- If alive, drown in oil
- If organic item (e.g. vegetable), likely to expand with water
- If need to syringe, aim postero-superiorly with clean warm water and Betadine
- If in doubt – refer to ENT

Acute otitis externa

- This is a skin infection of the external auditory canal
- Clinical features
 - Tenderness of EAM skin, tragus
 - Draining node on mastoid tip
 - Hearing issues is a late sign
 - Soothing dressing hurts (unlike otitis media)
- Subtypes
 - Diffuse, caused by *Pseudomonas*
 - Localised, caused by *Staph*. This is due to an infected hair follicle
 - Itchy, caused by *Aspergillus niger* or *Candida*.
- Management of diffuse
 - Usual analgesics
 - Topical steroid and ABx drops: Sofradex/Ciproxin HC
 - If severe pain
 - Topical steroid drops
 - Oral ciproxin or IV gent
 - One or two doses of systemic or oral steroids
 - Clean after 48 hours
- Management of localised: as above but needs oral Abx that are anti-staph
- Management of itchy: mop out ears, then use Locacorten-Vioform drops
- Prevention of otitis externa
 - Dry ears out after swimming with 'acidified alcohol' drops, aka Aquaeear, Ear clear drops for swimmers
 - Alcohol absorbs the water
 - Acid (e.g. vinegar) subdues *Pseudomonas*

Wax impaction

- Wax is made of skin, sebum and debris
- The best wax solvent is sodium bicarbonate eardrops, then hydrogen peroxide, or ear clear drops for wax removal. The least useful solvent is oil
- A lot of ear wax problem is due to skin problems like eczema, dermatitis and occasionally psoriasis, while overproduction is rare. Prevention with mild steroid cream can be useful.

Exostoses

- Exostoses occur from cold water exposure (swimming) and cause swelling of bone of ear canal
- Symptoms are otitis externa and wax
- Management = surgery

Acute otitis media

- This is an infection of the middle ear cleft (eustachian tube, middle ear and mastoid air cells)

- Bacteria involved are the same as acute sinusitis: *Strep pneumoniae* +++, *Moraxella catarrhalis*, *Haemophilus influenzae*. All acute complications due to strep pneumoniae
- Diagnosing of AOM is difficult, generally it is the rapid onset of signs and symptoms, such as otalgia and fever, of acute infection within the middle ear. Tympanocentesis is the most accurate method of diagnosis
- Symptoms of signs include
 - Otagia – pulling ear
 - Ear discharge
 - Fever
 - Recent irritability
 - Anorexia, vomiting, diarrhoea
 - Tympanic membrane full/bulging, opaque, reduced mobility
- Management
 - Watchful waiting for at least 24 hours
 - Antibiotics for at risk children (under 6 months, or under 2 with bilateral AOM, ear discharge, underlying cranio-facial anomalies, Aboriginal, immunosuppressed)
 - Amoxicillin recommended (40-90 mg/kg/day) for 5 days
- Complications
 - Subperiosteal abscess (in a posterior-superior direction)
 - Facial palsy
 - Sigmoid sinus thrombophlebitis
 - Intracranial abscess
 - Meningitis
 - Labyrinthitis

Acute mastoiditis

- Acute mastoiditis is inflammation of the mucosal lining of the mastoid air cells inside the mastoid process
- Swelling always pushes your ear downwards and forwards (i.e. it is located posterior-super). This is because this is the thinnest bone between the skin and mastoid air-cells
- Generally, occurs in <3-year olds
- Caused by *S. pneumoniae*
- Treatment
 - IV antibiotics and surgical drainage through skin and TM
 - Occasionally a cortical mastoidectomy is needed

Otitis media with effusion

- If effusion persists after 12 weeks, then is considered OME (occurs in 20%). This is also known as 'glue ear', so there is fluid in the middle ear with no signs or symptoms of inflammation
- The description of the fluid has no predictive value of severity
- Symptoms include blocked ear, reduced hearing, imbalance, sequelae (poor schoolwork, clumsiness)
- Treatment
 - Grommets, also known as ventilation tubes, pressure-equalising tubes, drainage tubes, MEDITs. The function is to break the suction and to ventilate the middle ear
 - These grommets last 6 months to 2 years (the larger the longer)
 - Adding an adenoidectomy to grommet insertion reduces re-insertion rate by >20%

- Grommet care
 - Avoid 'dirty' water, incl. baths, heated pools, freshwater
 - Risk Pseudomonas infection
- Sequelae of grommets or chronic infection: tympanosclerosis

Chronic suppurative otitis media

- This is an "open" otitis media, with perforation
- There are two types
 - Safe: permanent perforation in pars tensa
 - Unsafe: medial to TM, where skin can grow in the middle ear causing cholesteatoma
- Treatment
 - Repair of 'central' TM perforation: myringoplasty. Connective tissue provides the scaffold for fibrous tissue to grow in, use temporalis fascia, perichondrium or cartilage from the tragus

Cholesteatoma

- This is squamous epithelium in the middle ear cavity
- Treatment is always surgical: mastoidectomy

Hearing tests

- Audiology
 - 0-20 Hz is normal hearing
 - 20-40 Hz is mild hearing loss
 - 40-60 Hz is moderate hearing loss
 - 60-80 Hz is severe hearing loss
 - 80+ Hz is profound hearing loss
- Weber and Rinne's test
- Tympanometry helps diagnose OME
 - Uses tension of TM to bounce sound back
 - Reflected sound is minimal when the TM is floppy, and TM is floppy when air pressure in EAM = ME
 - No movement of the TM, or a low flat 'curve' = glue ear
 - ET dysfunction or "congestion" on the middle ear will show a negative pressure in ME = developing or resolving head cold

Lecture: The nose

Nasal examination

- To look at the nose internally, use a light source and lift nasal tip with finger
- Can use otoscope with speculum

Nasal foreign bodies

- Can present witnessed, or unwitnessed with unilateral nasal discharge with excoriation and epistaxis
- Urgent removal is needed with button batteries as it can lead to septal perforation within hours.
- To remove a foreign body, you need a metal ear curette, good lighting, position to child and decongestant or topical anaesthetic
- Management of septal perforation can be lifelong
 - Saline
 - Intermittent antibiotic cream
 - Vaseline

Epistaxis

- Most common in the anterior nose due to rich blood supply
- Usually > 2 years

- Due to digital trauma (picking their nose), winter, allergies and sprays
- Must exclude rare causes
 - Tumours, e.g. juvenile angiofibroma (males ~ 9 years)
 - Blood dyscrasia
 - Foreign body
- Management
 - If crusting and digital trauma, topical treatment (e.g. Bactroban) may be successful in approximately 70 %
 - Topical anaesthetic or decongest nose
 - Silver nitrate cautery
 - Post cautery management: treat dryness with topical antibiotic cream for a week, then petroleum jelly. Treat one side at a time and wait 4-6 weeks before moving onto the other side.

Neonatal nasal obstruction

- Neonates are obligate nasal breathers until 8-16 weeks, so bilateral nasal obstruction causes respiratory distress
- Neonates have anterior or posterior obstruction
- Investigation
 - Mirror misting test
 - Passage of nasal catheter
 - Nasoendoscope
 - CT scan
- The most common cause of neonatal nasal obstruction is choanal atresia, which is usually associated with CHARGE syndrome. This is most commonly unilateral.

Older children nasal obstruction

- This is usually in the anterior nose, due to allergies or posterior nose due to adenoids
- Workup
 - History: nasal obstruction, sleep, allergy
 - Examination: external appearance, adenoid facies, allergic salute, tonsils, throat
 - Investigations: lateral airway XR, CT, PSG, allergy testing
- Need to consider nose, adenoid and tonsils as a whole as obstructions occur in series
- Most common causes are allergy and adenoids, but there are other causes
- Adenoids
 - History of snoring, nasal discharge, nasal speech
 - Long facies (adenoid faces), mouth breathing, ear findings
 - View adenoid pad with otoscope
 - Lateral airway XR
 - Treatment: Adenoidectomy
- Allergy
 - 40-80% of children have parental atopy
 - Ask about seasonal or perennial symptom pattern
 - Treatment: avoidance, steroid spray, surgery is a last resort

Paediatric sinusitis

- Two subtypes: acute (< 8 weeks) and chronic (> 3 months)

- Predisposing factors: CF, ciliary disorders, immunodeficiency, anatomic abnormalities, environmental, genetics, asthma
- Multifactorial causes, incl. allergy, viral URTIs, day-care, reflux, air pollution
- Pathogens: *S. pneumoniae*, *H. influenzae*, anaerobes, Staph
- An acute viral URTI is the precipitating cause, with bacterial infection secondary to obstruction
- Symptoms
 - Nasal obstruction and congestion
 - Pressure with pain
 - Thick nasal discharge
 - Toothache
 - Fever
 - Cough
- Treatment
 - Antibiotics
 - Decongestant, saline nasal spray
 - Rule out reflex
- Management goals: control infection, reduce oedema, facilitate drainage, maintain patency of ostia and break the cycle so that it doesn't become chronic
- Surgical management is a last resort with adenoidectomy helping symptoms, or a conservative endoscopic sinus surgery
- Can also perform surgery of acute bacterial complications, where sinusitis has spread to surrounding organs

Nasal trauma and septal haematoma

- Paediatric nasal fractures are common when children are learning to walk or are playing sport
- The cartilaginous septum and nasal vault can bend with minor displacement
- X-rays are not always necessary except for medico-legal reasons
- Even though there is no deformity to the outer nose, fractures can still lead to a septal haematoma
- To diagnose a septal haematoma, there is
 - History of trauma, symptoms of nasal obstruction, new onset snoring
 - Examine nose and look at septum anteriorly
 - Early referral for nasal fracture needs to be reduced and drained within 7-10 days – can lead to septal abscess or septal perforation
 - Must see ENT surgeon early before fracture sets

Lecture: The throat

Throat anatomy

- Above soft palate – nasopharynx
- Above epiglottis – oropharynx
- Below epiglottis – hypopharynx
- Tonsils can come in all sizes, and Brodsky grade is used to grade the size (0 to 4)

Viral pharyngitis

- The 'common cold' or upper respiratory tract infection is the most common cause of sore throat in children
- Aetiology is usually viral, including rhinovirus, influenza virus, parainfluenza virus, adenovirus, coxsackievirus, echovirus, EBV and RSV
- Symptoms
 - Mild to moderate sore throat
 - Pharyngeal erythema

- Low grade fever
- Adenovirus-associated pharyngitis is more painful and can be associated with higher fevers and conjunctivitis
- HSV causes cold sores and gingivostomatitis in younger children

EBV pharyngitis

- This is also known as glandular fever or infectious mononucleosis
- Symptoms include
 - High fever, general malaise, headache, pharyngitis, dysphagia, odynophagia
 - Can have normal sized or hypertrophic tonsils, often with a grey exudate
 - Palatal petechiae
 - Large tender cervical lymphadenopathy
- Diagnosis by
 - Raised WCC
 - Raised transaminase, ALP, BR
 - Monospot test, though only 50% positive in first week, 90% positive after 1 month, children < 4 may not develop Ab
 - EBV specific serology
- Management is symptomatic
 - Avoid amoxicillin (causes rash)
 - NSAIDs
 - CS if airway obstruction

Herpangina

- Coxsackie virus causes vesicles which become ulcers and spread over the anterior tonsillar pillar, palate and posterior pharynx +/- cutaneous rash
- Hand foot and mouth is similar, and caused by Coxsackie A16 – results in high fever, oral and throat pain due to vesicular eruption in the mouth, rash or vesicles on the palms, soles and buttocks

Tonsillitis

- Group A beta-haemolytic strep is the most common cause of acute pharyngitis + tonsillitis
- Peak incidence at 5-6 years, rare in children < 3 years
- Symptoms
 - Fever
 - Sudden throat pain and odynophagia
 - Malaise, dysphagia, otalgia, headache, nausea and abdominal pain
- Signs
 - Oropharyngeal erythema
 - Enlarged tonsils with or without exudate
 - Swollen uvula
 - Palatal petechiae
 - Enlarged, tender cervical lymphadenopathy
- Scarlet fever is a variation of GABHS pharyngitis that cause a descending rash from the neck to the trunk and extremities (that develops 12-24 hours after the fever). It is associated with a strawberry tongue (enlarged papillae with or without a white membrane)
- Management
 - Antibiotics – penicillin or amoxicillin. First gen cephalosporin, clindamycin used if penicillin allergy
 - If any suspicion of peritonsillar abscess (fullness or swelling of soft palate) use anaerobe cover with metronidazole
 - Symptomatic treatment with paracetamol and NSAIDs

- Recurrent tonsillitis may be considered for tonsillectomy, if 7+ episodes in 12 months, 5+ each year for 2 years. Need to consider severity, abscesses, response to treatment and school absences
- Complications of tonsillectomy
 - Post-operative haemorrhage
 - Pain – avoid codeine if OSA
 - Velopharyngeal insufficiency
 - Injury to lips, tongue and teeth

Obstructive sleep apnoea

- This is sleep disordered breathing, which varies from partial obstruction of the airway causing snoring, to increased upper airway resistance to complete obstruction (apnoea)
- Incidence of snoring 8%, and OSA 1-4%
- Aetiology
 - Enlarged tonsils, adenoids and lingual tonsils (which enlarge after 2 years), relatively largest in 3-6 years, most common cause of OSA in children
 - Craniofacial anomalies (micrognathia, maxillary hypoplasia)
 - Laryngomalacia
 - Hypotonia
 - Obesity
- Untreated OSA can lead to
 - Neurocognitive and behavioural problems
 - Cardiovascular dysfunction – increased IR and higher DM
 - Growth restriction – altered GH secretion
 - Enuresis
- Diagnosed with polysomnography
 - Measures breathing, work of breathing, O2 saturation, CO2 levels through night
 - Apnoea index >1 / hours
 - Apnoea-hypopnoea index (AHI) > 1/hr. AHI <4 mild, 5-10 moderate, >10 severe
 - SaO2 below 92%
- Treatment
 - Adenotonsillectomy cures 60-80%
 - CPAP
 - Other surgery (rare): lingual tonsillectomy, tongue reduction, mandibular distraction, bariatric surgery, tracheostomy, UPPP
- Remember differences between adult and child anatomy
 - Child tongue is disproportionately large
 - Larynx is higher in neck
 - Narrowest portion at cricoid ring

Foreign bodies

- Choking accounts for 80% of deaths under the age of 1 year
- Most common in children <3 years
- Inhaled FB lodges in the bronchial tree, down the right main bronchus
- Ingested FB are twice as common but only 10-20% require removal, esp. coins, batteries, pins and jewellery
- Workup of airway FB
 - History: witnessed, cough, dyspnoea, wheezing, cyanosis or stridor
 - Examination: cough, decreased lung sounds and wheezes, fever, recurrent pneumonia, tachypnoea, hypoxemia or crackles, asymmetry of breath sounds or a prolonged expiratory phase

- Bronchoscopy
- Workup of oesophageal FB
 - History: witnessed ingestion or choking, dysphagia, vomiting, food refusal, chest pain
 - Examination: drooling or poor feeding
- Fish bones most commonly lodge in palatine tonsils, vallecular or pyriform fossa. This causes localised discomfort, and patients are unable to complete meal. May need a CT or laryngobronchoscopy
- Imaging
 - PA and lateral XR – only 10% of FB are radiopaque, notice double contour of button batteries on lateral film (will cause alkali burn in 4 hrs)
 - May see unilateral emphysema, hyperinflation or localised atelectasis if radiolucent FB

FOOD ALLERGY & IMMUNOLOGY

Lecture: Epidemiology and diagnosis

Epidemiology and definitions of food allergies

- These are common problems, but the prevalence is often overestimated (by parents to be 16%, but incidence is 3-9% by population-based studies).
- This varies significantly by age and inclusion of non-IgE mediated allergies
- Food hypersensitivity can be subdivided into food allergy (IgE or not) or non-allergic food hypersensitivity (including intolerances or enzyme deficiency)

Immunological response

- Allergens are processed by Mø and presented to T cells. The T cell may communicate with B cells, which may cause B cell to cause T cell switching to Th2 type and B cell switching to IgE production: sensitisation
- Circulating IgE binds to mast cells and basophils
- On re-exposure, IgE bound to mast cells recognises allergens, cross-links and releases histamine and mast cell mediators

IgE mediated food allergies

- 80% present in the first year of life.
- 10% of children < 1 year, 4-8% of children < 5 years, and up to 2% of adults
- Associated with atopic dermatitis, as 35% of moderate AD have IgE food allergy
- Appears in "high risk infants"
 - 1st degree relatives with atopy: 4x risk with parent, 10x risk with sibling
 - High concordance rate with identical twins, around 65% MZ 7% DZ but not 100%
- Foods that cause more than 90% of IgE mediated allergic reactions in children: milk, eggs, peanut, tree nuts and seeds, fish, shellfish, soy and wheat.
- Egg is most common, followed by peanut and cow's milk
- Symptoms
 - Onset < 2 hours, usually within 5-15 minutes
 - Unpredictable in severity (but not always consistent in severity)
 - Mild = localised facial erythema, urticaria/angioedema
 - Moderate = generalised urticaria/angioedema, or vomiting
 - Severe = generalised urticaria/angioedema, stridor or wheeze/collapse, floppy small infants
 - Prior exposure history not always present
- History enquired about
 - Timing of reaction
 - Type of reaction and severity of symptoms,
 - Likely allergen (dietary hx, prior exposure, reproducibility)
 - Risk factors

Lecture: Investigations

Skin prick tests

- These are generally reported as a wheal size – of maximum diameter in 2 directions, at 90 degrees to

each other. The flare is not considered a useful parameter

- Good NPV (>2 years), and fair PPV when combined with positive history ($S_n > S_p$)
- Size correlates with likelihood of reaction, NOT severity
- SPTs are good for monitoring tolerance to food that children were previously allergic to
- Not useful for assessing delayed food reactions, atopic dermatitis, eosinophilic oesophagitis

Serum-specific IgE (formerly RAST)

- This is a laboratory ELISA, which detects free IgE bound to allergen surfaces
- Level correlates with likelihood of reaction, NOT severity
- Poor NPV in low-intermediate risk subjects, validated for good PPV at high levels for egg, milk and peanut

Component resolved diagnosis

- These are a more specific serum specific IgE
- This detects IgE directed against single components (e.g. purified natural sources or recombinant)
- Types of CRD
 - Detect a single specific component of particular interest e.g. omega-5-gliadin in wheat-dependent exercise induced anaphylaxis
 - Small panels in peanut allergy
 - Microarray – not recommended outside research
- In general, higher S_p for determining allergy, but lower S_n , therefore good PPV but poor NPV
- Therefore, cannot determine those with negative tests don't have an allergy

Peanut CRD

- This is the most extensively studied and used
- Ara h1, 2, and 3, if positive most likely have clinical allergy and generalised reactions. Combination of 3 is more useful than a single component
- Ara h8 – mild symptoms, associated with pollen-food syndrome. This is seen in Europe
- Ara h9 – associated with peach allergy places, severe symptoms. This is seen in Mediterranean countries
- Ara h2 in particular has a high PPV for clinical allergy vs sensitisation
- However, CRD does not determine clinical reactivity or determine anaphylaxis risk

Other foods in CRD

- Omega-5-gliadin – wheat dependent EIA
- Gly m5 and Gly m6 – soy reaction severity
- Ovomucoid/Gal d1 – egg products vs egg
- Cor a8, 9 – systemic hazelnut allergy (Europe only)
- Cor a1 and Bet v1 – pollen food syndrome + hazelnut allergy (Europe only)

Provocation tests

- DBPCFC – this is the gold standard, usually a research tool as it provokes patient and parent anxiety
- Open food challenge – this helps establish tolerance or is for ruling in/out food allergy for children who have skin test+ but exposure is not known. This lasts for 2 hours and children are observed for a reaction
 - Not reliable in determining severity

Interpretation of IgE results

- Many atopic children have a positive allergy test, but no symptoms on exposure to the allergy
- Therefore, to diagnose an allergy, the child must have symptoms which occur as a result of exposure to the allergen
- 30-50% of children will have sensitivity on SPT without a clinical allergy

Lecture: Management**Food allergy management**

- Dietary avoidance
 - 70% of egg and milk allergic children can tolerate baked egg/milk e.g. cakes
- Dietitian input, e.g. calcium supplementation
- Understand clinically relevant cross-reactive foods
- Management plan
- Review for tolerance

Natural history of food allergy

- High propensity for outgrowing: milk, egg, soy and wheat
- Low propensity for outgrowing: peanut, tree nut, fish, mollusc and crustacea
- Depends on severity, number of food allergies, atopic history
- 30% tolerance at 3 years, 50% at 5 years, 75% of 10 years, 90% of 15 years

Cross reactive foods

- For peanuts, low reactivity to other legumes, soy. Most common cross reactivity is lupin and tree nuts
- For cow's milk, 90% of children will also be allergic to goat and sheep's milk
- For hen's milk, high cross reactivity
- For soy, low cross reactivity
- For shellfish, high for molluscs, lobsters, crustaceans but not fish commonly
- For fish, high but it is unpredictable
- For wheat, low cross reactivity to other grains (except in wheat-dependent EIA)

How do we predict severity of reaction?

- Older child
- Dose of allergen
- State of allergen e.g. raw or cooked
- Presence of asthma, inter-current illness, medication
- Exercise pre-ingestion
- Nature of allergen – nuts, shellfish, fish
- NOT the size of skin test – only predicts probability of reaction
- Type of epitope – number of allergenic epitopes, linear vs conformation epitope

What is anaphylaxis?

- Anaphylaxis is the most severe form of allergic reaction requiring urgent treatment
- It is any acute onset illness with typical skin features (urticarial rash or erythema/flushing, and/or angioedema) + involvement of respiratory +/- cardiovascular +/- persistent severe GI symptoms OR any acute onset of hypotension or bronchospasm or upper airway obstruction where anaphylaxis is possible even if typical skin features are not present

Lecture: Non IgE mediated food allergy**Non IgE mediated food allergies**

- There are no good proven allergy tests for a delayed reaction
- The best approach is therefore to make a provisional clinical diagnosis, then food elimination and rechallenge where indicated

Eosinophilic oesophagitis

- Histopathology: furrowing of mucosa, mucosal oedema, pale whitish patches, visible strictures
- Biopsy: >20 eosinophils per HPF, eosinophilic micro-abscess, proximal and distal lesions (different to reflux)
- Diagnosis ALWAYS require upper endoscopy and oesophageal biopsy
- Symptoms
 - Food refusal
 - Vomiting
 - Dysphagia
 - Abdominal pain
 - Bolus obstruction
- Treatment
 - Elemental diet – restricted to AAs, poorly tolerated in older children
 - Directed elimination diet based on results of allergy testing and removal of specific foods
 - Non-directed elimination diet where a standard 6 foods are eliminated – milk, soy, egg, wheat, fish, peanuts/tree nuts
 - Steroids – inhaled or budesonide slurry

Food protein induced enterocolitis syndrome

- Clinical features
 - Delay 1-4 hours after food
 - Profuse vomiting
 - +/- floppy, lethargic, pale, hypothermia, hypotension
 - No skin/respiratory symptoms
- Investigations
 - SPT/ssIgE negative
 - Neutrophilia, thrombocytosis +/- acidosis
- Epidemiology
 - Presents at 6-12 months when new foods are being introduced
 - Often misdiagnosed, delays in diagnosis and unnecessarily investigated
 - Most commonly due to rice, then cow's milk and egg
 - Triggers due to more than one food common
 - Usually outgrown by 2-4 years
- Treatment – supportive, IV rehydration + removal of foods

Lecture: Primary immunodeficiency disease in childhood**When should you consider a PID?**

- Unusual or opportunistic infections
- Disseminated infections
- Recurrent serious infections, e.g. sinopulmonary, fungal
- Infections in associations with FTT, chronic diarrhoea, unusual skin rashes, persistent oral Candida

Humoral deficiencies

- Characterised by recurrent bacterial infections (encapsulated organisms e.g. Pneumococcus) often in the ears
- Can be specific antibody deficiencies, e.g. IgA deficiency or combined (e.g. X-linked agammaglobulinemia, CVID)

Severe combined immunodeficiency

- Affects T cells usually but can affect B cells and NK cells as well
- Causes lymphopenia at birth, very rarely normal at birth (Omenn syndrome)
- Absent thymus on CXR
- Clinical presentation
 - Unusual and opportunistic infections, e.g. PCP, fungal, overwhelming viral infections
 - FTT, diarrhoea
 - Rash (maternal engraftment, Omenn's)
- Types
 - X-linked SCID (most common), often in defect in common gamma chain for receptors (IL2, 4, 7, 9, 15, 21) on NK and T cells
 - JAK3 (AR)
 - Adenosine deaminase deficiency
 - RAG1/RAG2 causes defects in VDJ recombination

Phagocytic cell deficiencies

- Defects in neutrophils: chronic granulomatous disease (CGD), poorly functioning neutrophils due to failure of neutrophils to undergo oxidative burst, X-linked, or AR
 - Gingiva a good place to assess neutrophil function
- Leucocyte adhesion disorders: high neutrophil count but inability to form pus and move into tissues

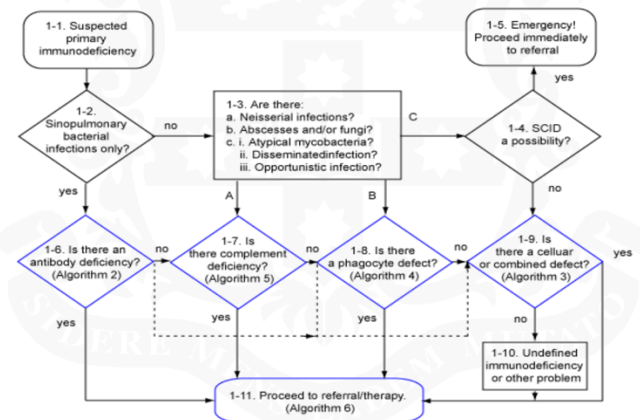
Complement deficiencies

- Deficiency in terminal pathway (C5-C9) (most common) causes susceptibility to Neisserial infections, due to inability to form MAC and lyse bacteria
- C2 deficiency causes recurrent bacterial infections resembling antibody deficiency, autoimmune disease

Workup of immunodeficiency disorder

- History
 - Chronic symptoms
 - Failure to thrive, diarrhoea
 - Unusual organisms, e.g. PCP, Aspergillus
 - Type of organism
 - Bacteria, esp sinopulmonary – Ab or complement
 - Opportunistic viral or fungal – T cell, SCID
 - Fungal, Staph, Pseudomonas, Nocardia – Neutrophil / CGD
- Investigations
 - FBC
 - Ig types
 - Functional antibodies – DTP titres
 - T, B NK cell subsets through flow cytometry

Practice parameter for the diagnosis and management of primary immunodeficiency (Algorithm 1)



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GASTROENTERITIS & FLUID Mx

Lecture: Managing children with gastroenteritis

Epidemiology of gastroenteritis

- Rotavirus is the most common cause
- It can account for up to 20% of all visits to ED, 6% of hospital admissions and 50% of children under 4 years old seek medical attention
- Majority of hospitalised children have no or mild dehydration and minimal acid base disturbances
- Prior to rotavirus vaccination (2007) it was estimated that 10 000 hospitalisations each year were due to rotavirus in children < 5 years, 5 000 visits to the GP and 22 000 visits to ED; one death due to rotavirus each year

Typical presentation

- Increased watery stools
- Vomiting
- Fever
- Infectious contacts

Causes of acute gastroenteritis

- Viruses (70%):
 - Rotaviruses (can be osmotic diarrhoea)
 - Enteric adenoviruses
 - Enterovirus
 - Small round viruses
 - Caliciviruses: norovirus (now most common cause of gastroenteritis, non-inflammatory small bowel diarrhoea)
 - Astroviruses
- Bacteria (15%):
 - Campylobacter jejuni, Salmonella, Shigella, Yersinia: bloody
 - E. coli
 - Cholera
- Protozoa: Cryptosporidium, Giardia, Entamoeba histolytica
- Helminths: Strongyloides stercoralis

History

- Vomiting: how frequent, bile, blood
- Stool output: number, blood, mucous, watery
- Intake: how much, type of fluid
- Urine output: wet nappies
- Lethargy, drowsiness
- Duration of illness
- Loss of weight
- Sick contact
- Previous medical history, e.g. short gut
- Family history, e.g. IBD

Examination

- Assess dehydration, as affects oral vs IV fluids to follow and length of treatment
 - The validity of individual clinical signs is low, but needs multiple signs
 - CRT correlates with degree of dehydration but affected by fever, high ambient temperatures
 - Marasmic patients may have false positive tenting

Description of dehydration	Dehydration (% of body weight)	Signs and Symptoms
No Clinical Signs		Reduced urine output Thirst No physical signs
Mild	3%	Reduced urine output Thirst Dry mucous membranes Mild Tachycardia
Moderate	5%	Dry mucous membranes Tachycardia Abnormal respiratory pattern Sunken eyes
Severe	10%	Above signs Poor Perfusion: Mottled, cool limbs/Slow capillary refill/Altered consciousness Shock: thready peripheral pulses with marked tachycardia and other signs of poor perfusion stated above

- Detect deterioration
 - Tone, interactiveness, consolability, look/gaze, speech/cry
 - Tachycardia, pulse volume, CRT, BP
 - Respiratory status, increased RR with acidosis
 - Neuro status: agitation, confusion, drowsiness
 - Renal status: oliguria or anuria
- Exclude sepsis, metabolic (DKA), surgical abdomen, IBD, complications (HUS)

Red flags

- Blood in the stool or vomit
- Bile in vomit
- Vomiting but no diarrhoea
- Headache
- Very high fever
- Young child
- Severe pain

Investigations

- In mild cases: no blood tests
- In moderate – severe dehydration consider
 - UEC
 - Blood sugar
 - FBC
 - Stool micro, culture, virology
- If considering other aetiologies: full ward test urine, blood culture, XR

Managing fluids

- Frequent reassessment of patient, adjust according to fluid and electrolyte status
- Minimum urine output:
 - Neonates and children < 30 kg: 1 mL/kg/hr
 - 30-60 kg: 0.5 mL/kg/hr
 - > 60 kg: 30 mL/hr

Dehydration in children: differences

- Children are at greater risk than adults
- Higher total body water and greater proportion of ECF
- Higher metabolic rates and fluid turnover, therefore vulnerable to fluid deficits
- Higher body surface area, higher insensible losses with illness
- Immature kidney functions leading to a reduced ability to regulate sodium and water
- Dependent on caregivers for food + water

Glucose in children: differences

- Hypoglycaemia is more common in infants and young children, due to low glycogen stores and high metabolic rate
- Hypoglycaemia can be an early sign of shock

Oral rehydration therapy

- ORT is recommended by AAP & WHO as the first line of treatment
- Gastrolyte is the most commonly used one here. Small amounts should be given at frequent intervals and the volume gradually increased until the child can drink as desired
- With ORT we know that glucose stimulates intestinal sodium transport. Although it may not reduce stool volume or duration of diarrhoea, we know we can adequately rehydrate a child with ORT
- Benefits include: no cannulation, no pain, physiological benefits, avoids complications of IV therapy
- Zinc supplementation can reduce duration and severity of diarrhoea in cholera

Lecture: Diarrhoea assessment and management

What is steatorrhoea?

- Fat digestion requires gastric acid, bile salts, pancreatic enzymes, an intact intestinal mucosa, chylomicrons and lymphatic system
- Small chain TAGs can bypass this system above and enter the portal venous system directly, which can be useful in therapy of fat malabsorption
- The most common cause of steatorrhoea in children is pancreatic insufficiency (usually due to cystic fibrosis)
- Other causes include
 - Inadequate bile salt action
 - Inadequate absorptive surface
 - Enterocyte defect, e.g. abetalipoproteinaemia
 - Defective lymphatic drainage

Assessment of fatty stools

- Fat globules – lipase deficiency, liver disease resulting in reduced bile salts
- Fat acid crystals – mucosal disease due to coeliac or short bowel syndrome

Cystic fibrosis

- Found in 1:2500 Caucasians, due to chromosome 7 mutation often in F508
- Presentation of CF
 - Meconium ileus and bilious vomiting as newborn
 - Chronic diarrhoea due to pancreatic insufficiency
 - Failure to thrive due to fat soluble vitamin deficiencies
 - Respiratory illnesses, e.g. S. aureus, then Pseudomonas colonisation

Steatorrhoea in chronic liver disease

- Extrahepatic biliary atresia
- Cholestatic jaundice
- Cirrhosis (leading to decreased bile salts)

Coeliac disease

- Found in 1:100

- Symptoms: FTT, diarrhoea, abdominal pain, vomiting, constipation, abdominal distension, irritability
- Need to test for coeliac disease in these patients: IDA, IgA deficiency, Down/Turner/Williams syndrome, first degree relatives with coeliac disease
- Diagnosis
 - IGA antibody to TTG – if high index of suspicion but low TTG consider IgA deficiency
 - Endomysial antibody (EMA)
- Treatment
 - Gluten free diet for life, which reverses GI symptoms, maximises growth and bone mineralisation
 - Lactose free diet in young children with severe disease
 - TTG can be useful to monitor response to treatment and dietary adherence. Measure at diagnosis, 6 months then annually

What causes watery diarrhoea?

- Osmotic diarrhoea: osmotically active particles which are not being absorbed leading to malabsorption, e.g. fructose or sorbitol excess, disaccharidase deficiency
 - Unabsorbed carbohydrate in LB ferment to short chain FAs, which the colon can absorb anionic forms of in exchange for bicarbonate → mild hyperchloraemic acidosis
- Abnormal water and electrolyte transport
- Mucosal diseases
- Infection
- Bacterial overgrowth syndromes

Refresher of carbohydrate digestion

- Dietary carbohydrates are made of starch, which are broken down by salivary and pancreatic amylase to tri-, oligo- and di-saccharidase
- Brush border enzymes break these down to monosaccharides. This includes sucrase (sucrose = glu + fru), lactase (lactose = glu + galac)
- Fructose is absorbed by facilitated diffusion through GLUT-5

Assessment of watery stools

- Presence of stool reducing substances: in breast-fed infants, up to 0.5% reducing substances is normal. Presence indicates osmotic diarrhoea
- Test stool pH
- Calculate osmotic gap ($280 - 2 \times (\text{stool Na} + \text{stool K})$). If > 100 , it is likely to be osmotic diarrhoea
- If NBM, secretory diarrhoea should continue with a normal osmotic gap and no stool reducing substances

Examples of osmotic diarrhoea

- Congenital sucrase-isomaltase deficiency may be suggested by into diet with onset of osmotic diarrhoea
- Disaccharidase deficiency can be caused by viral gastroenteritis, chronic giardiasis, coeliac disease, milk protein enteropathy, SB bacterial overgrowth, immunodeficiencies, autoimmune enteropathy etc
- Monosaccharide malabsorption is rare but can be a result of severe enteropathies (e.g. congenital villous atrophy, severe post-infectious enteritis). These are life threatening. They usually refer to defective glucose/galactose malabsorption due to mutations in SGLT1, with AR inheritance

Examples of secretory diarrhoea

- Congenital (rare) but can be due to congenital sodium diarrhoea and congenital chloride diarrhoea, due to disorders in Na/H exchanger and Cl/HCO exchanger. This diarrhoea occurs in utero leading to polyhydramnios → profuse diarrhoea at birth and electrolyte disturbances
- Infection (largest cause)

Bloody diarrhoea

- Causes
 - Infection – usually bacterial or parasitic
 - IBD – CD, UC, Milk protein intolerance
- Blood is not always obvious in the stool, so must do microscopy. Stool RBC indicates colitis, as does stool WBCs
- Excessive blood and protein loss can cause iron deficiency anaemia and hypoproteinaemic oedema; this is called protein-losing enteropathy which can be assessed by measuring stool alpha-1 antitrypsin levels
- Faecal calprotectin may differentiate IBD from IBS, it is a substance released by neutrophils into the GIT

HAEMATOLOGY

Lecture: Anaemia

Signs and symptoms of anaemia

- Pallor and lethargy
- Jaundice, or red/brown urine – in haemolysis
- Jaundice, organomegaly, painful skeletal changes, dactylitis – in thalassaemia or sickle cell disease
- Iron deficiency or B12 deficiency
- Bleeding, bruising, fevers – in decreased production of red cells
- Haematuria, melaena
- Oedema – in cow's milk enteropathy
- Failure to thrive – in chronic anaemia
- Can have no signs, if a chronic problem, e.g. iron deficiency, thalassaemia, congenital anaemia
 - More likely to have signs if acute anaemia (haemolysis, blood loss) – and children will be lethargic
- In neonates
 - anaemia is often due to haemolysis where jaundice and pallor occur on day one of life. This may be allo-immune haemolysis or inherited haemolytic anaemias (G6PD, hereditary spherocytosis)
 - Foetomaternal haemorrhage – Kelihauer Betke Test to look if there is foetal blood in maternal circulation
 - excess blood tests can precipitate anaemias
 - congenital anaemias will show a lack of reticulocyte response
 - beta-thalassaemia and sickle cell *do not* present in the neonatal period, only alpha-thalassaemia

Timing of symptoms in infants

- Lack of production
 - IDA most common cause, often due to excess cow's milk or blood loss
 - Congenital anaemia presents in first year, e.g. Diamond Blackfan Anaemia
 - Leukaemia
- Haemolysis
 - Congenital haemolytic disorders, e.g. hereditary spherocytosis, G6PD deficiency
 - Haemoglobinopathy, e.g. beta-thalassaemia and sickle cell disease typically present around 6 months of age
 - Autoimmune haemolytic anaemia, presents at any age, is extravascular so urine may not be changed

Timing of symptoms in adolescence

- Causes are as above
- However, in adolescent girl, need to target questioning about iron deficiency around menstruation

Examination for anaemia

- General observations – activity
- Pallor – palmar creases, conjunctiva
- Haemolysis – jaundice, scleral icterus
- Organomegaly
- Swelling of fingers and limbs (sickle cell disease)
- Bleeding, bruising and petechiae

Investigations

- NB: age related changes
 - noting Hb changes with age. 190 at birth, 110 around 2-6 months, and then over the next 12 years increases to adult ranges
 - MCV changes are similar, starting with 110 at birth, a nadir around 80 around 1 year, and then increases to adult ranges
- MCV
 - Normocytic: blood loss, haemolysis, anaemia of chronic disease, sickle cell
 - Microcytic: IDA, thalassaemia, sideroblastic anaemia
 - Macrocytic: B12/folate deficiency, bone marrow disorder, congenital anaemias
- Reticulocyte count
- Blood count
- Fe, Vit B12, folate
- LDH, Bilirubin – in haemolysis
- Coombs test – in autoimmune haemolytic anaemia
- Hb EPG – in thalassaemia
- Red cell EMA – hereditary spherocytosis
- Red cell ADA – Diamond Blackfan anaemia
- Bone marrow biopsy

Lecture: Bleeding disorders

Bleeding history

- Site, cause and duration of bleeding or bruising episodes
- Mucosal bleeding – nose bleeds, menorrhagia, bleeding from cuts around the mouth
 - If nose bleeds over 20-30 minutes or frequent – this is abnormal
 - If cuts around mouth over 5-10 minutes – this is abnormal
- Post-operative bleeding, e.g. circumcision, tonsillectomy or adenoidectomy, post-dental extraction bleeding
 - If these require blood transfusions or something to stop bleeding or re-presentation – this is abnormal
- Bleeding with immunisations – can present as a lump
- Bleeding from umbilical stump separation
- Spontaneous joint or muscle bleeding
- Family history

Clues to the diagnosis

- Duration of easy bruising or bleeding
 - Sudden onset or bruises/petechiae in a previously well child – ITP
 - Bruised easily since they were mobile in a boy – Haemophilia
- Family history
 - Haemophilia – most common X-linked
 - VWD – most common AD disease, most common in general
 - History of thrombosis and bleeding – dysfibrinogenaemia
- Site of bleeding
 - Mucosal bleeding – vWD, platelet function disorders
 - Excess bleeding from bruises or cuts – clotting disorders
- Timing of bruise/bleeding
 - Delayed bleeding often seen in F XIII deficiency

- Classic presentations
 - Bleeding from umbilical stump – F XIII deficiency
 - Spontaneous hemarthrosis in a boy – haemophilia

Investigations for bleeding disorders

- Platelet count and platelet morphology
 - Large platelet, e.g. ITP, May-Hegglin anomaly (w/doll bodies in neutrophils), Bernard-Soulier
 - Small platelets, e.g. Wiskott-Aldrich syndrome
 - Normal sized platelets, e.g. Glanzmanns thrombasthenia
- PT, INR – looks at Factors VII, V, X, thrombin, fibrinogen
- APTT – looks at Factors XII, XI, VIII, IX, X, V, Thrombin, Fibrinogen. May also be an inhibitor, e.g. lupus anticoagulant
- Fibrinogen level
- Mixing study – a prolonged PT or APTT which corrects on mixing 50:50 with normal plasma, indicates a clotting factor deficiency, e.g. haemophilia, VWD. If this doesn't correct this indicates an inhibitor, usually a transient lupus anticoagulant

Lecture: White cell disorders

Overview of white cell disorders

- Not enough white cells: congenital neutropaenia, immunodeficiency
- Too many white cells: leukaemia, malignancy
- Malfunctioning white cells: CGD, agammaglobulinaemia, neutrophil adhesion disorders

What sort of bacterial infections are concerning?

- Bacterial: skin pustules, recurrent mouth ulcers, recurrent otitis media, pneumonia, bacteraemia. Think neutropaenia, CGD, leucocyte adhesion disorder
- Viral infections: CMV, EBV, Varicella, rotavirus, measles. Think humoral or adaptive immunodeficiency if severe viral infections, e.g. recurrent or hospitalisation
- Fungal infections: Thrush can be common in bottle fed neonates, but otherwise not common. Think immunodeficiency.

History

- Recurrent infections
 - Onset, abnormal if from birth
 - Types, as multiple bacterial or fungal infections are abnormal, mouth ulcers are abnormal
 - Severity, e.g. sepsis, unusual severity common pathogens
- Lymphadenopathy
 - Small soft mobile cervical or inguinal nodes are common in infants and can be reactive
 - Large, widespread, hard and progressive nodes are concerning
- Hepatosplenomegaly
- Limbs may be due to bony pain

Investigations

- FBC, blood film

- NB: Neutrophils are high in the first few days and will be raised for the first 2 weeks. There is a nadir for the first 2 years after, until it raises to adult levels.
- NB: Lymphocytes are higher in children, especially in the first 2 weeks. Then it will be higher than adults up until 12 years of age.
- Toxic looking neutrophils (vacuolated) show a response to bacterial infections
- Reactive lymphocytes in infants can often resemble leukaemic blasts.

- T and B cell subsets
- Ig levels
- Antibody response to vaccinations
- Special tests: T cell activation testing, neutrophil function testing

Common causes of WCC abnormalities in healthy children

- Viral infections can lead to transient lymphopaenia or neutropaenia. IN a child who is otherwise well and recovers from illness, recheck these counts in 3-4 weeks
- Inappropriate 'normal' range used
- Atopy can lead to mild elevation in eosinophil count

Clues to the diagnosis in neutropaenia

- Neonatal: sepsis, alloimmune neutropaenia (from mother), congenital neutropaenia
- Infancy: viral infections, autoimmune, congenital neutropaenia
- Childhood: drug induced, viral leukaemia, marrow infiltration
- Duration can be a sign: if acute, viral or drug-induced, while chronic is congenital or autoimmune
- Associations
 - Vomiting in infancy – B12 deficiency
 - Malabsorption or loose stools – Schwachman syndrome, Pearson syndrome
 - Periodic mouth ulcers and fevers – cyclic neutropaenia
 - As part of a syndrome – cartilage hair hypoplasia, organic acidurias, GSD1b
 - Other autoimmune symptoms – SLE
 - Organomegaly or lymphadenopathy – leukaemia

IMMUNISATION

Lecture: Immunological basis of immunisation

Overview of immune system

- Remember T cells clear intracellular pathogens, while B cells clear extracellular pathogens via antibodies
- Clonal selection allows memory cells to quickly respond to an antigen (a second time) to form effector cells (plasma cells and cytotoxic T cells)

Role of vaccination

- A vaccination mimics the primary infection to induce a primary immune response by deliberate injection of weakened, modified or related antigen stimuli (that is not pathogenic)
- This produces memory cells and creates immunological memory, but doesn't cause disease in the host
- When you encounter the pathogen naturally, the secondary immune response is stimulated and the disease is reduced or not present

What is the ideal vaccine?

- Once only
- Oral
- Safely given at birth
- With lifelong protection
- However, the reality is multiple doses with different timing/spacing. This is affected by vaccine type, maternal antibodies, maturity of immune system and disease incidence and prevalence

Types of vaccines

- Live attenuated
 - Advantages: closest replication to natural infection providing long lasting protection
 - Disadvantage: cannot be used by immunocompromised persons, has some pathogenicity/needs good cold chain, effectiveness affected by circulating antibodies
 - Examples: MMR, VZV, yellow fever, BCG
- Inactivated: subunit, toxoid, polysaccharide, conjugate and recombinant. In these it is mostly a humoral response
 - Advantages: cannot cause disease and safe for immunocompromised people
 - Disadvantages: no amplification/replication, multiple doses required
 - Subunit: part of the bacterium or virus (e.g. pertussis, inactivated polio, influenza, HBV)
 - Toxoid: inactivated toxin that causes the disease (e.g. tetanus, diphtheria)
 - Recombinant: where DNA for an antigen is encoded into mammalian cells (e.g. HPV, HBV)
 - Polysaccharide: T cell independent immune responses that induces mostly IgM, causes poor immunogenicity (e.g. Mencevax or Pneumovax). There is no immune memory and no T cell response
 - Conjugate vaccine: Polysaccharide is conjugated to antigen, causing a T cell dependent response and is immunogenic in

under 2 year olds (e.g. Nimenrix, Prevenar, Hib)

Maternal antibodies

- Maternal antibodies can cross the placenta and provide some protection
- However, it can interfere with the vaccine response (e.g. measles, which is live)

How do we enhance immunogenicity?

- Repeat doses with appropriate spacing (sufficient time for memory development, but risk of insufficient protection against disease if too far)
- Add adjuvant (a depot), e.g. Alum

Lecture: Immunisation for special risk groups

What is a special risk group?

- These are groups of the population that are at increased risk for vaccine-preventable disease OR have altered immune systems

Aboriginal and Torres Strait Islanders

- Access to care is an issue, especially with remoteness
- There can be lack of culturally appropriate communication and service delivery; we need better community engagement & representation, AMSS, and Aboriginal Immunisation Health Care workers
- Reminder strategies may work
- Additional vaccines
 - Pneumococcal 13v at 6m
 - Pneumococcal 23v at 4-5 years and > 50 years
 - HAV at 12 and 18m (except NSW/Vic/Tas)
 - Influenza annual for all aged > 6mo
 - HBV
 - Meningococcal ACWY and B > 2 mo
 - BCG for neonates living in NT, QLD, SA
 - Japanese encephalitis for outer Torres Strait Islanders

Preterm infants

- Antibody responses are lower with increasing prematurity
- We must give routine vaccines at chronological, not corrected age
- Increased risk of apnoea: usually self-limiting but need to monitor hospitalised preterm infants up to 48h post vaccination. There is no increased risk of SIDS
- Additional vaccines
 - Pneumococcal 13v at 6mo (for GA < 28w or chronic lung disease)
 - Pneumococcal 23v at 4-5 years (for GA < 28w or chronic lung disease)
 - HBV at 12 months (for GA < 32w and/or BW < 2000g)

Immunosuppressed patients

- These patients may have an increased risk of M&M from vaccine preventable diseases
- There are issues with safety and efficacy, so additional doses may be required
- It is important to vaccinate pre-emptively when anticipating immunocompromise
- Examples of immunocompromise include: congenital immunodeficiencies, or secondary (asplenia, HIV, Ca, transplantation, HSCT, autoimmune diseases)

Corticosteroids and vaccination

- Low dose CS can receive live vaccines, but high dose or prolonged CS should defer live vaccines until at least 1 month after cessation
- Most biological DMARDs are contraindicated for live vaccines
- There are 'safe doses' for other drugs, e.g. MTX, but there may not be an adequate response
- Live vaccines are totally contraindicated if taking multiple immunosuppressants

Asplenia and vaccination

- The spleen plays a central role in opsonisation and removal of encapsulated organisms from bloodstream
- Asplenic patients are vulnerable to infection by encapsulated bacteria, including Pneumococcus, Meningococcus, Hib, influenza
- Live vaccines are safe

HIV and vaccination

- Age of vaccination is important and level of immunocompromise (e.g. CD4+ count, viral load, AIDS-defining illnesses)
- Live vaccines
 - BCG is contraindicated
 - If CD4 < 200/uL, AIDS-defining illness, and symptomatic, contraindicated
- Recommended vaccines
 - Influenza
 - Pneumococcal
 - Meningococcal
 - HAV
 - HBV
 - HPV

Solid organ transplant and vaccination

- Before transplantation, ensure up to date with all routine vaccines well before transplantation. This may mean you need to check serology for MMR and varicella
- Additional vaccines include: Pneumococcal, HPV, HAV (if liver Tx), travel vaccines
- After transplantation, inactivated vaccines are safe but best given > 6 mo after transplantation to maximise immune response.
- Live vaccines are contraindicated post-transplant
- Influenzavirus vaccine is indicated annually, with 2 doses in the first-year post-transplant

Haematopoietic stem cell transplant and vaccination

- No live vaccines until 24 months post-transplant, provided no ongoing immunosuppression
- There is a need to revaccinate, regardless of autologous or allogeneic, with post-vaccination serology after HBV and MMR to determine need for further doses
- Important to consider travel vaccines

Blood product recipients and vaccination

- Some live viral vaccines may be affected by circulating antibodies contained in immunoglobulin therapy or blood products
- The vaccines affected include MMR, varicella

Coagulopathy and bleeding disorders and vaccination

- Defer IM injection if INR > 3 or if Anti-Xa level >0.5 units/mL at 4h post-dose
- For patients with haemophilia, consider SC route, or to vaccinate soon after receiving clotting factor replacement
- Ice and immobilise if haematoma develops

Pregnancy and pre-conception and vaccination

- In pre-conception, consider HBV, MMR, varicella and influenza. Rubella and varicella are particularly important in pregnancy
- Avoid pregnancy for 28 days after live vaccine is given
- During pregnancy
 - Influenza vaccine
 - dTpa between 20-32 weeks, repeat every pregnancy
 - Other live vaccines are contraindicated
- Yellow fever virus is C/I in breastfeeding

Lecture: Adverse events following immunisation**Vaccine safety**

- Remember that vaccines are medication and all medications have side effects
- Adverse events can impact public confidence in immunisation

What is an adverse effect following immunisation?

- An AEFI is any untoward medical occurrence that follows immunisation, which doesn't necessarily have a causal relationship with the usage of the vaccine
- Examples: rash, injection site reaction and febrile convulsions
- Classified as
 - Vaccine product-related – febrile seizures
 - Vaccine quality defect-related e.g. manufacturing error
 - Immunisation error-related
 - Immunisation anxiety-related
 - Coincidental
- Common AEFIs
 - Pain, redness, swelling at injection site
 - Injection site nodule
 - Low grade fever
 - Mild nausea
 - Mild headache
 - Muscle aches, tiredness
- Serious AEFI
 - These are life threatening, requiring hospitalisation and can result in persistent or significant disability or in a congenital anomaly
 - These require interventions to prevent permanent impairment
 - Examples
 - Intussusception – rotavirus
 - Febrile seizures – influenza and MMR
 - GBS – influenza
 - Disseminated disease in immunocompromised patient

Injection site reactions

- These are tender for the first few hours, and pain is settled by the time redness appears

- There is no associated with systemic toxicity but can present with lymphadenopathy
- Movement is not affected

Rashes

- If the onset is earlier, they are more concerning. A delayed rash can be viral
- Ask about other anaphylactic symptoms, e.g. eye swelling, mouth and lip swelling, respiratory difficulty

Vaccine specific AEFIs

- BCG – ulcer at injection site (2-6 weeks)
- MMR – delayed reaction of fever, faint red rash, head cold/URTI symptoms, drowsiness, swelling of salivary glands, low platelet count. This occurs 7-10 days after vaccination
- VZV – mild pustular rash at injection site (4-26 following vaccination)
- DTPa – extensive limb swelling
- HBV – lump at injection site

Contraindications to vaccination

- Anaphylaxis to a previous dose of a vaccine or vaccine component
- Immunocompromised (live)
- Women who are planning pregnancy within 28 days

Allergies and vaccination

- Most vaccines do not contain food allergen
- Egg allergy: MMR vaccine cultured on chicken fibroblast cell cultures while influenza vaccine is on egg ovalbumin. Both can be given safely to egg anaphylactic patients

Management

- Report AEFI
- For the patient, refer to specialist immunisation clinics (children with previous AEFI, immunisation needs that can't be met, children with complex medical health conditions)

INFECTIOUS DISEASES

Lecture: Introduction to microbiology

Pathogen classification

- Bacteria/prokaryotes: single celled organisms with no membrane-bound nucleus, contain DNA and RNA
- Viruses: replicate within cells of other organisms and can be DNA or RNA-based
- Eukaryotes: single or multi-cellular organisms with a membrane-bound nucleus, such as fungi and parasites (protozoa, worms and flukes)

Diagnostic techniques with bacteria

- Direct microscopy with Gram stain
 - Gram positive organisms have a thick peptidoglycan wall takes up Crystal Violet = blue appearance
 - Gram negative organisms have a thin peptidoglycan wall surrounded by an outer membrane do not take up crystal violet = pink colour
- Culture-based techniques
- Genus & species identification using mass-spec (MALDI-TOF), which has superseded prior use of biochemical reactions
- Antimicrobial susceptibility testing
- Serology testing (e.g. for GAS, syphilis)
- Antigen detection
- Molecular techniques, e.g. directed and non-directed PCR tests
- Point of care diagnostic kits

Broad classification of bacteria

- Gram positive
- Gram negative
- Mycobacteria
- Other
 - Spirochetes e.g. Treponema pallidum (syphilis)
 - Rickettsia (typhus)
 - Chlamydia, Mycoplasma, Ureaplasma
 - Coxiella burnetii

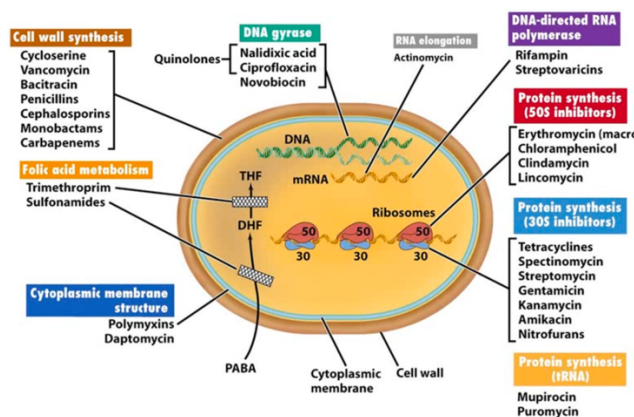


Figure 20-14 Brock Biology of Microorganisms 11/e
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Gram positive bacteria

- Gram positive cocci
 - Staphylococci
 - Streptococci pneumoniae, pyogenes (group A), agalactiae (group B)
 - Enterococcus

- Gram positive bacilli/rods
 - Listeria monocytogenes
 - Bacillus cereus, anthracis
 - Corynebacterium diphtheriae

Gram negative bacteria

- Gram negative cocci
 - Neisseria meningitidis, gonorrhoeae
 - Moraxella catarrhalis
- Gram negative bacilli
 - Enterobacteriaceae, e.g. E. coli, Klebsiella, Shigella, Salmonella, Yersinia, Enterobacter, Citrobacter, Proteus, Morganella, Serratia
 - Pseudomonas aeruginosa
 - Burkholderia cepacia
 - Bordetella pertussis
 - Haemophilus influenzae
 - Campylobacter
 - H. pylori
 - Kingella kingae – emerging pathogen as a cause of bone and joint infection in young children, but is a commensal organism in the oropharynx

Anerobic bacteria

- These are generally referred to as a separate group
- The disease manifestations include severe skin infections (e.g. abscesses, necrotising fasciitis), and colitis
- They can be Gm neg, Gm pos, rods or cocci
- Examples
 - Bacteroides spp
 - Clostridium difficile
 - Clostridium perfringens
 - C. botulinum
 - C. tetani

Mycobacteria

- Can be subdivided into
 - MTB complex: M. tuberculosis, M. bovis, BCG
 - Non-tuberculous mycobacteria
 - Slow growing NTM, e.g. M. avium
 - Rapidly growing NTM, e.g. M. abscessus
- Diagnostic techniques
 - Acid-fast stain and microscopy
 - Rapid PCR detection for TB
 - PCR for M. avium complex
 - Pan-mycobacterial PCR
 - Mycobacterial culture may take up to 6 weeks
 - Antigen tests and mycolic acid analysis can be used
 - Antimicrobial susceptibility testing (takes days – weeks)
 - Histopathology

Respiratory	Gastro-intestinal	CNS/eyes	Skin/ mucosa	Systemic
RSV	Rotavirus	HSV	HSV	HSV
Influenza	Norovirus	Parechovirus	VZV	EBV
Parainfluenza	Hepatitis A & E	CMV	HPV	CMV
Rhinovirus				HIV
Measles				Measles
				Parvovirus
				Hepatitis B & C
<----- Adenovirus ----->				Adenovirus
<----- Enterovirus ----->				

Common viral pathogens

- Viruses can show tropism or be broad across their organ systems affected
- Enterovirus is the most common cause of viral meningitis
- Enterovirus also causes hand, foot & mouth disease

Viral diagnostics

- Serology antibody tests (esp. IgM and IgG), including titre rise and avidity testing
- Culture is rarely testing. It requires specific cell-lines, is slow, requires specialised laboratories
- Antigen detection, e.g. in stool (adenovirus, norovirus, rotavirus) and respiratory (influenza and RSV)
- NAAT which has rapid turnaround time, with increased Sn and Sp and ability to measure viral load and resistance gene/mutation detection
 - Remember PCR tests can detect viral shedding and “dead” viral particles

Fungi

- Yeasts are single celled organisms that multiply by binary fission, e.g. Candida and Cryptococcus
- Moulds are multi-celled organisms that multiply by sexual or asexual reproduction, e.g. Aspergillus
- Dimorphic fungi can exist in yeast or mould phases, e.g. Histoplasma
- Diagnostic techniques
 - Direct: India ink, silver, lactophenol cotton blue stains
 - Culture is slow and require selective media. They are identified based on morphology
 - Confirmation based on mass-spec and molecular testing
 - Antifungal drug susceptibilities occur for yeasts and Aspergillus only
 - Targeted PCR for Aspergillus and PJP
 - Pan-fungal PCR
 - Antigen-based testing, e.g. Cryptococcal antigen in CSF/serum or Galactomannan assay for Aspergillus from serum or BAL or Beta-D-Glucan antigen for invasive fungal disease

Parasites

- Malaria is caused by Plasmodium genus
 - Diagnosis requires thick and thin blood films
 - Malaria antigen test
 - Malaria PCR
- Worms and enteric parasites, e.g. Giardia, pin worms (Enterobius vermicularis), Entamoeba histolytica
- Other less common parasites

- Toxoplasma gondii
- Strongyloides stercoralis
- Leishmania
- Trypanosoma cruzi
- Schistosoma

Diagnostic techniques

- Microscopy from wet prep (e.g. stool, urine, tissue)
- Molecular PCR tests
- Serology for specific pathogens

Lecture: Paediatric infections in Australia

Epidemiology of infections in children

- Children have a less than 1% mortality, due to improved access to and quality of neonatal health care, community awareness, sanitation and hygiene and immunisation

Neonatal infections

- Risk of bacterial sepsis is up to 30%, however early signs are non-specific including lethargy, poor feeding apnoea and fever/hypothermia
- Early onset sepsis arises within first 7 days: rapidly progresses to fulminant multi-system disease with pneumonia
 - Group B strep 40%
 - E. coli 25%
 - Then less commonly other GNR, S. aureus, Listeria, Strep pneumoniae, Neisseria meningitidis
- Late onset sepsis arises after first 7 days, and usually presents as a slowly progressive focal infection, most commonly meningitis
 - CoNS predominate
 - Still commonly GBS and GNB

Group B Streptococcus

- GBS are Gm positive aerobic diplococci that form a narrow zone of beta-hemolysis
- GBS is found in GI tract, vagina and urethra
- GBS can cause septicaemia +/- meningitis (up to 30%), pneumonia and focal infections such as osteomyelitis, septic arthritis and cellulitis
- Incidence of invasive disease is 0.42 / 1000 live births
- Maternal colonisation is up to 20% and can be constant or intermittent. Up to 70% babies are colonised if mother has GBS
- Further risk factors for infection
 - Pre-term birth
 - Prolonged rupture of membranes
 - Maternal fever or infection
- Nearly 50% of survivors of sepsis due to GBS have long term neurological sequelae
- Method and timing of routine maternal screening is controversial, but we know there is a stronger protective effect if screening > 35 weeks gestation
- Intrapartum antibiotics are given to maternal carriers of GBS or those with obstetric risk factors, which is effective in reducing neonatal colonisation and preventing early onset sepsis. This has helped reduce the incidence of early onset sepsis by 80%, but has no effect on late onset sepsis

Neonatal HSV

- HSV are large enveloped dsDNA viruses
- Neonatal HSV is a severe disease with high M&M even when antiviral therapy is administered

- Incidence is 6 in 100 000
- The infection can manifest in many ways, including localised skin, eyes and mouth, pneumonitis, CNS involvement, disseminated
- HSV should be considered in neonates with fever (esp. in first 2-3 weeks), with skin lesions, or with abnormal CSF findings (esp. with seizures)
- Greatest risk factor is if primary genital herpes occurs late in pregnancy
- However, 75% of infants who contract HSV infection are born to mothers with no history or clinical findings suggestive of HSV infections
- Recurrent skin lesions are common in surviving infants
- Treatment IV acyclovir, up to 3 weeks

Congenital infections

- CMV
- Toxoplasmosis
- Rubella
- Varicella
- Syphilis

RTI in infants – children

- Upper respiratory tract: rhinitis, pharyngitis, tonsillitis, sinusitis, otitis media
- Lower respiratory tract: bronchiolitis, viral-induced wheeze, pneumonia
- Most RTIs in children are viral in origin

Acute upper respiratory tract

- Acute URTIs are a severe health and economic burden, which are most prevalent among young children during winter months
- Children <5 years have a cold 23% of time, and most costs are attributed to carer's lost time at work
- Risk factors
 - Day care/preschool attendance
 - Overcrowded conditions at home
 - Exposure to cigarette smoke

Otitis media

- Otitis media can be caused by viral, bacterial or both
- Bacterial causes include *S. pneumoniae*, non-typable *H. influenzae* and *Moraxella catarrhalis*
- This is a significant issue in Aboriginal and Torres Strait Islander children, with studies suggesting affecting up to 90%
- Treatment
 - Pain relief
 - ?Antibiotics

Dental disease

- This causes the highest cause of morbidity amongst Australian children
- 55% of 6 year olds have decay in their primary teeth, while 50% of 12 year olds have decay in permanent teeth
- There is interplay between fermentable carbohydrates, saliva and oral plaque-forming bacteria, e.g. *Streptococcus mutans*
- Periodontal infection is important to identify, and can be polymicrobial, or *Treponema*, *Porphyromonas*, *Actinobacillus*

Pertussis/Whooping cough

- This is caused by *Bordetella pertussis*, a Gm neg coccobacilli
- Spreads via infected family members, and results in apnoea/death and disability in unvaccinated babies.
- Phases
 - Incubation 7-10d
 - Prodromal 1-2 weeks of rhinorrhoea and intermittent dry cough
 - Paroxysmal 1-6 weeks associated of cyanosis, apnoea, vomiting and sometimes classic inspiratory whoop
 - Convalescent stage of gradual recovery
- There is an acellular vaccine that controls disease but does not reduce mucosal carriage. Immunity is not lifelong, and so immunisation of pregnant women is needed to prevent early neonatal disease
- Treatment or contact prophylaxis uses intracellular penetrating antibiotics, e.g. azithromycin

Lower respiratory tract infections

- Bronchiolitis, which is a common cause of hospital admission in children <2 years.
 - Initially starts with an URTI
 - Caused by RSV most commonly
 - Management is supportive, with ABx, bronchodilators and CS showing no demonstrable effect
- Viral induced wheeze can overlap with bronchiolitis
- Pneumonia usually shows focal chest signs, most commonly caused by viruses

Measles

- Measles is an enveloped RNA paramyxovirus, genus Morbillivirus
- It is highly contagious and spread by respiratory aerosols or direct contact with respiratory secretions
- The average incubation period from exposure is around 10-12 days until prodrome
- Prodromal symptoms include fatigue, cough, coryza and conjunctivitis
- This is followed by a fever and maculopapular rash, which starts on the head and progresses down to the feet
- Children are infectious for the 24 hours prior to flu-like symptoms
- Measles is a vaccine preventable disease, with 1 dose of vaccine providing long-term immunity in most people. After a 2nd vaccine dose, 99% of people are immune

Slapped cheek disease

- This is caused by parvovirus B19, which is droplet spread
- There is a mild preceding febrile illness, with a rash developing in two stages: red + flushed cheeks, then lacy maculopapular rash 1-2 weeks later mostly on the arms and legs. This rash can be fluctuant and is brought on by hot baths
- Sequelae: bone marrow suppression and anaemia
- Risk to pregnant women
 - Bilateral and symmetrical joint pain
 - Foetal anaemia → hydrops fetalis
 - 50% of women are immune

Roseola infantum

- This is caused by HHV6, and manifests with a high fever for 3-4 days
- As the fever abates, a widespread maculopapular rash on the trunk develops. Lesions tend to be discrete, while in measles they tend to become confluent
- This may be associated with febrile convulsions during the prodrome, however, this is self-limiting
- Immunocompromised patients may develop encephalitis

Enterovirus

- Enteroviruses are a family of viruses, including polio, Coxsackie, echovirus and other
- They are spread through respiratory droplets and faecal-oral transmission
- Presentations can vary through the different organs

Impetigo

- Impetigo is a highly infectious bacterial infection caused either by GAS or S. aureus
- They present as a red macule that becomes vesicular, which burst to leave a honey-coloured crust. This is often around the mouth but can occur elsewhere
- GAS infection can cause rheumatic fever and rheumatic heart disease

Meningococcal disease

- This is a major disease that presents two main ways: meningitis or septicemia
- Most deaths occur in septicemic form
- Overall incidence of invasive meningococcal disease is low, but potentially increasing
- Presentation
 - Early maculopapular rash that may blanch
 - Characteristic petechial or purpuric rash, that is non-blanching. Bacteria can be isolated from these lesions
 - Arthritis, pericarditis and pleural effusions may occur as autoimmune phenomena 5-10 days after the acute infection
- Management
 - Initiate antibiotics
 - Notify PHU
- Prevention via vaccination

Lecture: Global perspectives of paediatric infectious diseases**IDs and NCDs**

- We are now faced with the “double burden” of disease in developing countries, of infectious disease and non-communicable diseases
- Biggest killers of children <5 is pneumonia and diarrhoea, aside from neonatal causes

Diarrhoea

- Annually, there are predicted to be 5 billion cases, with the greatest burden in children <5 years and in SE Asia
- Causes of gastroenteritis
 - Viruses are most common (70%), most commonly rotavirus
 - Bacteria
 - Fungi and parasites

- Prevention: breast feeding, sanitation, hand washing, refrigeration, food storage and preparation, vaccination

Pneumonia

- This is the leading cause of death in infants and young children, and the disease leading to most DALYs lost
- Key global strategies are the global action plan for prevention and control of pneumonia
 - Integrated management of childhood illness
 - Vaccination
 - Improved nutrition
 - Reduced cigarette smoke / air pollution exposure
 - HIV prevention or management – as children have mortality of 50% by 2 years

Paediatric HIV

- Paediatric HIV has a mortality rate of nearly 50% by 2 years of age
- Social stigma remains a major barrier, even though paediatric HIV is preventable
- New WHO guidance is that we treat ALL children <5 years of age, irrespective of immune status
- HIV transmission routes to children is mostly through mother to child (in utero, intrapartum or breastfeeding)

Tuberculosis

- It is estimated that 550 000 children have tuberculosis, though we know the true disease burden is higher (probably ~1 million per year)
- The burden of disease in children is dependent on community transmission, as well as exposure
- TB contribution to child mortality is greatly underestimated due to underdiagnosis, especially among deaths from pneumonia, malnutrition, meningitis and HIV

Emerging infectious diseases

- Increasing agricultural intensification, global travel and translocation of wildlife means there are increasing emerging infectious diseases
- 75% originate from animals
- There is a non-uniform geographic risk

MEDICAL IMAGING

Lecture: Introduction to paediatric radiology

Differences in children CXR

- Films taken in AP in young children, which magnifies the heart and mediastinum
- Upper mediastinum can look broad (< 2 years) because of the thymus, which shows a uniform density with no mass effect, and has a triangular shape (sail sign)
- Pleural effusions can create a lateral line of demarcation, and overtake the entire lung space
- Atelectasis has a more uniform density and causes a triangular appearance
- Round pneumonia is a rounded airspace change that is consolidation within the lung; the collateral channels are not well developed so the pneumonia doesn't spread as readily
- Foreign body aspiration leads to air trapping distal to the object, which is seen as increased lucency and on inspiration, little change in volume to lung

Patterns in child lung disease

- Diffuse – alveolar, interstitial, airways
- Focal – focal alveolar opacity, nodules and masses

Neonatal lung disease – what are we looking for?

- Respiratory distress syndrome or HMD, and looking for complications of process including air leak (pulmonary interstitial emphysema, pneumomediastinum)
- Transient tachypnoea of the newborn
- Meconium aspiration syndrome causes areas of increased density and collapse and hyperinflation; major differential is neonatal pneumonia
- Congenital lesions, e.g. diaphragmatic hernia (most commonly left-sided), oesophageal atresia
- Position of introduced lines and tubes

Lecture: Paediatric gastrointestinal radiology

Differences in children AXR

- Neonatal gas pattern will show stomach bubble and lots of gas filled loops of small bowel in a mosaic pattern
- Free intraperitoneal gas can be shown as
 - Football sign
 - Riglers sign (gas on both sides)
- NEC will show
 - Dilated bowel gas pattern, with bubbly lucency (gas in the bowel wall, mural gas)
 - Branching lucency over the liver (gas in the portal venous system)
- Examine for oesophageal atresias or duodenal atresia (in 3rd part of duodenum leading to double bubble sign)
- Hypertrophic pyloric stenosis can be reliably diagnosed on US, and is often >14mm
- Malrotation with volvulus will show a corkscrew appearance in contrast studies

NEUROLOGY

Lecture: Approach to children with headache

What is headache?

- Primary headaches are recurrent and episodic, e.g. tension and migraine. These are thought to be intrinsic to the nervous system with a genetic predisposition
- Secondary headaches are related to an underlying process, e.g. space-occupying lesion, neurocutaneous syndrome, infection, metabolic, endocrine
- It is important to do a thorough history, examination (esp. neck tenderness, TMJ stability, facial tenderness)
- Headaches are very common: 1.7 % of 7 year olds, 3.9% 7-15 year olds and 5.3% of 15 year olds report migraines

Red flags in headache

- Abnormal neurological exam
- No or incomplete recovery between episodes
- Severe, recent onset, not like anything else
- Early morning headache (evidence of raised pressure)
- Chronic progressive headache (sinister origin)
- Wakes them from sleep
- Associated with vomiting

Treatment of primary headache

- Avoid triggers if possible
- Acute: SNAIDs, hydration, sleep, anti-emetic. If severe and prolonged consider Imigran nasal spray
- Prevention (dictated by frequency, duration and impact)
 - Lifestyle conditions, e.g. sleep, hydration, diet, exercise, relaxation
 - OTC: Riboflavin, Mg, Coenzyme-Q10
 - Prescription: Sandimigran, propranolol, topiramate, valproate, TCAs, gabapentin; none are particularly effective in children

Common presentations of potentially serious problems

- Early morning headache with vomiting → SOL, raised pressure
- New onset torticollis → SOL of posterior fossa mass
- Bell's palsy in a child that is too young, not quite with it → leukaemia, SOL
- Afebrile focal seizure and decreased LOC → NAI
- Combination of CN dysfunction, cerebellar and long tract signs → brain stem lesion or mass

Lecture: Children who look different

Occipitofrontal circumference

- This is a marker of growth and alteration of head contents, including brain, CSF, blood, skull thickness
- Prior to fusion of cranial sutures, changes in volume are reflected in the HC
- It can be unreliable in the newborn until day 3-4 of life, because of moulding, cephalohaematoma, caput succedaneum
- Specific charts for certain conditions exist, e.g. prematurity, achondroplasia, NFA1, Downs syndrome
- HC is most relevant and important in first 2 years of life, prior to fusion of sutures

How to measure HC

- Tape just above the ears, trying to find the maximum head circumference
- Plot it on chart according to age

Aetiology of primary and postnatal-onset microcephaly

- Primary (at birth): chromosomal disorders, neurulation defects, migration defect, problems with how the brain was formed
- Secondary (post-natal): intrauterine disorders (e.g. TORCH), perinatal brain injuries, postnatal systemic disease
- Red flags
 - Head circumference that is rapidly decelerating
 - Not the same implications as for macrocephaly in terms of urgency

Aetiology of macrocephaly

- Is it isolated, or part of a syndrome?
- If there is no syndrome evidence, is there developmental delay?
- If the child is well, it is likely to be familial macrocephaly or isolated macrocephaly
- If the child does have developmental delay, there needs to be further scanning
- Red flags
 - Rapid increase of HC (>2 per month) must be fluid or blood
 - Change in consciousness, excessive drowsiness
 - Prolonged afebrile seizure
 - Developmental arrest or regression
 - Signs of raised ICP
 - New focal neurological deficit
 - Neurocutaneous stigmata, e.g. café au lait, ash leaf macules

Investigations

- These will be mostly dictated by the presentation (acute or insidious) and associated examination findings
- In general, a head USS is helpful in children < 6 months
- CT scanning of the brain should be used in acute settings due to radiation risk
- MRIB is the preferred modality for looking at brain parenchyma, but can need a GA after first few months of birth

Lecture: Children with motor difficulty 1

What does the gait examination tell us?

- Normal gait requires normal function of all motor systems
- Central: pyramidal (corticospinal tract), extra-pyramidal tracts, cerebellar
- Peripheral: muscle/NMJ, nerve, anterior horn cell, bone/joints
- The examination of children is age-dependent (usually has to be > 12 months), otherwise we need to do the 180-degree manoeuvre for the non-walkers
- 180 degree manoeuvre looks for head lag, sitting head control and posture, and head control and leg positioning in horizontal suspension. We also look at head control when prone and when bearing weight
- Assessed symmetry, stability, manoeuvres (toe walking, heel walking)

- Other gait manoeuvres
 - Hopping for asymmetric weakness
 - Gower's sign for proximal weakness
 - Squatting/rising from a low chair for proximal weakness
 - Frog manoeuvre (walking on outside of feet) for dystonia

Infant patterns

- In babies, being 'floppy' is often the first clue to an underlying problem
- Is the baby 'floppy strong'? This may be a central cause with central hypotonia
 - Most common, associated with impaired alertness, seizures, poor feeding, head circumference changes, dysmorphic, nystagmus, retinal abnormalities
 - No weakness and normal reflexes
 - Seen in HIE, meningitis, Down syndrome, hyperbilirubinemia
- Is the baby 'floppy weak'? This may be a neuromuscular problem
 - These babies may be alert and awake but have weakness and absent reflexes
 - Seen in spinal muscular atrophy

Clinical approach

- Assess type of motor difficulty
- Is it isolated or combined?
- Site of lesion
- Associated intellectual, social or speech issues

Cerebral palsy

- Cerebral palsy describes a group of developmental disorders of movement and posture, causing activity restrictions or disability that are attributed to disturbances occurring in the foetal or infant brain. The motor impairment may be accompanied by a seizure disorder, and by impairment of sensation, cognition, communication and/or behaviour
- Cerebral palsy can result from brain injury occurring during the prenatal, perinatal or post-natal periods (< 2 years, but debated)
- Most common cause of childhood disability affecting 2.5 per 1000 individuals in the industrialised world
- 70-80% acquired prenatally, and mostly from unknown causes; 6% caused by birth asphyxia
- 10-20% post-natally acquired, e.g. bacterial meningitis, viral encephalitis, hyperbilirubinemia, MVA, NAI, falls
- Neonatal risk factors include birth <32 weeks, BW < 2.5kg, IUGR, ICH, trauma
- Most children with cerebral palsy walk, as 60% are independent ambulators, 30% walk with an aid and 10% use a wheelchair
- Workup: history + risk factors + type, screen for ID/ sleep/eye, examination, neuroimaging (MRI)

Types of cerebral palsy

- Affected parts of body: diplegic, quadriplegic, hemiplegic
- Affected motor types: spastic (most common – 80-90%), ataxic, dyskinetic, mixed

Associated problems with CP

- Development: intellectual disability 50%, epilepsy 25%, behaviour disorder 25%, sleep disorder 20%

- Functional: no walking 30%, no talking 25%, blind 10%, deaf 4%
- Physical: pain 75%, hip displacement 30%, bladder control problems 25%, drooling 20%, requiring tube feedings 7%

Lecture: Children with motor difficulty 2

Proximal muscle weakness

- Differentials of progressive proximal weakness
 - Spinal muscular atrophy
 - Myasthenia gravis
 - Congenital myopathies – frog leg posture at birth
 - Limb-girdle muscular dystrophies
 - Myopathies – metabolic infectious, inflammatory, toxin
 - Endocrinopathies – thyroid, adrenal
- Clinical signs
 - Trendelenburg gait
 - Gower's sign
 - Difficulty in heel walking (no dorsiflexion)
 - Shoulder weakness
 - 180 exams shows floppy and weak

Duchenne muscular dystrophy

- This is the most common neuromuscular condition in children, caused by absent dystrophin
 - Reduced dystrophin is in a milder form of the disease called Becker muscular dystrophy
- X-linked recessive
- 1:3500 of live male births, and 1/3 is a new mutation while 65% have a family history
- Up to 20% have a gonadal mosaic, concentration of abnormal cells can vary from organ to organ
- Onset of symptoms at 3-6 years
- Clinical features
 - Gross motor delay, delayed walking (16-18 months)
 - Generalised developmental delay, which can lead to mild-moderate ID (20% have IQ < 70)
 - Progressive weakness, especially proximal (Gowers sign)
 - Incidental finding of raised AST/ALT on blood tests (<3 years)
 - Pseudohypertrophy of calf muscles
 - Spinal deformity later in life
 - Cardiopulmonary involvement

Investigations for DMD

- Blood
 - CK – 10-20x normal, present from birth, peak at 2-3 years
 - ALT and AST raised
 - LDH raised
- EMG – rarely used now but myopathic findings show low amplitude polyphasic motor potentials, with abnormal spontaneous activity
- Molecular testing – gold standard using the MLPA which covers 79 exons. 2-20% of carriers have weakness
- MRI muscle is not usually done for diagnosis but can be used to monitor muscle disease progression

Management of DMD

- Surveillance with an MDT

- Orthopaedic: scoliosis, hip dislocation, tendon release, contractures
- Cardiac – cardiomyopathy (100% at 18 years), conduction abnormalities, later heart failure
- Respiratory – neuromuscular sleep apnoea
- General – constipation, school support, depression
- Corticosteroid therapy can help muscle strength and prolong time to loss of ambulation, improved cardiac function and less scoliosis
- Gene therapy – not currently available, animal trials are underway
- Genetic counselling

Prognosis of DMD

- Wheelchair dependent around 12 years of age
- With the use of NIV and steroids, life expectancy is now 30 years
- Cause of death usually cardio-respiratory failure

Distal muscle weakness

- Differentials:
 - Spinal cord: demyelinating disorders, transverse myelitis, trauma, infarction, epidural abscess
 - Anterior horn cell: poliomyelitis, enterovirus
 - NMJ disease: tick paralysis, MG, botulism
 - Peripheral nerve: GBS, toxins, CIDP
 - Muscle: myositis, metabolic myopathies, periodic paralysis syndromes
- Clinical signs: Poor toe walking

Guillain-Barre syndrome

- GBS is an autoimmune-mediated demyelination of the peripheral nerve and is the most common form of acute inflammatory peripheral neuropathy.
- This usually presents as an ascending symmetrical paralysis, which progresses over hours or days (worst at 2 weeks)
- Clinical presentation
 - Pain in 50% of patients
 - Weakness of peripheral muscles
 - Weakness of respiratory muscles 15-30%
 - Ataxia, paraesthesia, urinary retention
 - Cranial nerves are not commonly involved, and are rarely bulbar
 - Autonomic changes: bradycardia, postural hypotension, arrhythmias
 - Ophthalmoparesis, ataxia and areflexia = Miller-Fisher variant

Investigations in GBS

- Culture or serology in stool, looking for antecedent infection (found in 2/3)
- CSF – usually normal opening pressure, elevated protein and lack of pleocytosis
- EMG – demonstrates demyelination +/- axonal loss, diminished NCS
- MRI with gadolinium contrast may show enhancement of cauda equina and nerve roots
- Most important DDx to rule out is spinal cord compression and so we can use an urgent MRI spine for this

Management of GBS

- Admit to hospital, assess ABC
- Assess autonomic instability
 - Paroxysmal hypertension
 - Cardiac arrhythmias
- If ambulant, mildly affected → watch and wait
- If non-ambulant
 - IVlg of 2g/kg over 3-5 days, risk of headache and viral transmission
 - Plasmapheresis is not commonly used in children
 - Steroids NOT affective

Inherited neuropathies

- CMT has a prevalence of 1 in 2500, with CMT1a the most prevalent of 60% (slow NCS in upper extremities + demyelinating neuropathy), while CMT2a is the second most prevalent in 6% (normal NCS with axonal degeneration).
- CMT1a and CMT2a are both AD, but there are other AR diseases
- Presentation
 - Child will walk on time, and then develop weakness or sensory loss in first 2 decades of life. Often has a high arch
 - Infantile onset will show walking after 15 months, the loss of independent ambulation by 20 years. Areflexia and foot deformity common, as well as scoliosis and congenital hip dysplasia. If severe, may be associated with facial weakness and tongue fasciculations
 - Adult onset develops symptoms as late as 40 years of age
 - Non CMT demyelinating neuropathies can present prenatally with hypotonia, arthrogryposis and respiratory insufficiency
- Investigations
 - NCS, noting that velocity increase with age up to 3 years of age
 - Needle EMG can distinguish between myopathies and neuropathies
 - NN biopsies can point to underlying genetic syndrome
 - Genetic studies
- Diagnosis
 - Is this an isolated neuropathy or is there CNS involvement?
 - Are there associated physical signs, e.g. cataracts?
 - Inheritance pattern?
- Treatment: PT, OT, orthotists, orthopaedic surgeons and pain management

Lecture: Unusual paroxysmal events 1

Work up of paroxysmal events

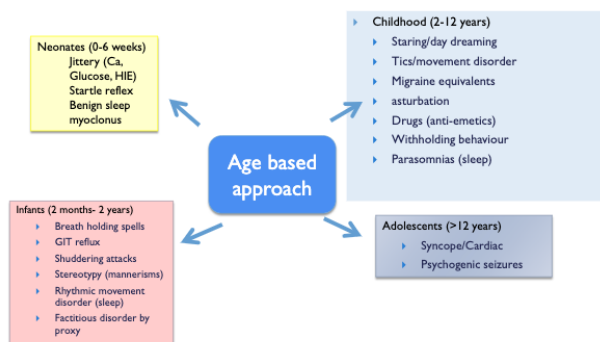
- Thorough history and examination
- Look for familiar patterns
 - Behaviour arrest
 - Unresponsive or aware during episode
 - Staring or eye deviation or nystagmus to one side
 - Movement/posture – extra movements or less movements
 - Breathing changes, colour changes
 - Automatisms
 - Post event

Terminology

- Myoclonic jerk – brief tonic contraction, lasting <0.1 sec
- Spasm – sudden brief bilateral tonic contraction of axial and proximal limb muscle with abrupt on/off, lasting 0.2-2 sec
- Clonic – repetitive flexion and extension, rhythmically recurring from 1-3Hz up to 5 Hz
- Tonic – persistent contraction of limb muscles lasting 2-10 seconds

Differentials to consider

- Benign neonatal sleep myoclonus
- Infantile masturbation – repetitive thigh adduction absorbed appearance often in car seat
- Stereotypies – often in learning disabled or autistic children, excited flapping
- Parasomnias – are night terrors, occur early in sleep (3-4 hrs)
- Sandifer syndrome – abnormal postures in kids with reflux +/- hiatus hernia
- Migraine or migraine variants
- Movement disorders
- Syncope
 - Benign – prolonged expiratory apnoea (blue breath holding), reflex anoxic (asystolic), where a painful stimulus leads to PNS stimulation and severe bradycardia, syncope, convulsive Valsalva
 - Malignant – LQTS, Hyperekplexia (stiff apnoea in neonate preceded by staccato cry)
- Pseudoseizures
 - Precipitated by stressful events, can be induced by suggestion
 - “Teddy bear” sign
 - Occur in wakefulness with witnesses
 - Lack of stereotypical characteristics
 - Asynchronous, asymmetrical with waxing and waning
 - No post-ictal confusion, can be more emotional
 - Eyes closed
 - La belle indifference
 - May have incontinence, tongue biting



Benign sleep myoclonus

- Onset within first few weeks of life
- Clinical features
 - Rapid, random, bilateral and asynchronous myoclonic jerk
 - Goes for seconds, but can up to 30 minutes in sleep, precipitated by bumping bed
 - Can occur in all stages of sleep

- Disappear completely and abruptly when infant is woken up; not seen when alert
- Do not stop on passive flexion
- Differentials: seizures, and jitteriness
- EEG is normal at baseline and during events
- Most disappear by late infancy ~7 months

Breath holding

- Common in 6 months and 2 years (up to 5 years)
- Clinical features
 - Trigger of fear, injury, and frustration
 - Crying then breath holding (expiration)
 - Become stiff, loss of awareness
 - Clonic jerks
 - Pallid type (reflex asystole): parasympathetic dysregulation, pale and limp with asystole
- Management
 - Must check ECG to rule out arrhythmias
 - Correct anaemia if present
 - Counselling and reassurance

Parasomnias

- Nightmares occur in younger groups, are shorter in duration, and occur in REM sleep in early AM, and usually able to be recalled
- Night terrors occur in older ages, last >5 minutes and can have autonomic features. It occurs in NREM III/IV sleep in the beginning of the night and are not able to be recalled

Features	Nightmares	Night Terrors
Age of Onset	2-5 years	4-8 years
Duration	<1-2 mins	>5 mins
Semiology	Cling, verbalise	Vary/autonomic
Stage of Sleep	REM	NREM III & IV
Time	Early a.m	First third of night
Recall	Usually able	Not able.

Febrile convulsions

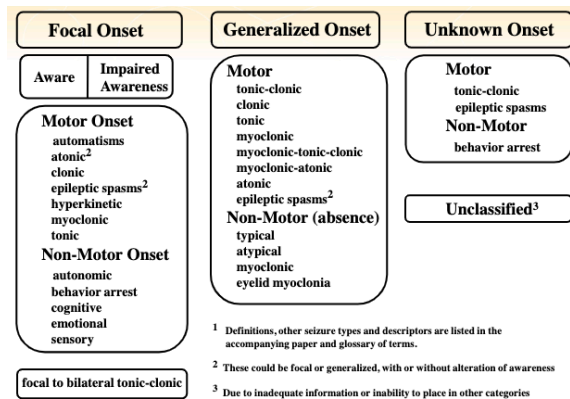
- These are epileptic seizures precipitated by fever
 - Usually high temperature > 38.5 degrees
 - Seizures may be the first symptom
 - Rule out intracranial infection or other CNS course
 - NO previous afebrile seizure
- Occurs in 2-5% of children < 5 years, 85% have first seizure by 4 years of age, with 60% of < 2 years
- Rare in < 6 months
- Up to 30% chance of 1st relative having febrile convulsions
- Clinical features
 - Brief, generalised, bilateral, clonic or tonic-clonic
 - Complicated if > 15 minutes, focal, reoccur more than once in a 24-hour period followed by Todd's paresis → 50% of risk of developing epilepsy
- Prognosis
 - 60-70% have one seizure
 - Most of the other have 2-3 episodes
 - Only 9% have > 3 episodes

- 75% reoccurrences occur in first 12 months
- Risk of afebrile seizure 2-5% (increased if first relative with epilepsy, and if complex febrile convulsions)
- Treatment
 - No evidence for antipyretics
 - If prolonged, think of buccal midazolam (0.3 mg/kg/dose)
 - Higher risk patients with RFs, consider anti-epileptic medication

Tics

- Tics are sudden, rapid recurrent, non-rhythmic motor movements or vocalisations
- They are distinct from stereotypies, and often have an urge associated with it
- Types:
 - simple (one muscle group)
 - complex (>1 goal directed, purposeless)
- If has occurred than 1 year, considered a provisional tic disorder of childhood, if lasts more than 1 year, it is a persistent tic disorder
- Tourette's syndrome is considered a combination of motor and tic disorders, nearly daily for >1 year, beginning before 18 years of age
- Epidemiology
 - 0.8% of population
 - Onset 3-8 years of age
 - M:F 4:1
 - Positive family history in 50%
 - Comorbidities include ADHD, OCD, ASD
- Symptoms may wax and wane over time, often peaking in severity prepubertally
- Most common type is blinking, typically alleviated by a physical task, worsened by stress and anxiety

Lecture: Unusual paroxysmal events



Infantile spasms

- West syndrome
 - Epileptic spasms
 - Interictal EEG of hypsarrhythmia
 - Behavioural regression or mental deterioration
- Occurs in 1/3500 births
- Onset in first year of life, 90% between 4-10 months
- Spasms are 1-2 seconds, repetitive, every 5-30 seconds up to 100x. These can be head nodding, shoulder shrugging more like, can be asymmetrical; they occur when drowsy
- Aetiology
 - 80% symptomatic: HIE, neurocutaneous syndrome, brain malformations, metabolic

and degenerative disease, infections, toxic (hypoglycaemia, lithium)

- Around 15-20% are cryptogenic, which has a more favourable course
- Treatment
 - Referral to paediatric neurologist, do not delay treatment as this can increase the chance of irreversible effects
 - Conventional anti-epileptics are ineffective
 - Need steroids, e.g. prednisolone, vigabatrin in TS, watch out for VFD
- Prognosis is best in cryptogenic cause especially if there were normal development prior to onset of infantile spasms

Tuberous sclerosis

- Periungual fibroma
- Ash leaf macule – hypopigmented lesion
 - Woods lamp can be used to see ash leaf macules
- Shagreen patch – CT nevus
- Adenoma sebaceum
- Calcification of cortical tubers on CT or MRI

Childhood absence epilepsy

- Absence seizures exhibit sudden suppression or marked decrease in consciousness with abolished awareness
- Onset 3-8 years
- F>M
- 10% of childhood epilepsy
- Clinical features
 - Brief, last 5-15 seconds
 - Abrupt onset and termination
 - May have subtle motor automatisms: lip smacking, chewing or fiddling
 - Occur multiple times a day
- Treatment: ethosuximide
- Prognosis
 - 65% remission, of those without remission, 40% develop JME
 - Usually normal development

Juvenile absence epilepsy

- Same as CAE, except onset >10 years
- Clinical features
 - Cluster in the hour following awakening
 - Longer staring episodes but less frequent than CAE
- 90% associated with generalised tonic-clonic seizures
- Can progress to become JME
- EEG will be the same, but may be a bit faster (4-6 Hz instead of 1-3 Hz)
- Treatment: ethosuximide + others if GTC

Juvenile myoclonic epilepsy

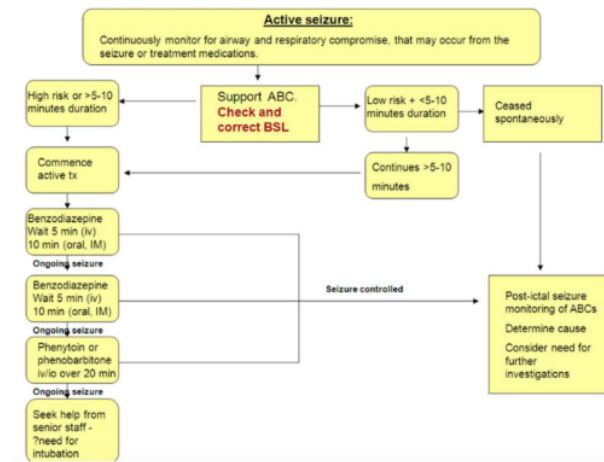
- These are the most frequent, well-defined idiopathic generalised epilepsy
- Onset of 12-18 years, mean 14 years
- Occur in 10% of adolescents with epilepsy
- Clinical features
 - Triad: absence, myoclonic jerks and GTC
 - Seizures mostly occur on waking with myoclonic jerks
 - Can be single or multiple
 - UP to 90% have GTC, and around 30% have absence seizures

- GTC seizures are provoked by sleep deprivation, alcohol intake, photic stimulation and arousals from sleep
- Treatment: lifelong, usually valproate
 - Lamotrigine or Keppra in women of childbearing age
 - Seizures made worse by phenytoin and carbamazepine

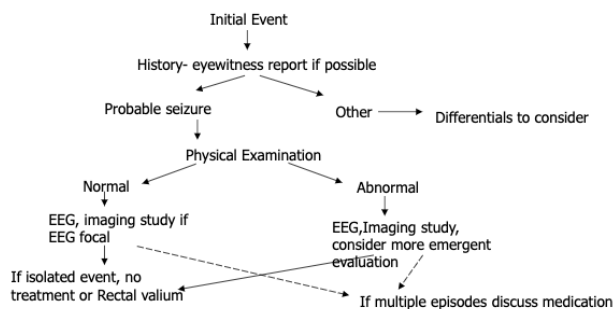
- If BSL < 3.5 give IV glucose 2mL/kg of 10% dextrose
- Treat reversible causes
- If febrile consider ABx and antivirals
- IM Benzodiazepines

Benign Rolandic epilepsy with centrottemporal spikes

- These are one of the focal epilepsies of childhood, with a strong genetic contribution
- Onset 5-10 years, M>F
- Clinical features
 - Brief 30-60 seconds, with no impairment in consciousness
 - Hemi-facial sensorimotor seizures
 - Oropharyngeal laryngeal ictal manifestation with hypersalivation, arrest of speech
 - Can have a secondary GTC in 1/3
 - Can have nocturnal GTC with focal features in up to 50%
 - Neurodevelopmentally normal
- Prognosis
 - Most remit <16 years, even in severe cases
 - Most patients < 10 years (25% only ever have one seizure)
 - Comorbidity with ADHD and language difficulty
- Treatment
 - Do not use antiepileptics in mild cases
 - Consider if frequent seizures, e.g. carbamazepine



Investigative approach to paroxysmal abnormal episodes



Status epilepticus

- This is a single seizure or cluster lasting > 30 minutes
- Start treating after 5 minutes of continuous seizure activity, otherwise may become refractory to treatment
- Bimodal distribution, peaks in toddlers (<5 years, esp. under <12 months) and elderly >55 years
- Mortality in children ranges from 3-11%
- Investigations
 - FBC, EUC, LFT, CMP, glucose
 - Blood/urine culture if febrile
 - Drug level if relevant
 - Urine tox screen if relevant
 - CSF if febrile, once stabilised and GCS>8
 - EEG
 - Imaging: CT if suspected trauma or blood, otherwise MRI
- Management
 - Primary survey

OBEESITY

Lecture: Management of obesity in childhood and adolescence

Clinical assessment of obesity

- Anthropometry: height, weight, BMI, waist circumference, WHR
 - WHR > 0.5 for children above 6 associated with increased cardio-metabolic risk
- General history: pregnancy details, birth weight, GDM, early medical history, ethnicity
- Weight history: history of obesity, interventions, dieting, impact
- Complications history: psychological history, menstrual history, exercise tolerance, knee/hip pain, constipation, enuresis, sleep disturbances
- Family history: ethnicity, BMI of first-degree relatives, family history of obesity, heart disease, diabetes, OSA, bariatric history
- Physical activity
- TV/screens history
- Dietary history, incl. snacking, sneaking food, binge eating, fast food, sweet drinks, breakfast consumption
- Sleep history, incl. weekends vs weekdays

Organ system	Physical findings of note
Skin	Acanthosis nigricans, skin tags, hirsutism, acne, striae, intertrigo, pseudogynaecomastia
Neurological	Benign intracranial hypertension, flat affect (depression)
Head and neck	Large tonsillar size, obstructed breathing
Cardiovascular	Hypertension (ensure appropriate cuff size), tachycardia
Respiratory	Exercise intolerance, wheeze (asthma)
Gastrointestinal	Hepatomegaly (NAFLD), abdominal pain (gallstones, reflux)
Musculoskeletal	Flat feet; groin pain and painful gait (slipped capital femoral epiphysis); tibia vara; lower limb arthralgia and joint restriction
Endocrine	Extensive striae, hypertension, buffalo hump, pubertal staging, reduced growth velocity
Other	Short stature, dysmorphism, developmental delay

Red flag features

- Short stature, particularly pre-pubertal
- Abnormal physical stigmata
- Developmental disability
- Visual disturbance or headache

Investigation

- We would investigate when: Older children, higher BMI, high risk family history, high risk ethnic groups, clinical suggestion of comorbidities
- Fasting blood tests
 - Glucose
 - LFTs
 - Lipids
 - Consider insulin
 - Consider TFT
- Consider referral for sleep assessment
- Other investigations: OGTT, Liver USS

Management

- Family based lifestyle interventions – decreases weight, improves LDL cholesterol/TAGs/insulin
 - Need to be developmentally appropriate

- Food intake: follow national nutritional guidelines, meal patterns, dietary intake, whole-of-family lifestyle change, involvement of dietitian
- Physical activity: increased activity, addressing screen time, parental involvement, involvement of exercise physio
- Goal setting
- Stimulus control, e.g. modifying or restricting environmental influences
- Self-monitoring and detailed recording
- Pharmacotherapy
 - Consider orlistat (lipase inhibitor) in obese adolescents in the context of a lifestyle change program, which needs to be weighed up against the potential for adverse events like bloating, faecal leakage, steatorrhea
 - Consider metformin in obese adolescents with evidence of clinical insulin resistance – off-label

Lecture: Prevalence, causes and complications of childhood obesity

Prevalence of obesity

- Most overweight children < 5 years are in low- and middle-income countries
- This has been increasing obesity prevalence, largely from 1980s in many Western countries; thought there has been a recent plateauing of prevalence in some European countries, Japan and Australia

Immediate complications of obesity

- Cardiovascular: HTN, dyslipidaemia, chronic inflammation
- Metabolic/endocrine: high insulin, DM, menstrual irregularities, early puberty, PCOS
- Respiratory: sleep apnoea, asthma
- Orthopaedic: flat feet, ankle sprains, fractures, slipped femoral epiphyses
- Gastro: gallstones, NAFLD
- Neurological: raised ICP
- Psychosocial: secondary disordered eating, teasing, discrimination, poor self-esteem. This is the most common cause to seek management

Medium- and long-term problems

- Adolescent weight status predicts adult health risks
- Risk factors for later CVD, diabetes, metabolic syndrome and persistent obesity, cancers
- Poorer psychosocial functioning in early adulthood
- Increased risk of death or complications from later cardiovascular or other disease

Causes of childhood obesity

- Environmental influences: medications, parenting style, psychosocial stress, smoking, infant feeding, gestational weight gain
- Genetic predisposition: genetic contribution to the variance in BMI within a population is 40-70%
- Epigenetic influences

How do we tackle obesity?

- School and childcare settings: healthy school canteens, PE teachers, train childcare staff, sponsorship of schools

- Food environment: regulate food marketing, subsidise fruit & veg to schools and low SES, soft drink tax
- Physical activity infrastructure: extend public transport, increase green space
- Workplaces: healthy canteens, subsidise active transport, promote stairwell use
- Other: social marketing campaigns

ONCOLOGY

Lecture: Oncological emergencies

Epidemiology of cancer in childhood

- 600-700 children aged < 15 years develop cancer in Australia
- Leukaemia is the most common malignancy
- 80% of children with ALL are cured, and up to 60% of children with non-lymphoblastic leukaemia. Cure rates for solid tumours vary between 20% and 95%, with an average of 65%
- Survival rates are highest if treatment is supervised by a tertiary paediatric oncology units
- Other childhood cancers include cerebral tumours, NHL, Wilms tumour, osteogenic sarcoma, Ewing tumour, germ cell tumour, retinoblastoma, neuroblastoma, rhabdomyosarcoma...

Clinical features of cancer

- Persistent fever or recurrent fever without identifiable cause
- Persistent pain (esp. bone pain, headache)
- Mass
- Purpura
- Pallor
- Strabismus or other eye changes
- Changes in coordination, behaviour or milestones
- Less common symptoms
 - Persistent limp
 - Progressive weight loss
 - Localised joint pain and swelling with fever
 - Asthma-like symptoms caused by subacute onset of upper airway obstruction
- In acute leukaemia, FBC may reveal low Hb, neutrophils, platelets or circulating abnormal blast cells

Emergencies at diagnosis

- Severe anaemia
 - Hb < 70 g/L may result in high-output cardiac failure
 - Transfusion with pRBC prompt, if WCC is < 50 x 10⁹/L
 - Products should be group-compatible, cross-matched, CMV-neg, irradiated
- Severe thrombocytopenia
 - Platelets < 10 x 10⁹/L is associated with life threatening bleeding; if there is a cerebral tumour platelets < 50 x 10⁹/L warrants transfusion
 - Products must be group-compatible, CMV-negative, irradiated
- Fever + sepsis
 - Fever may indicate bacterial septicaemia, and can be life-threatening in neutropaenic (< 0.5 x 10⁹/L)
- Acute onset weakness in LL or bowel/bladder dysfunction
- Acute thoracic inlet obstruction
- Very high circulating blast count
 - Counts > 300 x 10⁹/L may be associated with blood vessel sludging due to high blood viscosity and CVA
 - Risk is greater when thrombocytopenia is also present

Emergencies during therapy

- Febrile neutropaenia
 - This will inevitably occur, but the diagnosis must be presumed to be septicaemia until proven otherwise; needs broad-spectrum antibiotics
 - Fever occurring within 2 hours of a CVL flush is suggestive of CVL-associated sepsis
- Infectious diseases
 - Measles and varicella can be life-threatening in immunosuppressed children
 - If children are non-immune and exposed, they should be given passive protection with normal human Ig 0.5 mL/kg for measles, or VZIG for varicella or shingles
 - Children should not receive live vaccines until at least 6 months after chemotherapy has finished

Pain and symptom control

- Severe pain occurs in up to 50% of children with cancers
- Vomiting is commonly associated with chemotherapy
- Aspirin and NSAIDs should be avoided as they inhibit platelet aggregation, but paracetamol can be used for pain/fever

Long term follow-up

- Recurrences of the underlying cancer will rarely occur later than 5 years after completing the original treatment
- Follow up should continue via long-term survival clinics into adult life; cancer therapy can affect growth, puberty, fertility, intellectual function and psychological wellbeing

Lecture: An introduction to childhood cancer

Epidemiology of cancer in childhood

- 1 in 600 children develop cancer or leukaemia in childhood
- It is the 2nd most common cause of death in childhood, outside of neonatal period (following accidents, and before congenital anomalies)
- Over 75% of children with cancer are cured permanently

What sort of cancers occur in childhood?

- ALL 23% of cancer, CNS tumours 21%, the rest e.g. neuroblastoma, NHL, Wilm's
- Leukaemia is about 30% of childhood cancer
 - ALL is 80%
 - AML is 20%
- The really bad diseases
 - Metastatic sarcomas
 - Stage IV neuroblastoma over age 1
 - High grade glioma
 - ALL with bad chromosomes, or infant ALL
 - Recurrences

Acute lymphoblastic leukaemia

- This is the most common childhood malignancy and arises from a lymphoid progenitor
- Symptoms (bone marrow deficiency symptoms and bone marrow infiltration symptoms)
 - Anaemia, bruising, fever
 - Bone pain

- Adenopathy, hepatomegaly, splenomegaly
 - Malaise, anorexia
- Investigations
 - Anaemia
 - Thrombocytopaenia
 - Neutropaenia
 - +/- high WCC, circulating blasts
 - Blood count usually abnormal, but CAN be normal
- Differentials: ITP, other neutropaenias/anaemias, infectious mono, lymphadenopathy
- Good prognostic factors
 - Age 2-10 years
 - Lower WCC
 - Responds to steroids
 - No CNS disease (in CSF)
 - B-precursor
 - T (12;21)
 - Hyperdiploidy
- Poor prognostic factors
 - Very young or aged >10
 - High WCC
 - CNS disease
 - T cell precursor
 - Philadelphia chromosome or t(4;11)
- Treatment
 - 1 month of chemo, “induction” which causes 98% remission
 - 6 months ‘heavy’ chemo
 - 1.5 years oral ‘gentle’ maintenance chemo
 - Can be intrathecal or cranial radiotherapy used, especially for treatment of CNS
 - 75% cured

Acute myeloid leukaemia

- This causes 20% of childhood leukaemia and is more serious than ALL; best results are 60% cure
- Requires more intensive chemotherapy, esp. high-dose cytosine
- Causes prolonged marrow aplasia and needs inpatient stay

Brain tumours

- Most common ones in paediatric are medulloblastoma, cerebellar astrocytoma, ependymoma, brainstem glioma and supratentorial astrocytoma
- Neurofibromatosis type 1 associated with optic nerve glioma
- Treatment may include
 - Surgery
 - Radiation therapy: however, this can decrease neurocognitive function (worse if <5, worse if supratentorial) and cause hypopituitarism and is carcinogenic
 - Chemotherapy

Neuroblastoma

- This is a tumour of the adrenal gland or of sympathetic nerve trunks; most are abdominal (but can be thoracic, pelvic, cervical)
- Most are malignant with poor prognosis, and need intensive chemo, surgery, ABMT
- Some have favourable biology and readily cured, particularly <1 year
- Stage IV disease (over 12 months)
 - Presents with flank mass, bony metastases, anaemia

- Disseminated disease with worse prognosis, usually in preschool age
- Other presentations of neuroblastoma
 - Newborn with massive liver
 - Neck mass
 - Paraspinal chest mass
 - Spinal cord compression
 - Adrenal mass on antenatal USS

Wilm's tumour

- These are renal cancers of young children, usually 1-7 years
- Presentation
 - Flank mass, often painless
 - Haematuria
 - Metastases to lung and liver
- Treatment: surgery, chemo

Hodgkin's lymphoma

- These are lymphomas that occur in older children, e.g. teenagers
- Presentation
 - Neck mass, mediastinal mass
 - Constitutional symptoms
- Children tend to have the favourable histologies, so there are generally good outcomes with chemotherapy and radiotherapy

Non-Hodgkin's lymphoma

- T cell: older child, mediastinal mass, pleural effusion
- Burkitt's B cell: abdominal disease/mass, intussusception (in older children), marrow, CSF disease
- Large cell NHL

Bone sarcomas

- Typically occur in adolescents and young adults
- Ewing sarcoma occurs in the diaphyseal long bones and the axial skeletons, and can cause extra-osseous tumours. t(11;22) can lead to EWS-FLI1 fusion transcript. 70% survive if non-metastatic
- Osteosarcomas occur at the growing ends of long bones, e.g. distal femur, proximal tibia and proximal Humerus. Again, 70% survive if non-metastatic

Soft tissue sarcomas

- Rhabdomyosarcomas are most common, which have two subtypes: embryonal and alveolar
 - Sensitive to chemo and RT
 - Occur most commonly in genitourinary, parameningeal, extremities
 - 70% survive if non-metastatic
- There are other non-rhabdo soft tissue sarcomas, e.g. undifferentiated sarcoma, synovial

Chemotherapy

- Generally multi-agent protocols
- Usually 6-12 months of cycles of therapy, though 2 years of ALL
- Late effects of cancer treatment
 - Infertility
 - Secondary neoplasia: AML from alkylating agents, solid tumours from RT
 - Cardiac damage from anthracyclines
 - Hypopituitarism or neurocognitive damage from cranial RT
 - Ototoxicity, nephrotoxicity, others

OPHTHALMOLOGY

Lecture: Conjunctivitis in children

What is conjunctivitis?

- This is inflammation of the conjunctiva
- It is characterised by redness, watery eye and discharge (purulent, mucoid, stringy)
- Classified by
 - Infective
 - Allergic
 - Chemical
 - Neonatal
 - Conjunctivitis associated with systemic disease

History of conjunctivitis

- Unilateral or bilateral
- Type of discharge?
 - Watery/mucoid – usually viral
 - Mucopurulent
 - Purulent – can be bacterial
- Systemic illness – sore throat
- Itchiness – itchy eyes in allergic conjunctivitis
- Vision

Examination

- VA – normal, unless associated with keratitis and mucous
- Conjunctival changes – papillae? Follicles (viral) ?
- Keratitis
- Skin rashes – associated Steven-Johnson syndrome
- Lymphadenopathy – often associated with viral

Laboratory investigations

- Usually not necessary, can make clinical diagnosis
- Conjunctival swab/scraping for Gm stain, culture and PCR
 - Neonatal conjunctivitis
 - Chronic or recurrent
 - Suspected HSV 1 or 2
 - Atypical reaction

Neonatal conjunctivitis

- Day 1 – gonococcal
- Day 4-6 – Strep, Staph, Haemophilus
- Day 8-12 – Chlamydia
- Day 7-14 – Herpes simplex
- Differentials: Congenital nasolacrimal duct obstruction (in 6% of infants, usually resolves)

Management

- Neonatal
 - Systemic antibiotics/treatment
 - Treat the mother
- Bacterial
 - Topical chloramphenicol or tobramycin
 - Eye drops 2-4 hourly during day, eye ointment at night
- Viral (usually adenovirus)
 - Can have a form called pharyngoconjunctival fever – may need conjunctival swab for diagnosis (not mx)
 - Self-limiting condition
 - Use lubricant eye drops

- Infection control measures
- Primary herpes simplex blepharoconjunctivitis
 - Examine cornea after fluorescein
 - Oral acyclovir
 - Topical acyclovir ointment
 - Do NOT use steroid eye drops which means virus will enter corneal stroma and cause a reaction
- Acute allergic blepharoconjunctivitis
 - Systemic antihistamines
 - If recurrent, topical olopatadine HCl 0.1%
 - Topical steroids – do not prescribe, just administer
- Vernal conjunctivitis or vernal keratoconjunctivitis
 - Present with subtarsal conjunctival papillae, limbal papillae (Trantas dots), can involve cornea
 - These can be chronic, recurrent and debilitating and can lead to visual loss (corneal scar, glaucoma)
 - Cold compressions
 - Topical eye drops
 - Mast cell stabiliser and antihistamine – topical olopatadine (Patanol)
 - Steroid eye drops for short term use
 - Lubricant eye drops
 - Systemic antihistamines not useful
- Stevens-Johnson syndrome
 - This is a systemic condition that affects mucous systems in the skin and eye
 - Needs systemic treatment
 - Needs ocular treatment
 - Lubricants
 - Topical antibiotics if secondary infection
 - Daily symblepharolysis with glass rod
- Chemical conjunctivitis
 - Alkali can penetrate the cornea, whereas acids cause superficial ulceration of the cornea
 - Tap water at home
 - Primary care
 - Local anaesthetic
 - Irrigate to neutralise for 30 minutes with NS
 - Particles of lime need to be removed
 - Instil antibiotic eye drops
 - Refer to emergency unit

Lecture: Strabismus and amblyopia

What is strabismus and amblyopia?

- Approximately 2% of children have strabismus and 2% have amblyopia
- Strabismus (e.g. cross-eyed, lazy eye) is the misalignment of the two eyes, where only one is looking at the object
- In the first 4 months of life, babies may cross and uncross eyes intermittently; it is only of concern if it is constant
- Differential diagnosis
 - Pseudostrabismus
 - True strabismus:
 - Infantile esotropia

- Accommodative esotropia – occurs in children who are hyperopic, around 2-3 years
- Intermittent exotropia
- Constant exotropia
- Paralytic strabismus – CN III, IV and VI palsies
- Associated with restrictive problems, e.g. Duane syndrome and Brown's syndrome
- Associated with intra-ocular or intracranial pathology

Types of true strabismus

- Infantile esotropia
 - Both eyes turning in, cross-fixating
 - This manifests during first few months of life, need to be referred early to fix binocularity
- Accommodative esotropia
 - Occurs in long-sighted children, where eyes turn in during accommodation
 - Treated with glasses
- Intermittent exotropia
 - Child tends to close 1 eye in sunlight, or may be more obvious when tired or daydreaming
- Constant exotropia
 - Some may be associated with neurological problem
 - Can be due to sensory loss of vision, e.g. sensory loss
- Paralytic strabismus
 - This is an emergency, and is sudden-onset and often presents with diplopia
 - Causes are congenital, viral, trauma, increased ICP
 - Investigate with MRI
- Restrictive strabismus
 - This is associated with Duane syndrome (abnormal innervation to LR) or Brown's syndrome (where SO doesn't pass through trochlear)
- Associated with intra-ocular pathology
 - Retinoblastoma
 - Congenital cataract
 - Toxoplasmosis

Tests for strabismus

- Observation and inspection for strabismus, abnormal head posture
- Visual acuity
- Corneal light reflection
- Cover/uncover test
- Ophthalmoscopy – red reflex, optic disc

When to refer the child?

- Concern about vision
- Sudden onset or constant strabismus
- Restricted eye movements
- Unequal red reflexes
- Suspected underlying eye pathology or intracranial pathology

Risk factors for childhood strabismus

- Family history
- Premature birth or low birth weight
- Drug or alcohol use during pregnancy

- Refractive error
- Intraocular pathology, e.g. cataract, RB
- Neurological disorders
- Craniofacial disorder

Amblyopia

- Reduction in VA that occurs as a result of interruption of the normal visual development during the “sensitive or critical period” of visual development (first few years of life)
- If detected during this period, it is potentially reversible
- Types
 - Unilateral: form deprivation, strabismus, anisometropia
 - Bilateral: ametropic, form deprivation
- Aetiology
 - Strabismus
 - Refractive – high ametropia and anisometropia
 - Deprivational – cataract, ptosis
- Management
 - Assess VA in all preschool children
 - Commence treatment during sensitive period of visual development (0-8 years)
 - Correct refractive error
 - Correct impediment to visual development
 - Stimulate visual development in amblyopic eye (e.g. patching, atropine drops)

Lecture: Leukocoria

Leukocoria

- Diagnosed by looking for the red reflex, which is routine at neonatal examination of newborn
- May be unilateral or bilateral
- Differential diagnosis
 - Retinoblastoma
 - Cataract
 - Persistent foveal vasculature (failure of the primary vitreous to regress)
 - Coat's disease, an exudative renal condition
 - Endophthalmitis (infective)

Retinoblastoma

- 1 in 20 000 live births
- Most (75%) present before age 3
- Bilateral in 25%
- Germline mutation in 25%, while new somatic mutation in 75%
- Presents with leukocoria and strabismus
- Early diagnosis is better for prognosis and treatment
- Treatment
 - Laser, cryotherapy, chemotherapy
 - Or enucleation

Congenital cataracts

- Aetiology
 - Familial 25%
 - Intrauterine infection in first trimester, syndromes, metabolic disorders 25%
 - Idiopathic 50%
- Investigation
 - Genetic studies
 - Specific investigations related to possible aetiology
- Management

- Surgery if vision significantly affected, do it in first 4 weeks if unilateral and in first 3 months if in first 3 months
- Correct aphakia after surgery with contact lenses (0-2 years) or intraocular lenses/contact lenses (>2 years)

Lecture: Assessing vision and strabismus in children

Vision assessment

- Birth – red reflex test
- 4 years – VA
- In preverbal children
 - Fix and follow
 - Ability to make eye contact
 - Teller Acuity, Cardiff Acuity
- Verbal vision tests
 - Kay Picture Test
 - Sheridan Gardiner Test
 - Snellen Test
- Corneal reflection (for strabismus), where we look for symmetrical light reflections; may be hard with large epicanthal folds
- Cover/uncover test

ORTHOPAEDICS: LIMP

Lecture: Development dysplasia of the hip

Epidemiology

- This is a condition that is nearly always present at birth
- Occurs as a wide spectrum of severity; can go from dislocation to dislocatable
- Incidence 4/1000 live births
- F>M 6:1

Risk factors

- Family history – major risk factor
- Breech presentation at birth
- First born
- Packing – lack of amniotic fluid
- Race – Canadian Inuit higher

Clinical features

- Asymmetrical skin creases – not a hard sign, 30% of normal children can have this
- Shortening
- Limited abduction in flexion – this is the easiest clinical sign to demonstrate and most important
- Barlow's test – performed to dislocate the hip
- Ortolani's test – performed if you think the hip is dislocated
- Other features of foetal packing – plagiocephaly, torticollis, postural foot deformities

Investigations

- Under 6 months – ultrasound if suspected DDH (FHx, breech, abnormal sign)
- Over 6 months – X-ray

Treatment

- Closed reduction and brace
 - Pavlik harness for infants
 - Abduction brace older children
- Reduction under GA and brace in spica cast
- Open reduction under GA and pelvic osteotomy and spica

Lecture: Transient synovitis of the hip

Clinical features

- Preschool child
- Limp with hip pain on movement
- Often after an URTI (in 50%)
- Systemically well

Epidemiology

- Commonly occurs in children 2-5 years of age
- M>F 2:1
- Differential
 - Transient synovitis
 - Septic arthritis – child systemically unwell, with fever, dehydration and lethargy
 - Perthes disease – systemically well, X-ray changes will help diagnose

Treatment

- Transient disorder with no long-term sequelae, often recovers in 2 weeks

- Use ibuprofen as analgesia

Lecture: Perthes disease

Epidemiology

- This is idiopathic AVN of the femoral head in childhood
- Occurs in 4-10 years
- Males > females 4:1
- Incidence 1 in 1000, and 10% of these they are bilateral (often not at the same time)

Clinical features

- Muscle wasting esp. quadriceps
- Limited hip abduction
- Positive Trendelenburg test

Management

- Investigation: X-ray showing fragmentation, collapse and bone cyst
- Treatment
 - Observation
 - Operative containment if femoral head subluxed (uncovered) by doing a pelvic osteotomy
- Prognosis
 - Late onset, whole head involvement of AVN and lateral subluxation are poor prognostic factors
 - 50% of adults who had Perthes will develop OA

Lecture: Slipped capital femoral epiphysis

Epidemiology

- This condition occurs as children go through their growth spurt, therefore 10-13 years in females, 12-15 years in males
- Males > females 3:2
- Incidence has increased more than 200% over the last 30 years due to the prevalence of childhood obesity

Contributing factors

- Growth spurt
- Obesity
- Trauma – rarely isolated cause
- Endocrinopathy – rarely

Clinical features

- Obesity
- Pain on hip, knee or thigh
- Fixed external rotation deformity, leading to limp and out-toeing gait

Management

- Investigate: AP and frog lateral view of X-ray
- Treatment: screw fixation of femoral capital epiphysis
- Outcome: Increased risk of OA

ORTHOPAEDICS: FRACTURES

Lecture: Fractures in children

Differences between paediatric and adult skeleton

- Anatomical differences
 - Physis (growth plate) for longitudinal growth
 - Thick periosteum for appositional growth
 - Apophyses are protuberant growths from separate ossific centre
- Biomechanical differences
 - Physis is a relative weak point in children's bones, so they are more likely to incur fractures than dislocations or sprains (as ligaments are stronger than growth plates)
 - Young bone is also more porous so tolerates more deformation rather than fracture; the bone may fail in compression as well as tension, causing buckle fractures
- Physiological differences
 - Children's bone typically heals faster than adults (physeal injuries in 3 weeks – Salter-Harris)
 - Non-union is an uncommon complication in a child
 - Non-anatomic reduction can be acceptable (as children's bone can still remodel)

Approach to fracture

- History
 - Age
 - Mechanism of injury
 - Time of injury
 - Witnesses
 - Treatments prior to presentation, e.g. splint
- Examination
 - Look – limb position, swelling and deformity, skin abrasions or lacerations
 - Feel – palpation of bony tenderness, joint effusion, vascular status (pulse, capillary refill), neurological status
 - Move
 - Neurologic status
 - Tendon injury
- Emergency treatment (if needed)
 - Reduction should occur if deformity is obvious and causing pressure to skin or neurovascular compromise is present
- Investigations
 - Plain radiograph

Classification of children's fractures

- Complete: spiral, oblique, butterfly, transverse
- Incomplete: buckle/Torus, greenstick, plastic deformation
- Epiphyseal injuries
 - Avulsion of ligamentous attachment
 - Osteo-chondral fractures
 - Compression fracture
- Physeal injuries (Salter-Harris)

Lecture: Management of fractures in children

Initial emergency treatment

- Indications for emergency reduction include

- Obvious clinical deformity causing pressure on the skin
- Obvious deformity with neurovascular compromise: loss or alteration of sensation, loss of motor function, pulseless limb

Initial treatment for all patients

- Appropriate analgesia
- IV fluids
- Reduce and splint injured limbs
- Elevate injured limb
- Ice
- Regular neurovascular observations
- Reduction, which relieves pain, tension on soft tissue structures including nerves and blood vessels and bleeding
- Maintenance of reduction
 - Slings
 - Broad arm – for clavicle and scapula fractures
 - Collar and cuff – for humeral fractures
 - Plaster back-slab
 - Below elbow or knee are for distal injuries only: Salter-Harris, buckle, wrist, hand and foot
 - Above elbow or knee are for more proximal fractures
 - If in doubt, an above elbow/knee backslab is always the safest option
 - Skin traction
 - NOT for SCFEs
 - Thomas splint traction: for femoral #s in children
 - Gallow's traction: for femoral #s in babies

Open fracture management

- Primary survey
- Initial cleaning or irrigation of wound before reduction
- Early administration of IV ABx, and targeted if soil contamination, freshwater contamination or community MRSA
- Tetanus booster or Ig as appropriate
- Surgical debridement is urgent (not emergent) <24 hours

Lecture: Long term implications of fractures

Remodelling of fracture

- The more proximal the injury, the less remodelling occurs
- Occurs to some extent in all fracture healing
- Sufficient to accommodate initial deformity if more than 2 years of growth remains, fracture is within metaphysis and deformity is in the plane of movement of the joint
- Remodelling with NOT accommodate the deformity if # is in the diaphysis, if there is gross angulation, shortening or rotation, or deformity is at right angles to plane of movement of the joint
- Remodelling will not restore articular congruency

Overgrowth

- Overgrowth is a phenomenon that results from increased blood flow to the physis next to the fracture

- This stimulates growth plate resulting in increased longitudinal growth. Coupled with the loss of “tether” of the periosteum can result limb length difference
- Is more relevant following femoral fracture, where it can result in 1-2 cm differences
- Limb length differences may be dealt with by epiphysiodesis, which is destruction of growth plate on the long leg. Needs to be carefully timed to allow short leg time to grow but not overgrow

Complete physal arrest

- This is permanent growth plate damage leading to premature closure of the physis
- Any limb length difference depends on growth remaining in the opposite limb
- Common following Salter-Harris Type V injury
- More likely in children approaching skeletal difference
- Mx: depending on predicted difference, e.g. epiphysiodesis or leg-lengthening procedures

Partial physal arrest

- Permanent but localised growth plate damage can cause a “bony bar” crossing the physis at the site of injury
- This acts as a tether at one point in the physis
- Angular growth may occur depending on the amount of growth remaining and location of the bar within the physis
- More common in SH Type IV and some SH type III injuries
- Management’
 - Depends on size and location of bony bar and predicted deformity
 - ‘Guided growth’ analogous surgical technique that allows for a tether on one sided of the physis to allow for gradual correction of angular deformity at the level of the physis
 - Epiphysiodesis
 - Epiphysiolysis
 - Osteotomy

PAIN MANAGEMENT

Lecture: Pain management in children

One voice approach

- One voice heard during procedures
- Need for parent/carer involvement
- Educate parent/carer before procedure
- Validate child with words
- Offer position of comfort
- Individualise approach
- Choose appropriate distraction techniques
- Eliminate unnecessary people from the room

Chronic pain

- Defined as pain persisting for 3 months
- Estimated 1 in 4 children suffer from chronic pain
- Often results in sleep disturbance, mood disturbance and loss of function

Measurement tools

- CRIES
 - Crying
 - Desaturating
 - Tachycardia/hypertension
 - Expression/Irritability
 - Sleeplessness
- FLACC < 4 years
 - Face – frown, clenched jaw, quivering chin
 - Legs – kicking, legs drawn up
 - Activity – arched, rigid, jerking
 - Cry – crying steady, screams, complaints
 - Consolability – difficult to console or comfort
- Modified FLACC for cognitively impaired
- Visual analogue tools > 7 years

Routes of administration of drugs

- Oral: can be not palatable
- Rectal: useful for NBM, well tolerated in young infants
- Transdermal: more for palliative than acute
- IV: rapid and titratable, preferred for opioid, useful for NBM
- Intranasal

Paracetamol

- 15 mg/kg/dose q6hrly, maximum 4 doses/day, 1g/dose in children over 3 months
- Dosing in hospitalised patients needs to be reviewed every 48 hours
- Should not be used regularly long term due to risk of hepatotoxicity

NSAIDs

- These have antipyretic, analgesic and anti-inflammatory properties; and work by inhibition of PG synthesis at COX site of prostaglandin-H2 synthetase
- Ibuprofen is most widely used in children > 3 months, 10 mg/kg/dose q8h
- Diclofenac (non-selective NSAID), used in juvenile idiopathic arthritis, used in children >10 kg 1 mg/kg/dose BD or TDS
- Naproxen, 5 mg/kg/dose
- Contraindications

- GI ulceration
- Thrombocytopaenia
- Clotting abnormalities
- Renal impairment
- Severe asthma is a relative contraindication

Opioids

- Oxycodone is widely used and has good oral bioavailability
 - 0.1-0.2 mg/kg/dose q4h PRN > 1 month
- Morphine has poor oral bioavailability
 - <1-year 0.1-0.2 mg/kg/dose q4h
 - >1-year 0.2-0.5 mg/kg/dose q4h
- IV opioids usually given with PCA (over 7 years) or NCA
- Adverse effects
 - Constipation, nausea, vomiting
 - Respiratory depression
 - Itching
 - IV naloxone 5 mcg/kg is reversal of choice

Adjuvant analgesics

- These are commonly prescribed with primary analgesics, for evidence of neuropathic pain
- Examples: pregabalin, gabapentin, amitriptyline, carbamazepine

Medications to consider for procedures

- Emla/topical anaesthetic agent
- Sucrose < 6 months
- Oral medications: analgesia, midazolam, ketamine
- NO

RENAL**Lecture: UTI in children****Epidemiology of UTI**

- These are common; 8% of girls and 2% of boys by 7
- 5% of febrile children in ED have UTI
- For the first 6-12 months, more common in boys (then girls afterwards)
- Up to 1/3 of children have recurrence, and most within 1 year
- Causes are E. coli (80%), and other causes include Klebsiella, Enterobacter, Proteus and Pseudomonas
 - Pseudomonas associated with anomalies in urinary tract
 - Gm+ organisms are uncommon, probably reflects a contaminant
- About 30% of children with UTI have VUR

Diagnosis of UTI

- Symptoms
 - Infant: poor feeding, vomiting, jaundice, fever
 - Older: dysuria, abdominal pain, wetting, haematuria, fever
- Urine sample
 - Types: Clean catch, in-out catheter, suprapubic tap
 - Nitrite+ leukocyte+
 - WBC > 100x10⁶/L
 - Single organism >10⁷ (catheter) or >10⁸ (voided)

Treatment of UTI

- Afebrile (cystitis) – treat with Keflex/Bactrim for 3 days
- Febrile (pyelonephritis) – treat with Keflex/Bactrim for 7 days
- If child is < 1 month, or particularly unwell – treat with IV ampicillin + gentamicin until appropriate for oral switch

Follow up investigations

- Under 6 months
 - Typical UTI: ultrasound within 6 weeks
 - Recurrent UTI: US in infection, DMSA 4-6 months after, MCUG
- Over 6 months
 - Typical UTI: nothing
 - Recurrent UTI: USS within 6 weeks, consider DMSA 4-6 months following acute infection
 - Avoid MCUG

Preventative measures

- Regular fluids and voiding
- Avoid and treat constipation
- Avoid caffeinated drinks
- Cranberry supplements
- Correct wiping
- Circumcision in boys
- Prophylactic antibiotics for 6-12 months if recurrent UTIs

Lecture: Nephrotic and nephritic syndromes**Differences between nephrotic and nephritic**

- Nephrotic: oedematous, proteinuria and can have some haematuria
- Nephritic: haematuria, HTN, red cell casts, raised CRT and can have some oedema and proteinuria

Nephrotic syndrome

- 2-7 per 100 000 children new cases per year
- 80% present before age of 6
- 80% relapse, about ½ frequent relapsers or steroid-dependent
- 80% steroid responsive
- 80% have minimal change nephrotic syndrome
- Classical triad
 - Hypoalbuminaemia →
 - hypercholesterolaemia
 - Heavy proteinuria
 - Oedema
- Treatment of first episode
 - Oral prednisone 60 mg/m² for 4-6 weeks, then wean
 - Education – test urine
 - Fluid restriction and no added salt
 - Pneumococcal vaccine, flu vaccine
- Treatment of relapse
 - Shorter course of prednisone (2 months)
 - Second line agents for frequent relapses or steroid-dependent, e.g. cyclosporin, cyclophosphamide, MMF, tacrolimus, rituximab
- Potential complications
 - Infection due to urinary loss of IgG, decreased complement, particularly encapsulated organisms
 - Thromboembolic due to increased clotting factors synthesis and loss of coagulation inhibitors
 - Drug side effects

What may not be minimal change?

- Renal impairment
- Significant hypertension
- Age < 1 year (may be congenital) or > 10-12 years (may be membranous or FSGS)
- Macroscopic or persistent microscopic haematuria
- Red cell casts in urine
- Family history of nephrotic syndrome
- “Syndromic” child
- Steroid resistance

Acute nephritis

- Clinical features
 - Coke-coloured urine
 - Some weight gain
 - Hypertension, raised crt, raised K
- Investigations
 - UEC, LFT, CMP
 - FBC + blood film
 - ASOT (recent Strep), anti-DNaseB (recent Strep)
 - C3/C4 (low in post-infectious GN, but recovers)
 - UA, urine MCS
 - Renal USS
 - +/- ANA, ANCA, anti-GBM, CK, renal biopsy

- Differentials
 - Post-infectious glomerulonephritis
 - IgA nephropathy (urine brighter red)
 - HSP
 - Membranoproliferative glomerulonephritis
 - Rapidly progressive GN
- Treatment
 - Conservative
 - Some may need diuretics (can help with HTN), anti-hypertensives
 - Immunosuppressants
 - Management of AKI

Lecture: Imaging of the renal tract

Renal ultrasound

- Looks for structural anomalies, e.g. pelvis dilatation, cysts, hydronephrosis, masses, stones
- Size of kidneys, position of kidneys
- Bladder outline

MCUG

- Gold standard for VUR and posterior urethral valves
- These are quite invasive, requiring catheter and contrast; this is usually used for children < 2 years

DMSA

- Static scan to look at differential function (e.g. 50% of left and 50% of right); values always add to 100
- Gold standard for picking up scarring

MAG-3 (DTPA)

- Functioning scan; amount of time on scan can tell us whether there is an obstruction
- Gold standard for picking up obstruction
- Can assess GFR

CT scan

- For complex anatomy that USS picked up
- Gold standard for kidney stones

Lecture: Hypertension in childhood

Measurement of BP

- Cuff size of bladder: width > 40% of mid upper arm circumference, length 80-100% of mid upper arm circumference
- Child sitting, legs uncrossed, cuff at heart
- Hypertension is SBP or DBP > 95th centile
- Consider 24 hr ambulatory BP monitor for children at least 5 or 6 years

Causes of hypertension in childhood

- Renal
 - Parenchymal disease: PCKD, scarring, GN
 - Obstructive uropathy
 - Liddle's syndrome
- Vascular
 - Renal artery stenosis
 - Coarctation of aorta
 - Takayasu's arteritis
- Endocrine
- Metabolic syndrome
- Central
 - SNS abnormalities
 - Raised ICP
- Iatrogenic: drugs, IV fluids

Infants	Young children	Older children	Adolescents
Renal vein or artery thrombosis	Tumours eg Wilm's or neuroblastoma	Renal parenchymal disease	Metabolic syndrome
Coarctation of aorta	Renal parenchymal disease	Renovascular disease	Essential hypertension
Congenital renal abnormalities	Vascular eg Renal artery stenosis or coarctation of aorta	Endocrine	Renal parenchymal disease
Bronchopulmonary dysplasia		Essential hypertension	Endocrine

Investigations of hypertension

- NB: need to be tailored
- FBC UEC, glucose and lipids if overweight
- Renin and aldosterone
- Urine catecholamines
- Urinalysis
- Renal doppler, abdominal USS
- Angiogram
- Echo and ophthalmal review

Treatment of hypertension

- Lifestyle changes – weight loss, low salt, exercise
- ACEIs or ARBs – NB teratogenic
- CCBs
- Beta-blockers
- Alpha-blockers
- Diuretics
- Centrally acting agents

RESPIRATORY

Lecture: Acute respiratory infections

Epidemiology

- These account for 50% of all illnesses < 5, 30% 5-12 years
- 95% of these are predominantly URTI, while LRTI are seen in earlier life and boys
- URTI includes
 - Coryza
 - Pharyngitis
 - Tonsillitis
 - Otitis media
 - Sinusitis
- LRTI includes
 - Laryngotracheobronchitis (croup)
 - Epiglottitis
 - Acute bronchitis
 - Acute bronchiolitis
 - Pneumonia
- Vast majority are viral (90%), while bacterial is less common: GBS, Sp. Pneumoniae, H. influenzae
- Mycoplasma pneumoniae most common in school age children

Differential diagnoses of cough

- Asthma
- Recurrent viral bronchitis
- Specific bronchitis
- Suppurative lung disease e.g. CF, bronchiectasis
- Focal lesions of the tracheo-bronchial tree e.g. tracheomalacia (brassy)
- Pulmonary aspiration
- Upper airway obstruction
- Smoking
- Habit cough

Lecture: Bronchiolitis and croup

Bronchiolitis

- Most common cause of admission to hospital in <1 year
- 1% in-hospital mortality
- Clinical features
 - Cough
 - Coryza
 - Wheezing, crackles
 - Increased WOB
- Usually in children <12 months, peaks 3-6 months
- More common in boys
- Peaks in winter
- Risk factors: lack of breast feeding, previous bronchiolitis
- Causes: RSV (75%)
- Severity risk factors
 - < 3 months
 - Prematurity < 32 weeks
 - Cardio-respiratory compromise
 - Aboriginal or Torres Strait Islander background
- Do not give steroids, beta-agonists, adrenaline, chest physio, antibiotics
- 10-14 day natural history
 - 50% fully recover by 14 days
 - 10% prolonged cough/wheezing

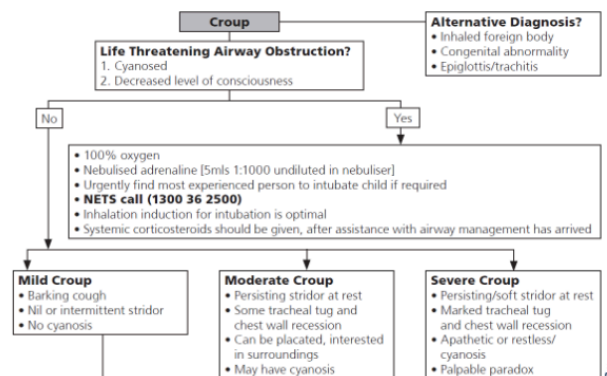
- Not a risk for asthma
- RSV immunoprophylaxis (palivizumab) is \$\$ and only given to highest risk children

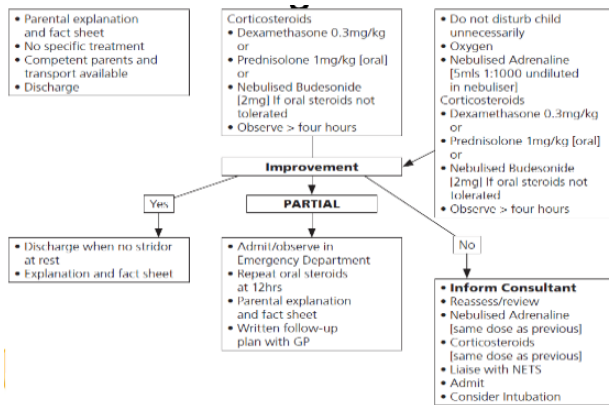
Initial Severity Assessment			
Treat in the highest category in which any symptom occurs			
Symptoms	Mild	Moderate	Severe and Life Threatening
Appearance	Well	Mildly Unwell	Unwell
Respiratory Rate	Mild Tachypnoea	Moderate Tachypnoea	Apnoeas Severe Tachypnoea Greater than 70 Bradypnoea Less than 30
Work of Breathing	Normal	Mild to Moderate	Moderate to Severe Grunting
Cyanosis	No Cyanosis	No Cyanosis	May be Cyanosed or Pale
Oxygen Saturation Oxygen Requirement	Above 95% in Air	90- 95% in Air	Less than 90% in Air Less than 92% in O2
Heart Rate	Normal	Mild Tachycardia	Tachycardia greater than 180
Feeding	Normal or Slightly Decreased	Difficulty feeding but able to take more than 50% of normal feed	Difficulty feeding taking less than 50% of normal feed
		Contact paediatrician	Get senior help then Call NETS 1300 36 2500

Treatment			
Oxygen	No	Give O2 to maintain saturation at or above 95% and or to improve the work of breathing	Maintain oxygen saturations greater than 95% Ensure high inspired oxygen via high flow delivery device if required
Hydration	Recommend smaller more frequent feeds if required	Smaller more frequent feeds Consider NG feeds	IV fluids and NBM
Investigations	Nil required	Nil required	Consider – CXR and Blood Gas / BSL
Observation & Review	hourly	Continuous SaO2 monitoring Minimum hourly observation	Continuous cardio respiratory and SaO2 monitoring Constant observation
No or Poor response to Treatment		Check diagnosis Escalate treatment	Get Senior Help Consult PICU via NETS Consider CPAP May need intubation
Disposition	Likely to go home	Likely to admit Decisions around hospitalisation of infants with SaO2 between 92 & 95 % should be supported by clinical assessment, phase of the illness & social & geographical factors	Transfer to an appropriate paediatric unit via NETS If in a children's hospital, may need PICU

Croup

- 5% admitted to hospital, 1% intubation, 0.5% in-hospital mortality
- Clinical features
 - Hoarse voice
 - Barking cough
 - Inspiratory stridor
- Usually in children <6 years, peaks 2 years
- More common in boys
- Peaks in autumn (variable)
- Risk factors: FHx
- Causes: parainfluenzae (75%)
- Give: systemic corticosteroids, nebulised adrenaline
- 48-72 hours of symptoms
 - Almost all recover
 - Frequently recur





Lecture: Cystic fibrosis

Epidemiology

- AR disorder
- 1 in 3200 of northern European descent, with a carrier frequency of 1:20 of Caucasians

Genetics

- CFTR gene defect on long arm of chromosome 7
- Most common deletion is the F508del in 85% of Caucasian CF
- Six functional classifications of CFTR mutations
 - Class I – no synthesis
 - Class II – folding defect (F508del)
 - Class III – channel opening defect
 - Class IV – ion transport defect
 - Class V – decreased protein synthesis
 - Class VI – decreased half-life of protein

Pathophysiology

- CFTR is a cAMP-dependent protein and plays a crucial role in transport of chloride, bicarbonate and sodium across the cell membrane
- Mutations in CFTR lead to abnormally thick and viscous secretions, causing intraluminal obstruction
- Pathology in CF is thought to arise from “obstructive tubulopathy”, i.e. ductal or glandular obstruction within the affected organs due to an inability to hydrate macromolecules within the ductal lumen

Diagnosis

- Newborn screening
 - Measure immunoreactive trypsinogen (IRT) as a biomarker of CF-related exocrine pancreatic disease
 - If elevated, CFTR mutation analysis is performed
 - Refer to CF centre
- Sweat chloride test
 - Primary diagnostic test
 - Measures amount of sodium and chloride in sweat

Clinical features

- Bronchiectasis, small airways obstruction – most M&M
- Chronic sinusitis and polyps
- ABPA – Allergic bronchopulmonary aspergillosis
- Pancreatic insufficiency
- Diabetes related to pancreatic insufficiency
- Biliary cirrhosis
- Meconium ileus

- More predisposition to GORD and intussusception
- Heat shock of sweat glands
- Infertility or subfertility
- Congenital absence of vas deferens
- Risk of low bone mineral density

Common respiratory pathogens in CF

- Childhood
 - S. aureus
 - H. influenzae
 - Pseudomonas aeruginosa
- Adulthood
 - Pseudomonas aeruginosa
 - S. aureus

Management principles

- Airway clearance
 - Chest physio
 - Exercise
 - Mucolytics – hypertonic saline, inhaled recombinant DNase
 - Bronchodilators
- Antibiotics: oral, inhaled (tobramycin for Pseudomonas), IV
- Nutrition
 - Pancreatic enzymes
 - Supplemental Vit A, D, E and K
- Emerging therapies
 - Ivacaftor for Class III gating mutations, potentiator that increases Cl⁻ secretion of CFTR
 - Lumacaftor – corrector drug
- Psychosocial aspects
 - Parental feelings of disbelief, shock, anger, anxiety
 - Feeling different from peers
 - Transitioning: frustration, non-adherence
 - Coping with progression

RHEUMATOLOGY

Lecture: Musculoskeletal symptoms in children

What is arthritis?

- Joint swelling or effusion
- OR tenderness/pain on movement, restricted ROM, increased warmth

Septic arthritis

- Cardinal features
 - Pain, swelling and redness of the joint
 - Fever
- Management
 - Surgical emergency
 - Joint aspirate and culture
 - Blood culture

Osteomyelitis

- Features
 - Bone and joint pain, pinpoint bone tenderness
 - Fever
 - X-rays can be normal early, so bone scan or MRI helpful

Reactive arthritis

- Features
 - Occur 1-3 weeks after infection at another site
 - Joint aspirate sterile
- Causes
 - Bacterial – GAS, enteric (e.g. salmonella)
 - Viral – rubella, measles, EBV, CMV

Transient synovitis

- Occurs in children of pre-school or early school age who present with sudden onset of limp, with early morning stiffness
- Constitutionally well, and often follows viral infection

Malignancy

- Systemic: leukaemia, lymphoma, neuroblastoma can cause bone pain, but polyarticular arthritis
- Local: osteoid osteoma, eosinophilic granuloma, sarcoma
- Consider malignancy if fever, weight loss, night pain, lymphadenopathy, anaemia, hepatosplenomegaly, pain out of proportion, metaphyseal lucencies on X-ray

Mechanical conditions

- Overuse syndrome e.g. little league elbow, swimmer's shoulder
- Avascular necrosis, more commonly an idiopathic cause (than steroids), in boys, no morning stiffness
- SCFE can cause secondary AVN, preferentially externally rotated
- Apophysitis, pain worse with activity
 - Sinding-Larsen-Johansson syndrome
 - Osgood-Schlatter – tibial tuberosity
 - Sever's disease – achilles heel insertion

Types of pain

- Hip pain

- 2-6 years – transient synovitis
- 4-10 years – AVN of femoral head
- 10-14 years – SCFE
- Pain syndromes
 - Chronic regional pain syndrome
 - Generalised pain syndrome, e.g. fibromyalgia
 - Treatment: physio, psychological support
- Growing pains
 - No relation to growth
 - Affects up to 1/3 in calf, thigh, shins bilaterally at night
 - Age 3-10 years
 - Normal physical exams and laboratory tests

Lecture: Juvenile idiopathic arthritis

JIA classification criteria

- Onset < 16 years of age
- Duration > 6 weeks
- Exclude other causes of arthritis
- Subtypes
 - Systemic arthritis (Still's) – 10%
 - Oligoarthritis (<4 joints) 50%
 - Polyarthritis (>5 or more joints) 30%

Systemic JIA

- This often mimics malignancy
- Features
 - Fever (quotidian)
 - Weight loss
 - Evanescent salmon-coloured rash, correlates with fever spikes
 - Arthritis
 - Hepatosplenomegaly
 - Lymphadenopathy
 - Serositis – pericarditis
 - Anaemia, high platelets, ESR and CRP
- Remission in 50%

Oligoarticular JIA

- Age < 5 years
- Girls > boys
- Large joints affected, usually knees
- ANA+ (80%), RF –
- Complications
 - Knee flexion contracture
 - Quadriceps atrophy
 - Leg length discrepancy
- Remission in 50-60%

Uveitis in JIA

- Most commonly associated in oligoarticular JIA (20%)
- Associated with ANA+
- Asymptomatic
- Does not parallel disease course
- Needs screening eye exams

Polyarticular JIA (RF-)

- Occurs in young girls > boys
- Affects small joints, large joints, neck, TMJ
- Uveitis occurs in 10%
- ANA+ in 50%
- Complications
 - Cervical spine arthritis
 - TMJ arthritis

- Growth disturbance
- Remission in 40%

Polyarticular JIA (RF+)

- Occurs in older girls
- Affects small and large joints, symmetrically
- ANA+ sometimes
- Rheumatoid nodules over pressure points in 30%
- Poor prognosis if not treated, no remission

Seronegative spondyloarthropathies

- Older boys >> girls
- Peripheral arthritis, leading later to axial arthritis
- Enthesitis
- 20% remission

Psoriatic arthropathy

- Clinical features
 - Psoriasis
 - Dactylitis
 - Arthritis
- Treatment: same as other forms of arthritis
- No remission

Non-articular manifestations

- Of enthesitis-related arthritis and juvenile spondyloarthropathy
- Psoriasis
- Iritis, conjunctivitis
- GI – IBD, mouth ulcers

Lecture: Non-juvenile arthritis

Types of vasculitis

- Large vessel: GCA, Takayasu (aorta)
- Medium vessel: Kawasaki disease, polyarteritis
- Small vessel: ANCA+ vasculitis, HSP

Henoch-Schonlein purpura

- This is the most common vasculitis of childhood, and affects the skin (leukocytoclastic infiltration)
- Affects
 - Skin
 - Kidneys
 - Joints
 - Gut
- Criteria (2/4)
 - Palpable purpura
 - Age < 20 years at onset
 - Bowel angina (after meals, or bowel ischaemia with bloody diarrhoea)
 - Granulocytes in walls of arterioles or venules on biopsy
- Long term outcome is good
- Follow up needs to check BP and UA
- Can cause swelling at the scalp, scrotal swellings (serositis of the tunica vaginalis)

Kawasaki disease

- This is a self-limited vasculitis of unknown aetiology, which is the second commonest vasculitis of childhood
- 80% of cases occur < 5 years
- Male > female
- Seasonal peaks
- Clinical features (need 4 of 5)

- Prolonged fevers
- Red injected conjunctiva
- Cracked red lips / mucositis
- Lymphadenopathy
- Rash
- Red palms and soles

- Most common cause of acquired heart disease in the developed world
- Major consequence is coronary artery aneurysm
- Early diagnosis and treatment reduces the complication rate from 25% to 5%

Dermatomyositis

- Rash on extensor surfaces (elbows, knees)
- Perioral rash
- Gottron's papules
- Nailfold capillary dilatation
- Calcium deposition in soft tissue
- Progressive muscle weakness
- Associated with malignancy in adult population, not paediatric population

SLE

- 20% of cases begin in <18 year olds
- More common Blacks, Hispanics, and Asians
- F>M 4-5:1
- Average at diagnosis 11-14 years

SLEEP DISORDERS

Lecture: Normal sleep in children

Sleep in infants and children

- Daytime sleep up to 3 years old
- Newborns spend up to 20 hours of every day asleep, which reduces to 11 hours by 6 months and 10 hours by 2 years
- Adolescents sleep for 8 hours, with a decrease in SWS

Stages of sleep

- Wakefulness: alpha, 8-13 hz
- Stage 1: theta, cessation of eye movements, mixed frequency
- Stage 2: spindles, K complex (nearly 50% in this stage)
- Stage 3-4: delta waves, slow waves, 0.5-2 Hz
- REM: low voltage, mixed frequency

Sleep architecture

- First half – predominantly SWS
- Second half – predominantly REM
- Duration of REM sleep reduces from 50% in newborns to 25% in adolescents

Symptoms of sleep disorder

- Disorders of duration of sleep
 - Excessive sleeping in day
 - Frequent daytime sleep
 - Insomnia
- Disorders of timing in sleep-wake cycle
 - Delayed sleep phase disorder
 - Abnormal or reversed circadian rhythm in children with blindness or neurological disease
- Sleep related behaviours
 - Movement disorders
 - Parasomnias: night terrors, nightmares, sleep walking and talking, enuresis, teeth grinding (bruxism)
 - Sleep related seizures
- Snoring
 - Prevalence of snoring is 10%
 - Primary snoring is on a milder spectrum of sleep disordered breathing: without gas abnormalities, apnoeas or frequent arousals. 60% of children will outgrow this by age 2

Obstructive sleep apnoea

- Prevalence of 2-5%
- Most common cause in children is enlarged adenoids and tonsils, so peak age is 3-5 years
- Gold standard investigation is PSG
- Nocturnal symptoms
 - Chest recession
 - Paradoxical breathing
 - Observable apnoea
 - Abnormal sleeping
 - Nocturnal enuresis
 - Snoring
- Daytime symptoms
 - Difficulty waking up
 - Morning headache
 - Poor school performance, attention span, hyperactivity

- Intrusive naps
- Frequent sore throat and dry mouth
- Examination
 - Mallampati classification (grades 1-4) – describes orientation of tongue with palate and posterior pharynx. Class 3-4 are at high risk of OSA
 - Tonsillar size (grades 1-4)
 - Lateral X-ray to identify adenoids, which can be a frequent cause of blocked nose (Sn 60%)
 - Lateral airway
 - Sleep diary
 - Pulse oximetry and morning VBG
 - Actigraphy
 - Video or audio tapes
 - Polysomnography

Lecture: Sleep disorders in children

Parasomnias

- Night terrors
- Partial confused arousals
- Nightmares
- Head banging
- Sleep walking and talking
- Teeth grinding

Night terrors

- Begins in 18 months, often in clusters, disappears for several months and then recurs
- Occurs in first 1/3 of sleep, where transition from NREM to REM
- Prevalence 6%
- Clinical features
 - Child inconsolable, often doesn't recognise parents
 - May bolt upright, stare, be diaphoretic, have irregular breathing
 - Lasts up to 10 minutes
 - NO recall in the morning
- Causes: genetic, sleep disordered breathing, acute triggers (fever, sleep deprivation, viral illness)
- Evaluation
 - Diagnosis by history
 - Exclude treatable causes of night terrors e.g. disrupted sleep, sleep apnoea
 - Review sleep hygiene if history of deprivation (less frightening TV, physical activities, consistent bedtime routine)
 - Refer if new onset in school age child, more than 3/week, lasts >30 min, affecting daytime performance
- Management
 - Patient education and reassurance
 - Do not awaken the child
 - Do not discuss in morning
 - If more resistance – scheduled awakenings, where you wake the child 10-15 minutes before and keep them awake for 5 minutes. Follow the procedure for 5-7 nights
 - Consider pathology if school age child and referral to specialist

Partial confused arousal

- This is a milder version of night terrors
- Occurs during transition from SWS to REM sleep (like night terrors)

- Child is confused, and may appear agitated or dazed for a period of 5-15 minutes
- Diagnosis
 - Careful history
 - Two-week sleep log or video
 - Overnight PSG if other sleep disorder suspected
- Management
 - Parent education and reassurance that they will outgrow by school years
 - Do not awaken the child
 - Do not discuss in morning
- Positive family history in up to 50% of patients
- Management
 - Reschedule sleep time by moving it early ½ hour each week
 - Improve sleep hygiene
 - Establish a morning routine
 - Incorporate regular exercise
 - Use relaxation techniques to manage trigger factors
 - Melatonin can be considered in refractory cases

Head banging

- Incidence 3.3-15%
- M > F 3:1
- Starts at 8 months and stops before age 4
- Usually occurs at bedtime
- Prior to onset head banging, most head bangers display other rhythmic habits, mostly body rocking

Nightmares

- These occur in the last 1/3 of the night during REM sleep
- Prevalence 10-50%
- Onset 3-6 years
- Features
 - Child often cries, is consoled
 - Child can recall episode
- Treatment
 - Give comfort at time of episode
 - Minimise night-time discussion
 - If stressor can be identified, decrease exposure

Obstructive sleep apnoea

- Pathophysiology
 - Enlarged adenoids and tonsils
 - Structure of the airway
 - Loss of neuromuscular tone in sleep
 - Nasal receptors maintain pharyngeal tone, which play a role in ventilatory drive and produces NO which is a pulmonary vasodilator
- Sequelae
 - Increased work of breathing – FTT, decreased appetite
 - Intermittent hypoxia
 - Neurobehavior in children – inattention, hyperactivity, irritability, daytime sleepiness
 - Pulmonary hypertension
 - Systemic hypertension
 - Sleep fragmentation – poor concentration, executive function, aggressive outburst
- Having daytime and nighttime symptoms makes the diagnosis of OSA more likely
- Management
 - Observe and reassess every 6 months
 - Nasal steroids and LTRA
 - Weight management
 - Adenotonsillectomy
 - CPAP
 - Tracheostomy

Delayed sleep phase disorder

- Prevalence of 7-16% in adolescents

TRAUMA

Lecture: Paediatric trauma

Epidemiology of paediatric trauma

- Trauma is the most common cause of death and mortality
- The spectrum of traumatic injuries is unique to children, and the most common ones causing death are road traffic injuries, then drowning, while the most common injuries are falls/sports then MVAs
- Blunt injuries are more common, and multiple injuries are more common

Primary survey

- Airway
 - Obligate nasal breathers from 4-6 months
 - Larger tongue in proportion to mouth
 - Pharynx is smaller
 - Epiglottis is larger and floppier
 - Larynx is more anterior and superior
 - Narrowest at cricoid
- Breathing
 - Ribs are more horizontal
 - Fewer Type 1 fibres – more fatigable
 - Thin chest wall
 - Infant RR 40-60, tidal volume 7-1 mL/kg
 - Hypoventilation is the most common cause of cardiac arrest in a child
- Circulation
 - Decrease in 25% of blood volume → shock, first manifests with raised HR, as hypotension only occurs with 45% of loss of volume
 - Systolic BP = $80 + 2 \times \text{age}$
 - Diastolic 2/3 systolic
 - Mx: 3x crystalloid boluses 10-20 mL/kg, then blood
- Disability
 - Anterior fontanelle window
 - AVPU
 - Modified GCS where best verbal response is 5 (appropriate, social smile), 4 (cries but consolable)
- Exposure
 - Warm the patient
 - Common sites missed: scalp, neck, hands, back, perineum
- Urinary output
 - Up to 1 year 2mL/kg/hr
 - Toddler 1.5 mL/kg/hr

Specific trauma

- Head trauma is the cause of death in most multi-trauma patients
- Chest trauma is rare, but sensitive to tension pneumothorax and underlying lung contusion (due to elastic ribs)
- Spinal trauma is rare but SCIWORA (Spinal Cord Injury without Radiographic Abnormality) can occur

Age group	Resp. rate	Heart rate	Systolic BP
Term baby	40-60	100-170	50
3 months	30-50	100-170	50
6 months	30-50	100-170	60
1 year	30-40	110-160	70-90
1-2 years	25-35	100-150	80-95
2-5 years	25-30	95-140	80-100
5-12 years	20-25	80-120	90-110
>12 years	15-20	60-100	100-120

PAEDIATRIC SURGERY REVIEW

Lecture: Congenital surgical problems

Gastroschisis

- This is herniation of the bowel via a small defect in the abdominal wall, just right of the umbilicus. The gut is also malrotated and susceptible to volvulus
- Incidence 1:4000 live births
- Associated with young mother, maternal smoking and recreational drug use
- Children tend to be premature and growth delayed
- Aetiological theories include rupture of the umbilical ring amnion and herniation of bowel, failure of the mesoderm to form in the body wall, abnormal involution of the right umbilical vein, disruption of the right vitelline artery and abnormal abdominal wall folding
- Usually detected on antenatal US where loops of bowel are free floating in amniotic fluid
- Associated anomalies are uncommon, but include intestinal atresia and undescended testes
- Postpartum management involves resuscitation, nasogastric suction and surgery. Oral feeding takes a month to establish

Exomphalos/Omphalocele

- This is a failure of the viscera to return to the abdominal cavity following physiological herniation in the pre-natal period. Herniated viscera include bowel and often liver, covered with amniotic membrane
- Incidence 1:4000-6000 live births
- Aetiological theories include persistence of the primitive body stalk, failure of the bowel to return to the abdomen, failure of migration of lateral body folds and body wall closure
- Usually detected on antenatal US
- Associated anomalies are seen in 50%, and associated syndromes include Beckwith-Wiedemann, Trisomy 13, 18 & 21, pentalogy of Cantrell
- Postpartum management depends on the size of the defect and anomalies

Tracheo-oesophageal fistula and oesophageal fistula

- Incidence 1: 2500-3000 births
- Unknown aetiology, though associated with chromosomal anomalies in 10% of cases, including trisomy 13, 18 & 21.
- Associated anomalies affect 50-70%, including MSK, cardiac, limb, anorectal, neural tube, etc
- 50% of anomalies are part of VACTERL or CHARGE syndrome
- Classification system is used – Gross
 - A: Pure oesophageal atresia
 - B: Oesophageal atresia with proximal fistula
 - C: Oesophageal atresia with distal fistula (most common)
 - D: Oesophageal atresia with proximal and distal fistulae
 - E: H type fistula with no oesophageal atresia
 - F: Oesophageal stenosis
- Usually diagnosed post-nasally, if diagnosed antenatally may be due to anhydramnios
- Features: difficulty breathing, excessive salivation, regurgitation or aspiration after birth

- NG tube will fail to progress after T2 to T4
- Management
 - Bronchoscopy to look for fistulae and other anomalies
 - Surgery via thoracotomy or thoracoscopic approach – difficulty with lengthening the oesophagus. May need the Foker technique (creating a neo-oesophagus)
- Prognosis is good but depends on other abnormalities (e.g. cardiac)

Congenital diaphragmatic hernia

- This is a developmental discontinuity of the diaphragm allowing the abdominal viscera to herniate into the chest
- Herniated viscera include gut, spleen and left lobe of liver. The viscera enter the thorax, taking up the space of the developing lung, and can cause pulmonary hypoplasia. This results in pulmonary hypertension postnatally.
- Incidence 1: 2000-5000
- Mortality of approximately 50% including in utero deaths
- Associated anomalies in 50%, and syndromes, e.g. trisomy 13, 18 & 21, Frey, Beckwith-Wiedemann syndrome
- There are two main types of congenital diaphragmatic hernias
 - Posterolateral Bochdalek hernias, which account for over 90% - due to failure of the pleuroperitoneal membrane to close. The right one closes first, which is why these are usually left sided
 - Anteromedial Morgagni hernias, which account for 2% - usually in older children
- The diagnosis is typically made antenatally on ultrasound, at around 24 weeks gestation. Hernias on the right can be obscured by the liver.
- Postnatally, most neonates are stable for around 24 hours. Ensuing respiratory deterioration is largely dependent on the degree of pulmonary hypoplasia and pulmonary hypertension
- Clinical features include
 - Scaphoid abdomen
 - Asymmetrically distended chest
 - Deviation of trachea away from defect
- Only prognostic factor is the degree and severity of other abnormalities and whether the child was able to immediately oxygenate after birth

Hirschsprung's disease

- This is a developmental disorder of the intrinsic component of the ENS characterised by the absence of ganglion cells of the myenteric and submucosal plexuses of the intestine in continuity from the anus to a variable distance proximally
- Incidence 1 in 5000 live births
- 80% confined to rectosigmoid area
- Associated syndromes include trisomy 21, neural crestopathies, MEN type 2, congenital central hypoventilation syndrome and NFT. There is also an association with neuroblastoma
- The aetiology is unknown but likely heterogeneous
- Often presents in the neonatal period with failure to pass meconium, abdominal distension, food intolerance and bilious vomiting

- Older children present with chronic constipation, abdominal distension, and failure to thrive
- AXR shows distended loops of bowel
- Diagnosis by suction rectal biopsy, which reveals absence of ganglion cells and hypertrophied nerve trunks in the myenteric and submucosal plexuses. Calretinin staining that is positive in normal bowel is negative in Hirschsprung's disease.
- Surgery
 - Remove aganglionic bowel
 - Bring normal innervated bowel to the anus whilst preserving sphincter function (needs dentate line)
 - Swenson, Soave, and Duhamel procedures
- Long segment Hirschsprung's disease is a transition zone beyond the mid-transverse colon, which can cause operative challenges

Necrotising enterocolitis

- This is a condition of the neonatal bowel characterised by a multifactorial exaggerated inflammatory response due to an initiating event in the context of an immature intestine and immune system
- Triggering event is some kind of physiological stressor but is not often identified
- Affects 1 in 1000 live births, 90% of whom are preterm
- Mortality is 30%, 50% in those requiring surgery
- Pathophysiology is unclear, however following initiation, the intestinal epithelial barrier is disturbed. There is bacterial translocation and exaggerated immune response causing a systemic response
- Severity measured using modified Bell's classification systems
 - Stage 1: suspected, apnoea, bradycardia, temperature instability, mild abdominal distension, occult blood in stool and increase gastric residuals. AXR show distended loops of bowel
 - 2 → 2a → 2b → 3a and 3b, patient presents with or develops metabolic acidosis, oliguria, coagulopathy and shock. Gastric aspirates increase, abdominal distension worsens with abdominal wall oedema and erythema and AXR shows gaseous distension, focal or widespread pneumatosis, portal venous ileus, pneumoperitoneum
 - Lab findings: high WCC, low plt, metabolic acidosis and rising CRP
- Non operative management
 - Bowel rest
 - Gastric decompression
 - IV fluid resuscitation, ABx
 - NICU admission
- Operative management
 - Clinical deterioration despite optimal non operative care
 - Perforation
- Any segment of bowel can be affected, but the pattern goes: any segment of SB/LB → multiple isolated SB lesions → necrotising enterocolitis totalis
- Consequences
 - Short gut syndrome
 - Long term TPN
 - Intestinal failure
 - Death
- Gestational age is the main predictor of outcome

Lecture: Abdominal pain

Abdominal pain

- 20% of all children present with abdominal pain, however only 5% require hospitalisation or intervention
- 10% of these children have recurrent abdominal pain

Work up of abdominal pain

- Take a thorough history and examination
 - Remember posture (legs flexed at hip in peritonism)
- Some investigations may be useful. X-ray is often not useful, and USS can be useful in a non-obese child. Laboratory tests done usually look for inflammatory markers, liver function tests and pancreatic enzymes
- Repeated clinical examination has the highest sensitivity, more so than any other investigation

When is it a surgical abdomen?

- Bilious vomiting
- Bloody stool
- Localised tenderness
- Distension and guarding
- Mass
- Inguinoscrotal swelling

Causes of abdominal pain by age

- In the first three months, the most common cause of abdominal pain is infantile colic, with attacks of screaming, unable to be consoled, drawing of legs up. This may be secondary to gas, oesophagitis
- In first few years of life
 - Congenital abnormalities
 - Incarcerated inguinal hernia
 - Intussusception
 - Intestinal volvulus
 - GI perforation
- In older children
 - Acute appendicitis
 - Trauma
 - Testicular torsion or rupture of ovarian cyst
 - Pancreatitis
 - Meckel's diverticulum
 - Cholecystitis
 - Primary peritonitis
 - Intestinal worm infestation
 - RAP*
- Remember non-surgical causes of abdominal pain: DKA, hyperthyroidism, Addison's disease, hypercalcemia, lead poisoning, porphyria, pneumonia

Appendicitis

- Most common in the second decade of life
- Classic triad: pain before vomiting, fever is uncommon
- Time course
 - 0-24 hrs – central colicky pain, tender RIF
 - 36-48 hrs – localised pain in RIF
 - >48 hrs – risk of perforation

Intussusception

- This is uncommon under 3 months and over 6 years, otherwise should consider lead point causing intussusception
- 75% idiopathic

- Treatment
 - Non-operative fluoroscopy
 - Surgery if perforation, failed reduction, sick child

Lecture: Head and neck lumps in children

Where is the problem?

- Midline
 - Midline cervical lymph node
 - Midline dermoid cyst: Can connect intracranially
 - Thyroglossal duct cyst or thyroid gland lesion: Skin coloured cyst, will move with tongue protrusion
 - MAIS: atypical mycobacterial infection causes a collar stud abscess, erythematous. Associated with positive Mantoux test
 - Plunging ranula – best examined from inside the mouth
- Lateral
 - Lateral cervical lymph node
 - Lateral cervical lymphadenitis: Enlarged painful swelling, diffuse redness
 - Lymphatic malformation, lymphangioma or cystic hygroma: skin coloured cyst, can compress on airways
 - Lateral external dermoid cyst: appears between 8 – 20 months, attached to deeper structure, often appears at the angle of the fronto-nasal process and the maxillary process. These can occasionally occur elsewhere, e.g. in the midline or in the post-auricular area
 - Branchial remnant, e.g. fistula (can sometimes have clear fluid discharged) or cyst (found in older children, necrosed tissue after recurrent URTIs)
 - Sternomastoid 'tumour'

What do we need to know about the lump?

- Location
- Association with underlying structures and overlying tissue
- Association with regional lymphadenopathy
- Protrude tongue to see if it moves – related to deep cervical fascia
- Transillumination

Lecture: Inguino-scrotal disorders

What type?

- Inguinal-scrotal
 - Inguinal hernia: reducible, non-reducible
 - Hydrocoele: primary, secondary
 - Undescended testis: normal line of descent, ectopic
- External genitalia
 - Acute scrotum: testicular torsion, torsion testicular appendage, epididymo-orchitis
 - Penile pathology: phimosis, hypospadias
 - Ambiguous external genitalia

Cases presented

- Hydrocoele: Positive transillumination + can get above the lump

- Undescended testis: If not descended in first 3 months, unlikely to descend by itself, needs surgical intervention between 6-12 months of age.
- Torsion of the appendix testis: Common in prepubertal boy, history of minor trauma unrelated to pathology, erythema of the scrotum, one tender spot
- Testicular torsion: More common in pubertal boy, with severe pain + vomiting
- Varicocele: Bag of worms appearance, more painful after standing, always left side
- Phimosis: May respond to topical steroids if mild, but usually circumcision needed
- Hypospadias: 1 in 200 males, sometimes with family history (10-15%), more common in ART, chordee (bend of penis) contributes to severity

Lecture: Vomiting in children

When is it a surgical cause of vomiting?

- Bile stained – yellow coloured
- Bilious – green
- Persistent
- Projectile
- Blood stained
- Loss of weight or FTT
- Unwell child
- Associated mass

Vomiting in the neonate

- Mucous vomits during first day is common – swallowed blood and mucus
- Other causes are usually medical, surgical causes are usually bilious
- *Malrotation* often presents in first week with bilious vomiting, without abdominal distension and pain. A contrast study is the gold standard to identify duodeno-jejunal junction, which should be level with pylorus and in the midline.
- *Bowel atresia* is most common in the small bowel, but will present with abdominal distension, bilious vomiting and failure to pass meconium. This needs nasogastric decompression with surgery.
- *Hirschsprung disease* is also known as congenital aganglionic megacolon with the absence of ganglion cell in the large bowel (functional obstruction). These children present with abdominal distension, bile-stained vomiting and not having passed meconium in first 24 hours. AXR will show multiple dilated large bowel loops with air, and diagnosis is demonstrated by histological rectal suction biopsy.

Vomiting in infancy

- Infectious are the commonest cause of vomiting, e.g. tonsillitis, otitis, pneumonia, meningitis, UTI
- Gastro causes may include
 - *Reflux*: Common, referred to as regurgitant non-bilious vomiting, sometimes with blood. The infant has no weight loss and is not distressed with vomiting.
 - *Pyloric stenosis*: Most common cause of surgical vomiting, mostly affects MALES 75%. 1 in 450 incidence, with 20% with a family history (polygenic inheritance). The infant presents with projectile non-bilious vomiting and hungry. Visible gastric peristalsis or an olive mass may be seen in

the abdomen. Diagnosis is confirmed by ultrasound.

- Malrotation
- Gastroenteritis
- Malabsorption
- Obstructed hernia
- Food allergy

Vomiting in the older child

- Appendicitis is the most common cause
- There are many other causes